GCP Basic

(Google Cloud Platform)



강사 : 고병화

GCP 개요

- 클라우드 서비스란
- 서버나 스토리지와 같은 자원을 클라우드에서 제공하여 소프트웨어 개발 및 실행을 할 수 있게 해주는 서비스
- 구글(GCP), 아마존(AWS), 마이크로소프트(Azure) 에서 유료로 제공

- 클라우드 서비스 분류
- (1) IaaS (Infrastructure as a Service)
  : 서버나 OS, 스토리지,네트워크 등의 시스템 인프라를 On-Demand로 제공하는 서비스,
  (Compute Engine, Cloud Virtual Network, Cloud DNS)
- (2) PaaS (Platform as a Service)
  : 시스템 인프라와 애플리케이션 개발에 필요한 미들웨어를 묶어서 제공, 개발 환경
  개발에 필요한 라이브러리는 물론이고, 실제 운용 환경에 대한 배포나 시스템 감시 등의
  기능을 제공 , (App Engine, Cloud SQL)
- (3) SaaS (Software as a Service)
  : 이미 만들어진 애플리케이션 기능을 서비스로 제공, 애플리케이션 프로그램
  (BigQuery, AutoML)

GCP 개요

GCP가 제공하는 주요 서비스

- 컴퓨트(Compute)
- GCE (Compute Engine) : VM 서비스
- GKE (Kubernetes Engine) : 컨테이너 오케스트레이션
- GAE (App Engine) : 운용이 쉽고 확장성을 가진 애플리케이션 플랫폼

- 스토리지/데이터베이스
- Cloud Storage : 오브젝트 저장소
- Cloud SQL : RDBMS의 관리형 서비스
- Cloud Datastore : NoSQL 데이터베이스

- 네트워킹
- Cloud Virtual Network : 가상 네트워크
- Cloud Load Balancing : 부하 분산기
- Cloud DNS : DNS의 관리형 서비스

GCP 개요

GCP가 제공하는 주요 서비스

- 빅데이터
- BigQeury: 대규모 데이터 분석에 맞는 데이터웨어 하우스
- Cloud Bigtable : 레이턴시가 낮고 높은 처리량을 가진 NoSQL 데이터베이스
- Cloud Spanner : 강한 견고성(Strong Consistency)과 가용성을 겸한 분산형 관계형 데이터베이스
- Cloud Dataflow : 스트리밍 처리와 배치 처리를 위한 관리형 데이터 처리 서비스
- Cloud DataProc : Spark/Hadoop의 풀 매니지먼트 서비스
- Cloud Pub/Sub : 실시간으로 신뢰성이 높은 메시징과 데이터 스트리밍 서비스를 제공
- 머신러닝
- Cloud Vertex AI : 머시러닝 통합 분석 서비스
- Cloud Vision API : 이미지 분석 서비스
- Cloud Speech API : 음성 텍스트 변환 서비스를 제공 하는 API
- Cloud Translation API : 텍스트 번역 서비스를 제공하는 API
- Cloud Natural Language API : 자연어 분석 서비스를 제공하는 API
- Cloud Machine Learning Engine : TensorFlow를 사용한 머신러닝 서비스
- 운용 관리도구
- Cloud Console : 통합 관리 콘솔, 대시보드 제공
- Cloud Shell : 브라우저에서 이용할 수 있는 명령 라인 콘솔
- Stackdriver : 통합 감시 서비스
- Cloud IAM : 사용자 접근 관리

GCP 개요

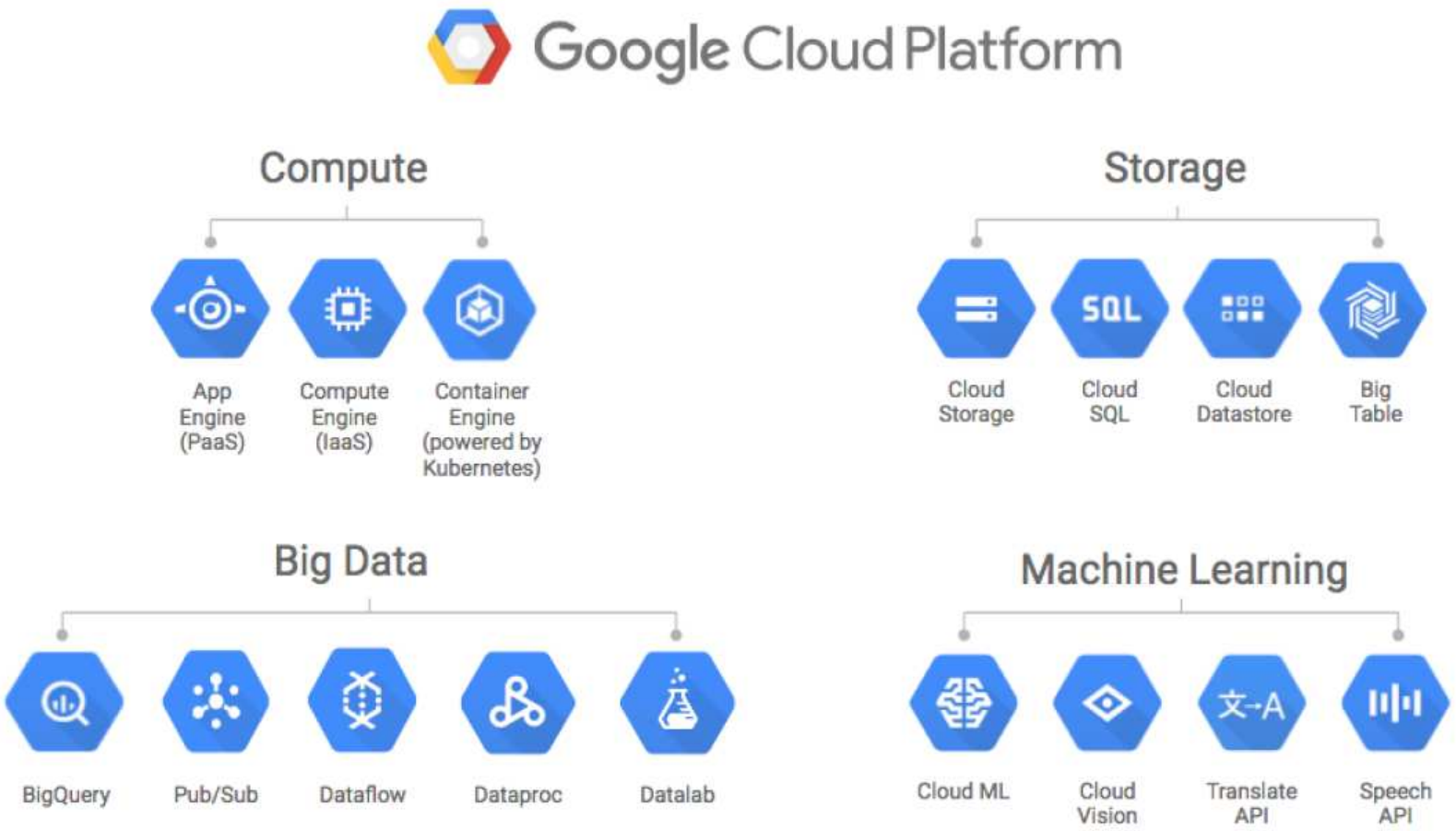
- GCP 와 AWS/Azure 서비스와 비교

Amazon Web Services	Microsoft Azure	Google Cloud Platform
Amazon EC2	Azure Virtual Machines	Google Compute Engine
AWS Elastic Beanstalk	Azure App Services	Google App Engine
Amazon EC2 Container Service	Azure Container Service	Google Kubernetes Engine
Amazon DynamoDB	Azure Cosmos DB	Google Cloud Bigtable
Amazon Redshift	Microsoft Azure SQL Data warehouse	Google BigQuery
Amazon Lambda	Azure Functions	Google Cloud Functions
Amazon S3	Azure Blob Storage	Google Storage
AWS Direct Connect	Azure ExpressRoute	Google Cloud Interconnect
AWS SNS	Azure Service Bus	Google Cloud Pub/Sub
AWS Cloudwatch	Application Insights	Stackdriver Monitoring

GCP 서비스 유형
https://cloud.google.com/docs/overview/cloud-platform-services

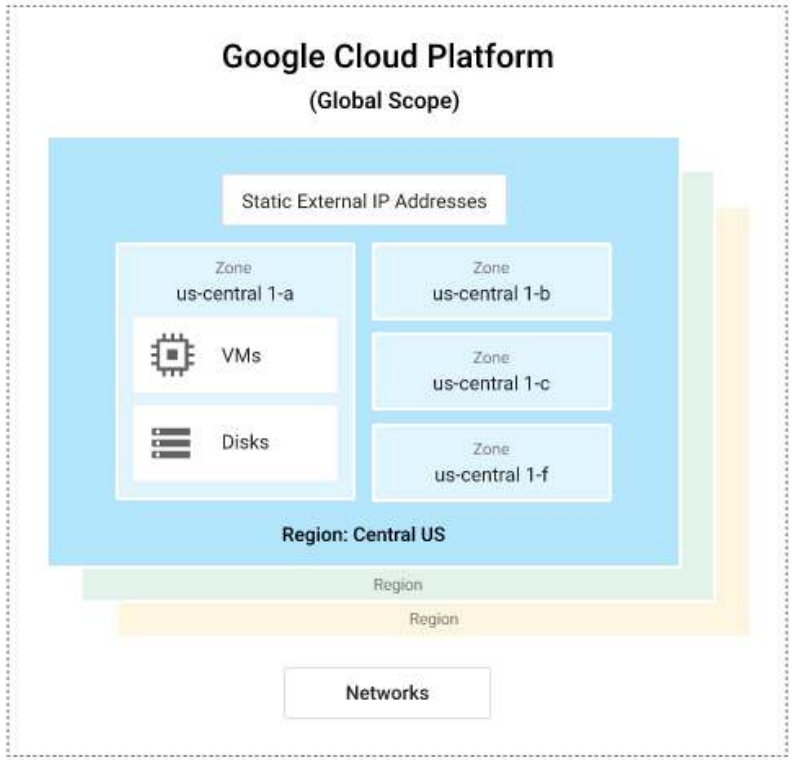


GCP 서비스 유형


Google Cloud Platform
Compute
App Engine (PaaS)
Compute Engine (IaaS)
Container Engine (powered by Kubernetes)
Storage
Cloud Storage
SQL
Cloud SQL
Cloud Datastore
Big Table
Big Data
BigQuery
Pub/Sub
Dataflow
Dataproc
Datalab
Machine Learning
Cloud ML
Cloud Vision
文→A
Translate API
Speech API

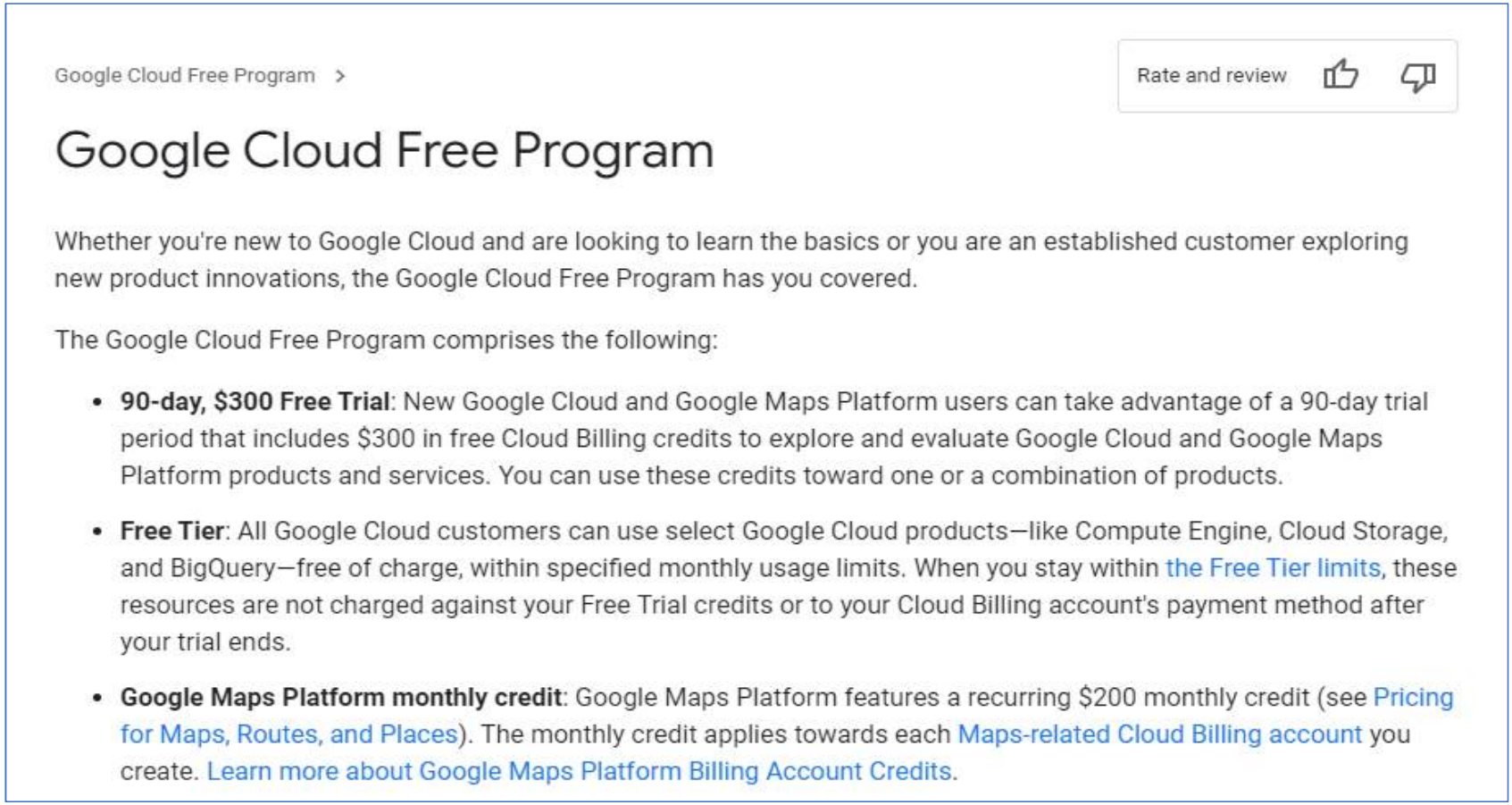
GCP 데이터 센터
https://www.google.com/about/datacenters/locations/

- Global/Region/Zone 리소스



GCP 무료 크레딧 신청

https://cloud.google.com/free/docs/gcp-free-tier



Google Cloud Free Program >

Rate and review

Google Cloud Free Program

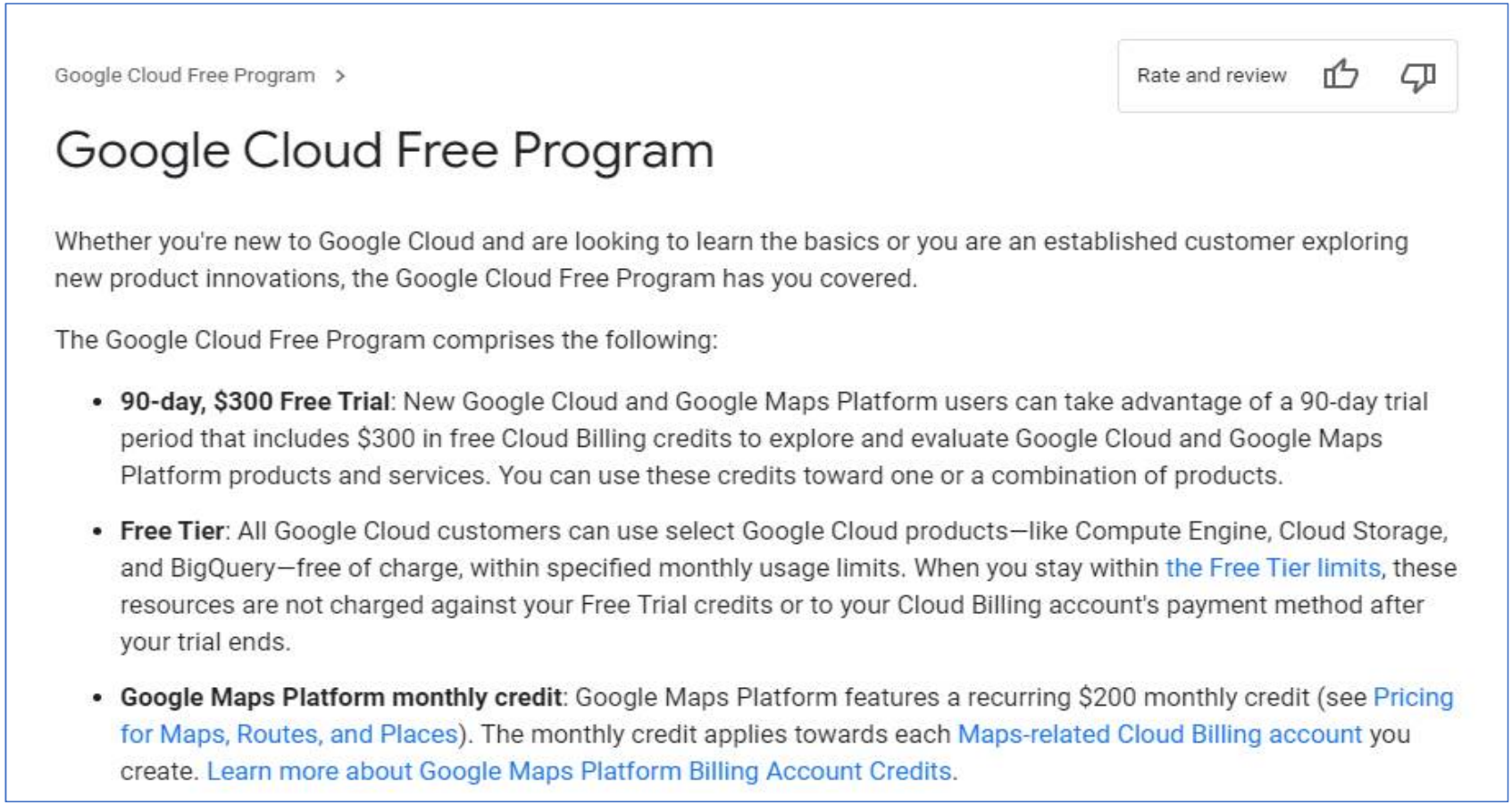
Whether you're new to Google Cloud and are looking to learn the basics or you are an established customer exploring new product innovations, the Google Cloud Free Program has you covered.

The Google Cloud Free Program comprises the following:

- **90-day, $300 Free Trial**: New Google Cloud and Google Maps Platform users can take advantage of a 90-day trial period that includes $300 in free Cloud Billing credits to explore and evaluate Google Cloud and Google Maps Platform products and services. You can use these credits toward one or a combination of products.
- **Free Tier**: All Google Cloud customers can use select Google Cloud products—like Compute Engine, Cloud Storage, and BigQuery—free of charge, within specified monthly usage limits. When you stay within the Free Tier limits, these resources are not charged against your Free Trial credits or to your Cloud Billing account's payment method after your trial ends.
- **Google Maps Platform monthly credit**: Google Maps Platform features a recurring $200 monthly credit (see Pricing for Maps, Routes, and Places). The monthly credit applies towards each Maps-related Cloud Billing account you create. Learn more about Google Maps Platform Billing Account Credits.

GCP 무료 크레딧 신청

https://cloud.google.com/free/docs/gcp-free-tier



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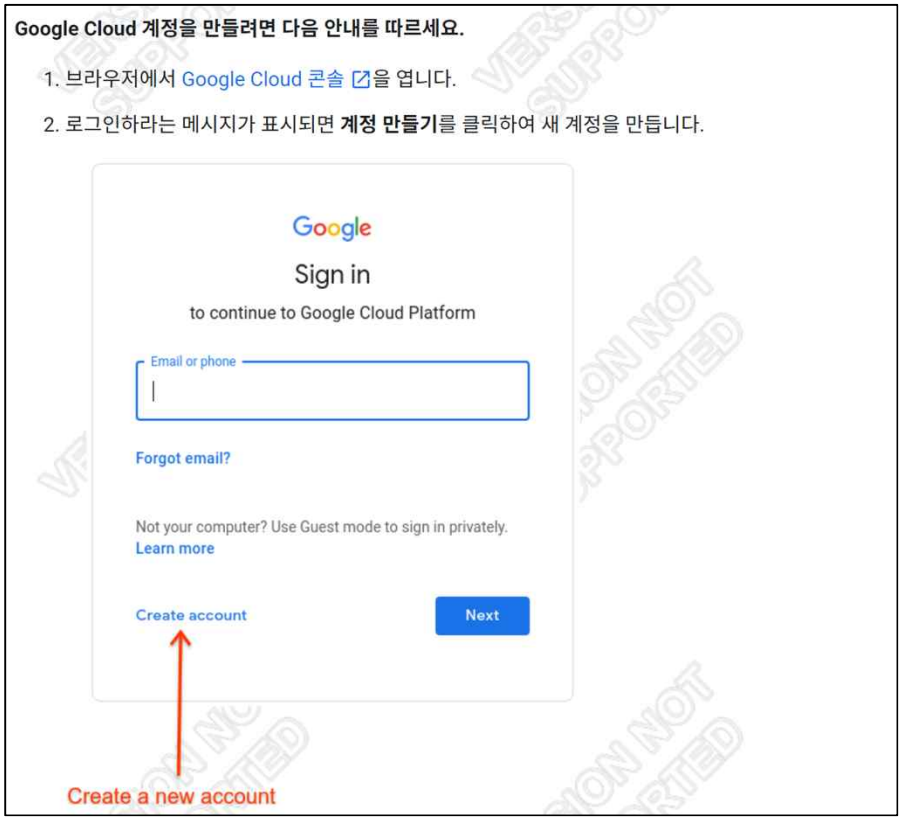
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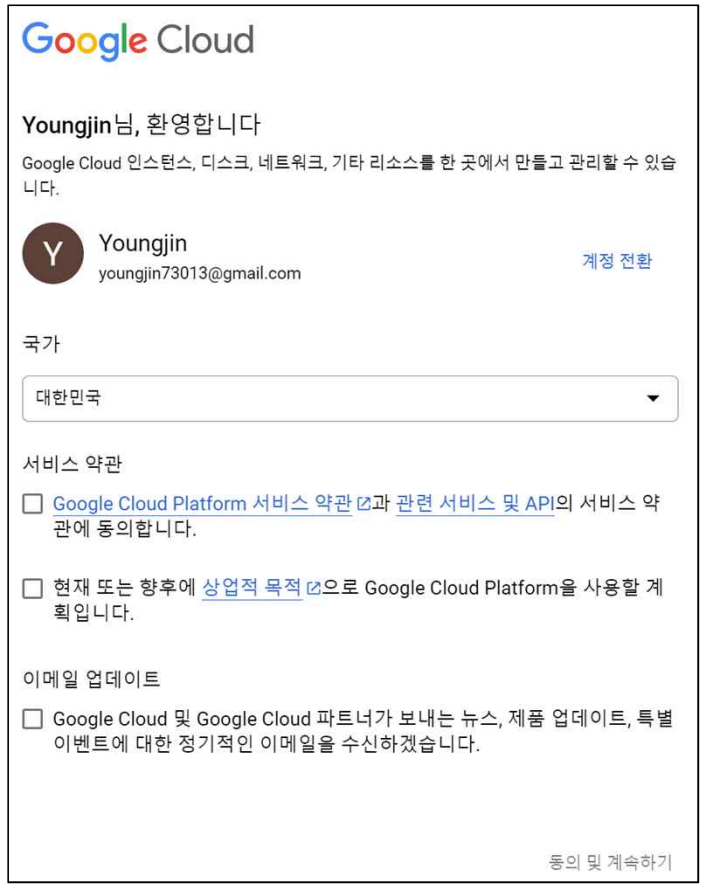
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GCP 계정 생성하는 방법

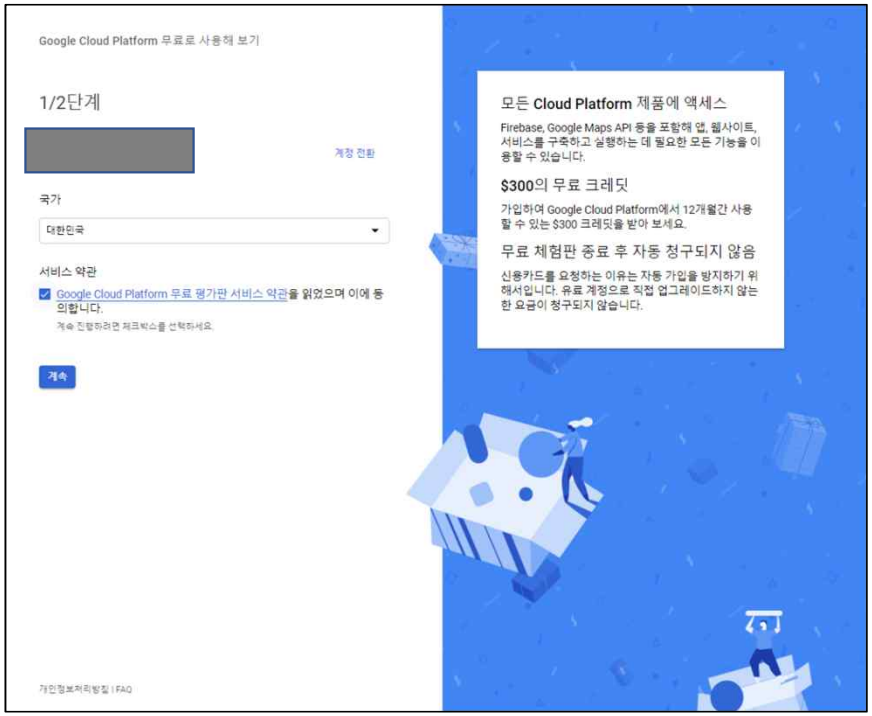
https://cloud.google.com/
https://console.cloud.google.com/?hl=ko



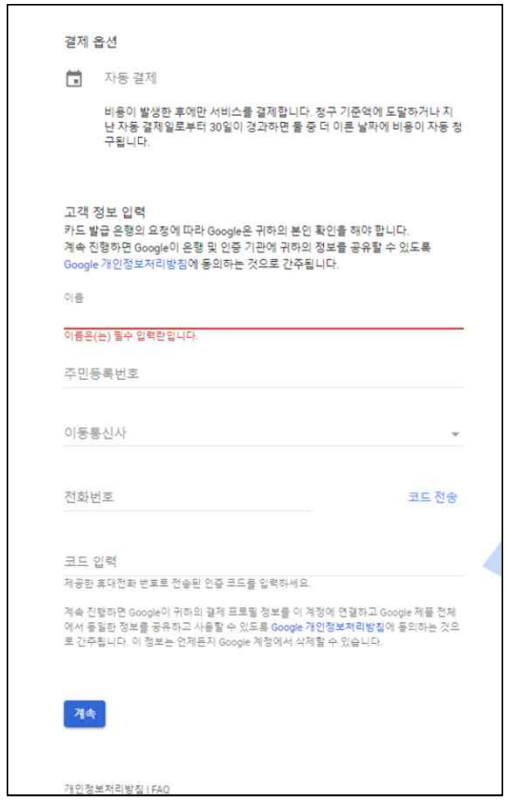


결제 정보 입력 및 무료계정 활성화

https://sidepower.tistory.com/19

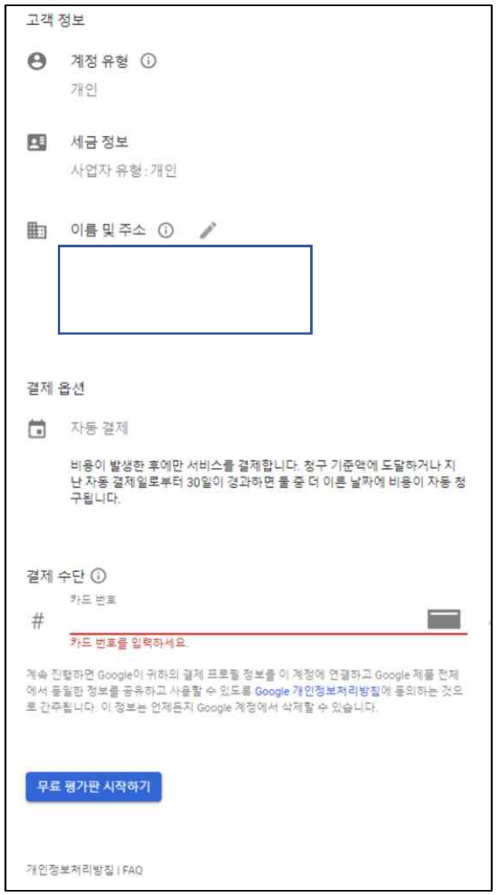






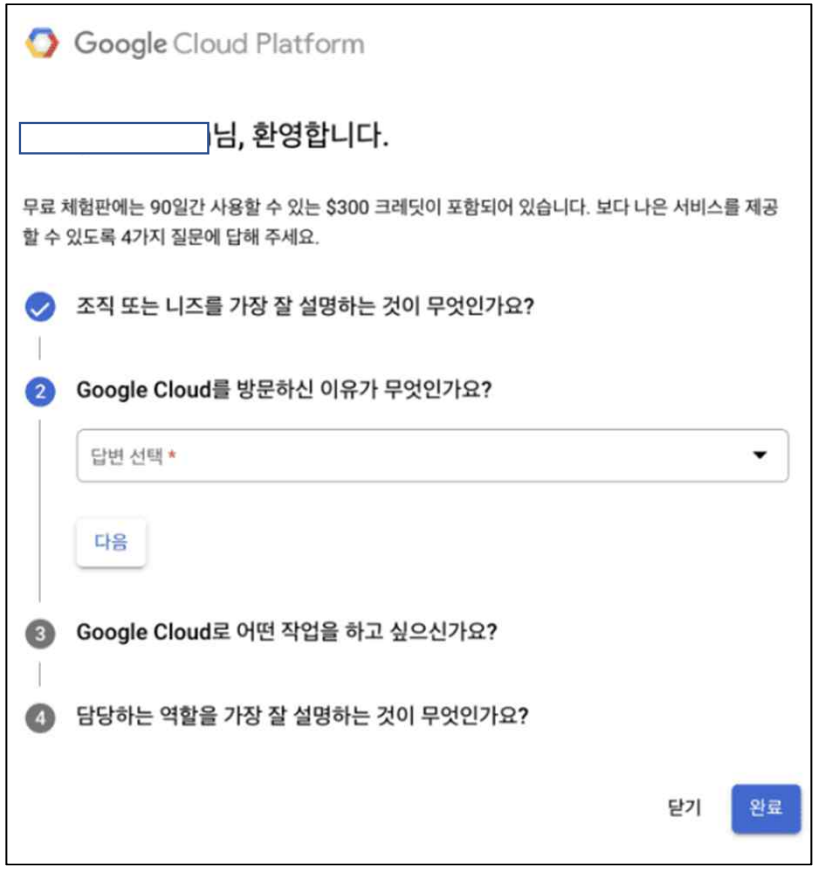
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https://sidepower.tistory.com/19



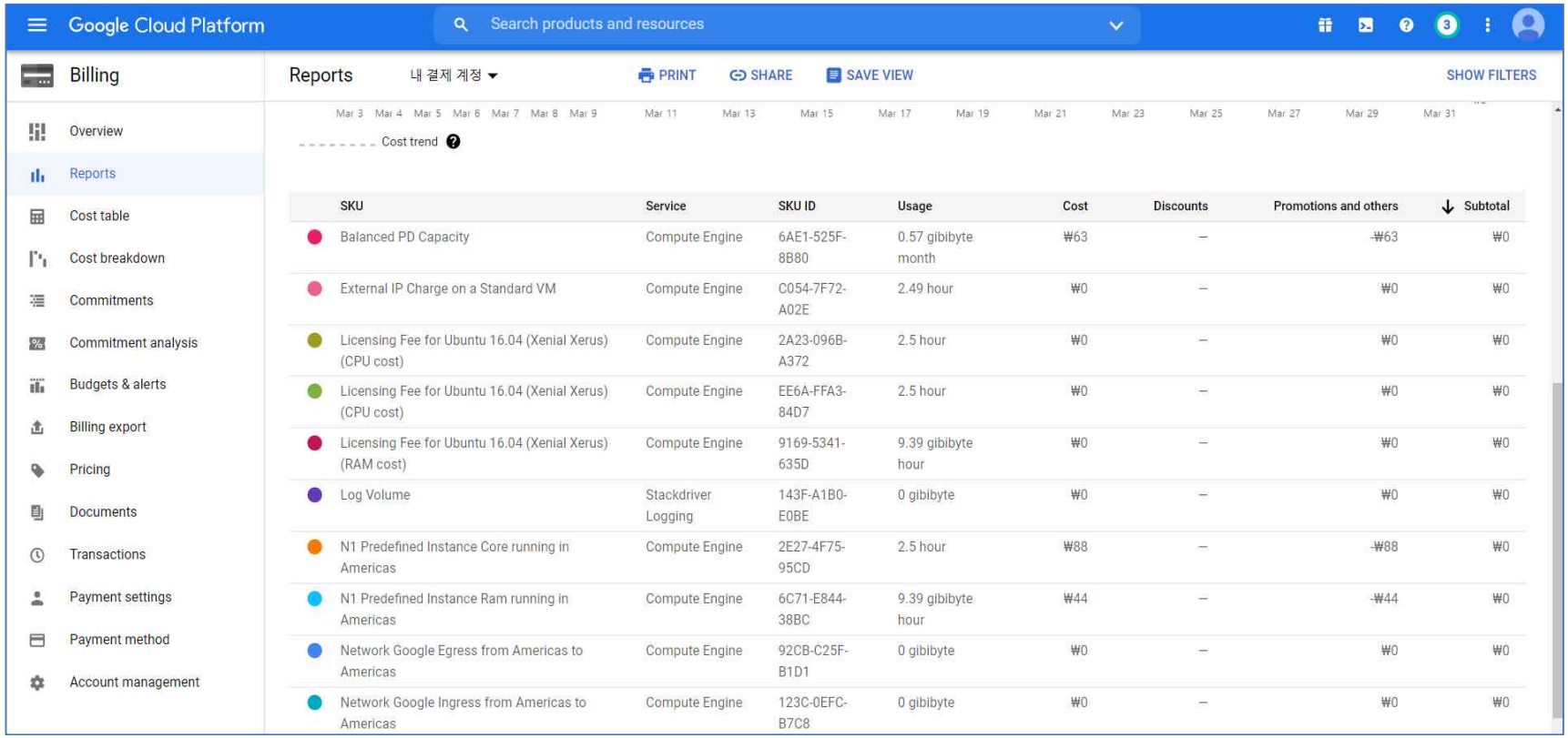
본인 인증이 완료되면 카드를 등록합니다.
본인 인증한 명의의 카드만 등록이 가능합니다.

카드 번호와 관련 정보를 입력하고, 무료 평가판 시작하기 버튼을 클릭하면 무료 구글 클라우드 계정 사용이 가능하게 됩니다



Billing Reports

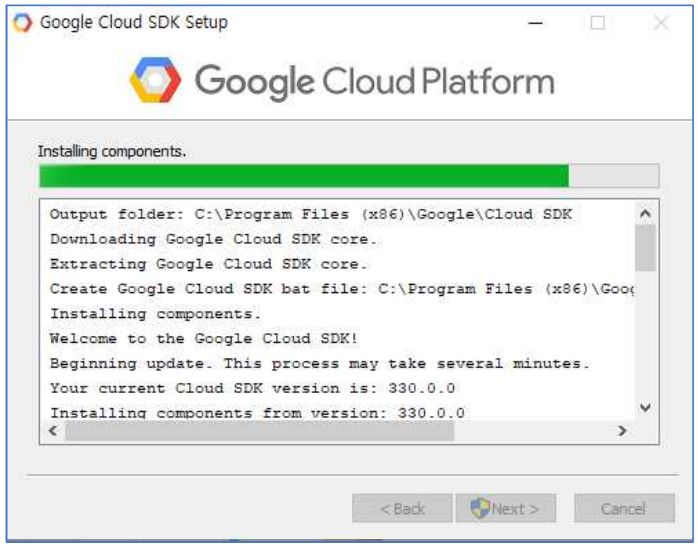
: QuickLab 무료 사용 예시

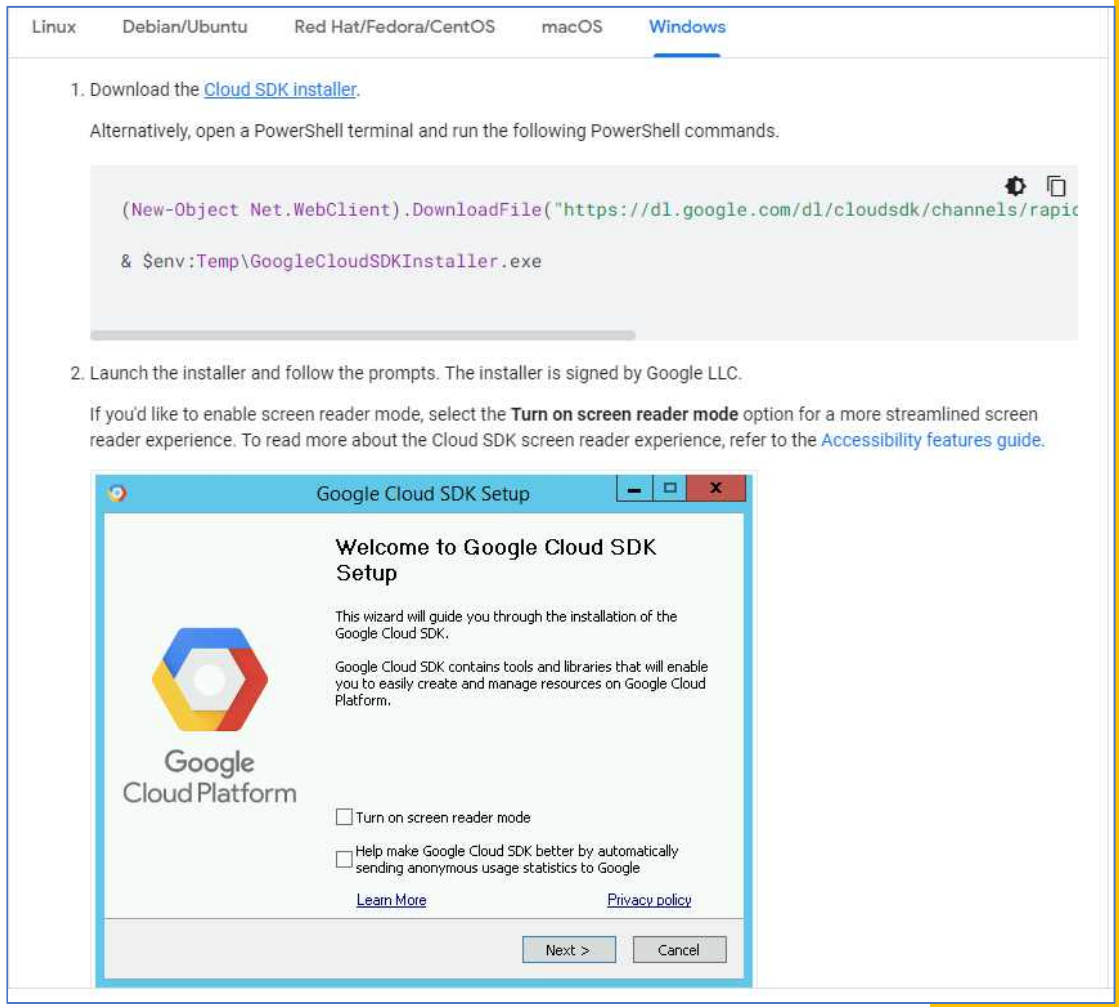


SKU	Service	SKU ID	Usage	Cost	Discounts	Promotions and others	Subtotal
Balanced PD Capacity	Compute Engine	6AE1-525F-8B80	0.57 gibibyte month	₩63	–	-₩63	₩0
External IP Charge on a Standard VM	Compute Engine	C054-7F72-A02E	2.49 hour	₩0	–	₩0	₩0
Licensing Fee for Ubuntu 16.04 (Xenial Xerus) (CPU cost)	Compute Engine	2A23-096B-A372	2.5 hour	₩0	–	₩0	₩0
Licensing Fee for Ubuntu 16.04 (Xenial Xerus) (CPU cost)	Compute Engine	EE6A-FFA3-84D7	2.5 hour	₩0	–	₩0	₩0
Licensing Fee for Ubuntu 16.04 (Xenial Xerus) (RAM cost)	Compute Engine	9169-5341-635D	9.39 gibibyte hour	₩0	–	₩0	₩0
Log Volume	Stackdriver Logging	143F-A1B0-E0BE	0 gibibyte	₩0	–	₩0	₩0
N1 Predefined Instance Core running in Americas	Compute Engine	2E27-4F75-95CD	2.5 hour	₩88	–	-₩88	₩0
N1 Predefined Instance Ram running in Americas	Compute Engine	6C71-E844-38BC	9.39 gibibyte hour	₩44	–	-₩44	₩0
Network Google Egress from Americas to Americas	Compute Engine	92CB-C25F-B1D1	0 gibibyte	₩0	–	₩0	₩0
Network Google Ingress from Americas to Americas	Compute Engine	123C-0EFC-B7C8	0 gibibyte	₩0	–	₩0	₩0

GCP 클라우드 SDK 설치
https://cloud.google.com/sdk

- 우측화면에서 GoogleCloudSDKInstaller.exe을 다운 받아 설치(windows용은 GUI방식)





GCP 클라우드 SDK 설치
https://cloud.google.com/sdk

- 설치 후 오른 상단 화면에서 1번

  [1] Re-initialize this configuration

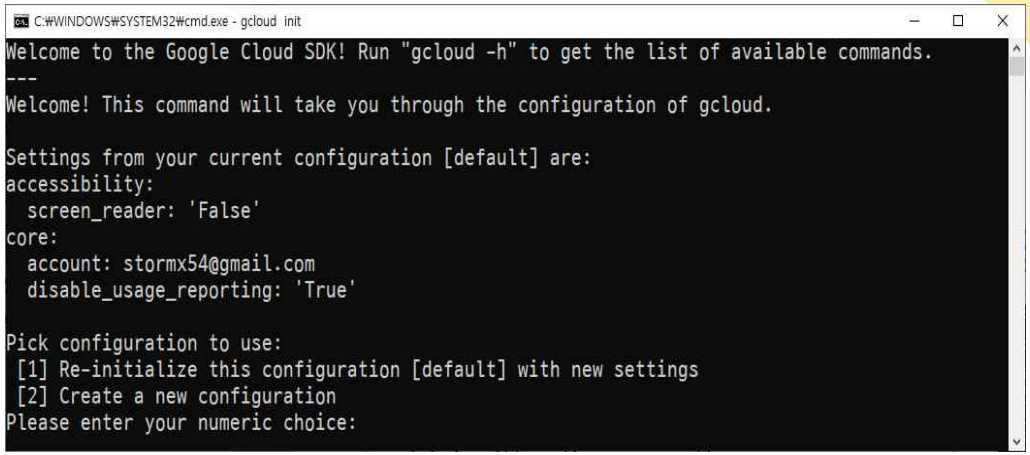
  [default] with new settings

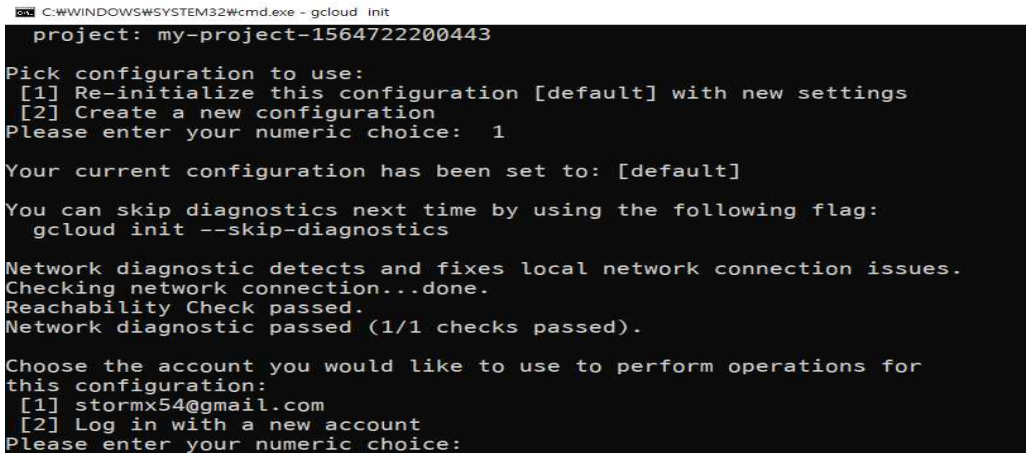
  를 선택하고
- 다음 메뉴에서 2번

  [2] Log in with a new account

  를 선택해 준다
- 브라우저에 계정 선택 화면이 나타나면 원하는 GCP로그인할 계정을 입력하거나 선택해준다

다음 화면에서 '허용'버튼을 눌러 준다

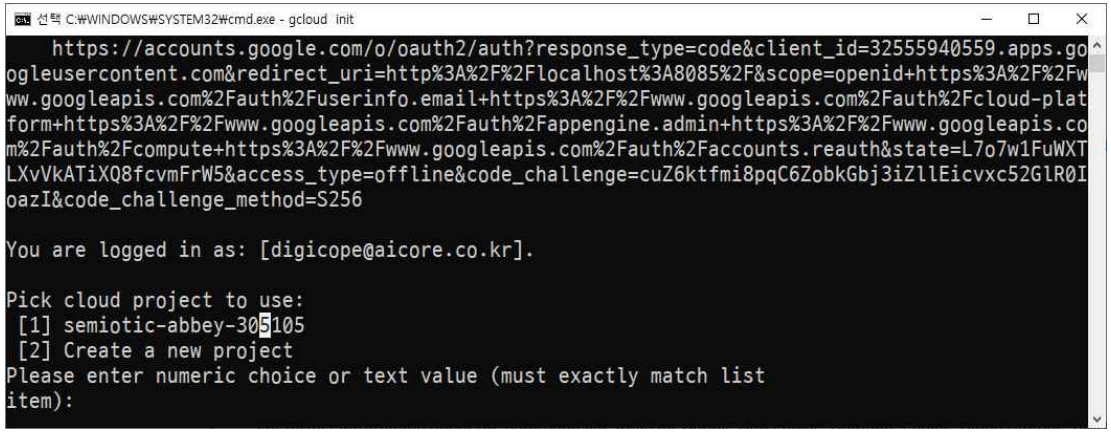




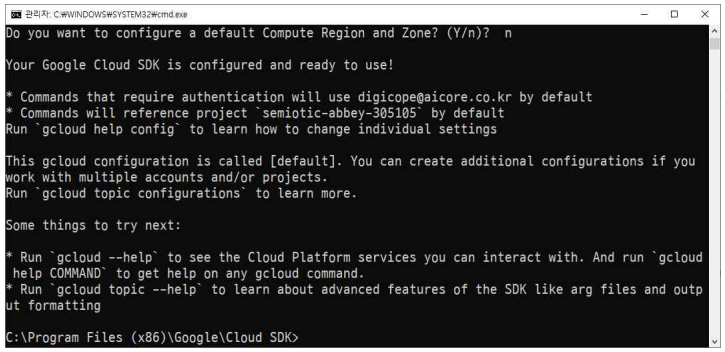
GCP 클라우드 SDK 설치

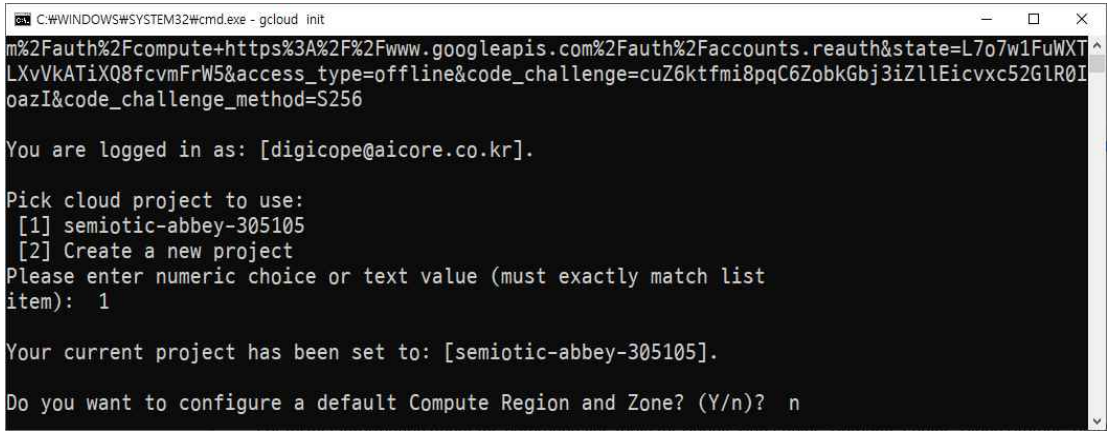
https://cloud.google.com/sdk

- 다음 화면에서 원하는 프로젝트 번호(예를 들면 1번)을 눌러 준다
- 우측 화면에서 2번을 선택하면 새로운 프로젝트를 만들 수 있다
- Do you want to configure a default Compute Region and Zone? (Y/n)? 선택에서 'n'를 입력한다



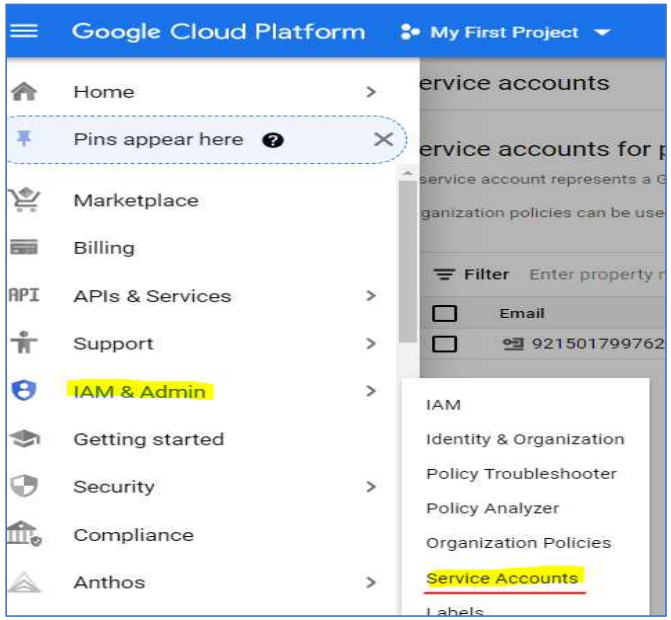
- 아래와 같이 설치가 완료된다



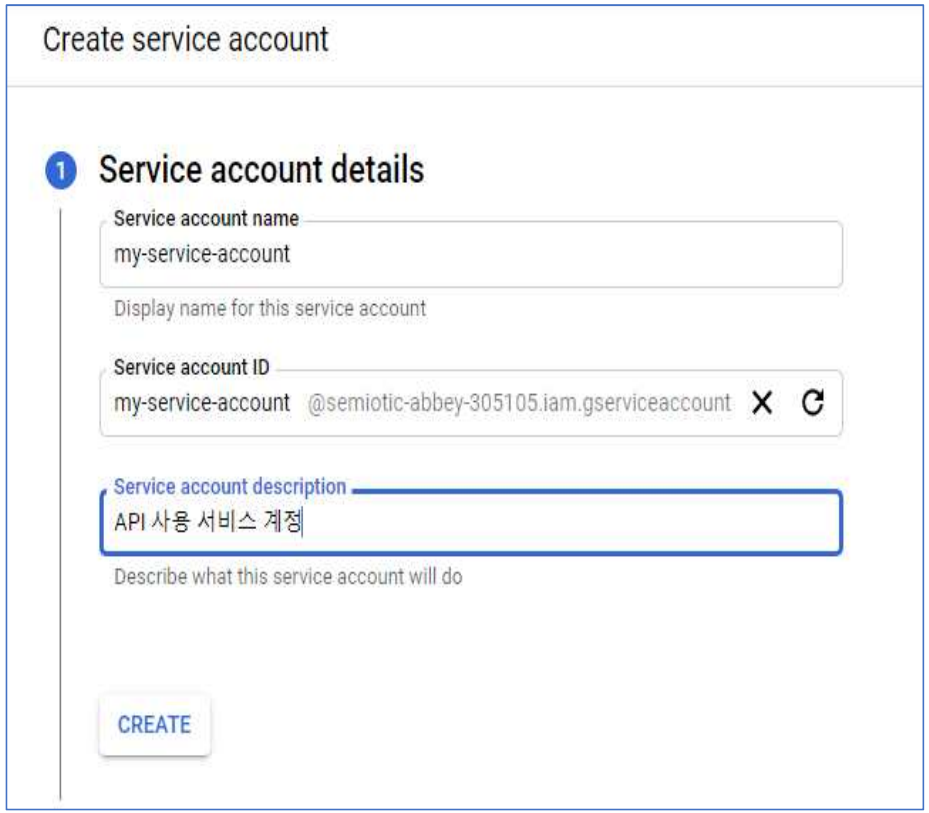


GCP 서비스 계정 키 설정

- GCP 콘솔에서 메뉴 IAM & Admin의 Service Accounts 를 선택한다

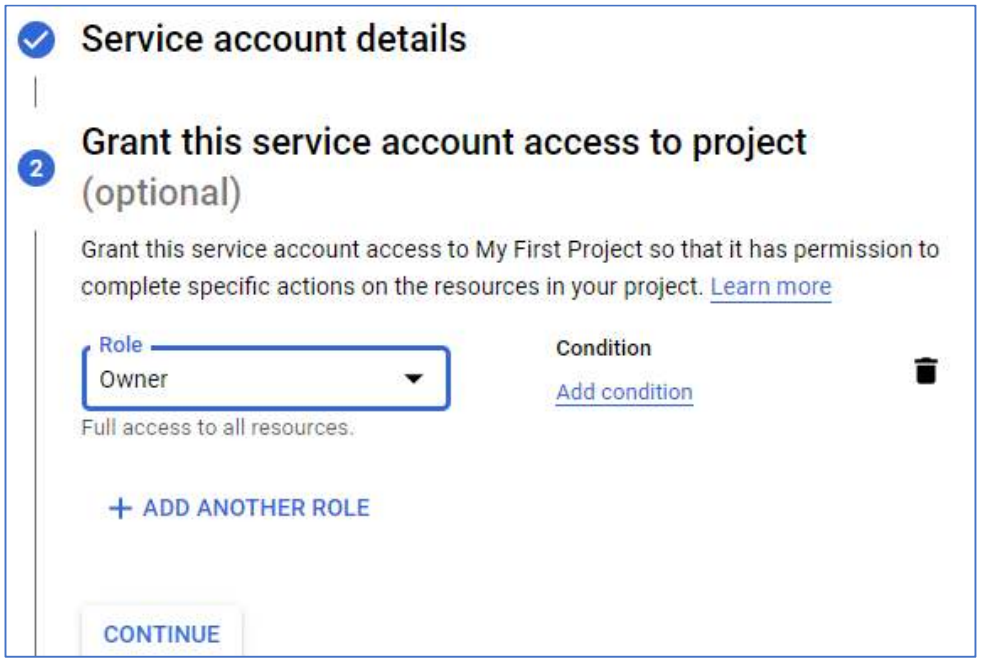


- CREATE SERVICE ACCOUNT를 누르고 우측과 같이 서비스 계정 이름과 서비스 계정 설명을 입력하고 CREATE을 누른다

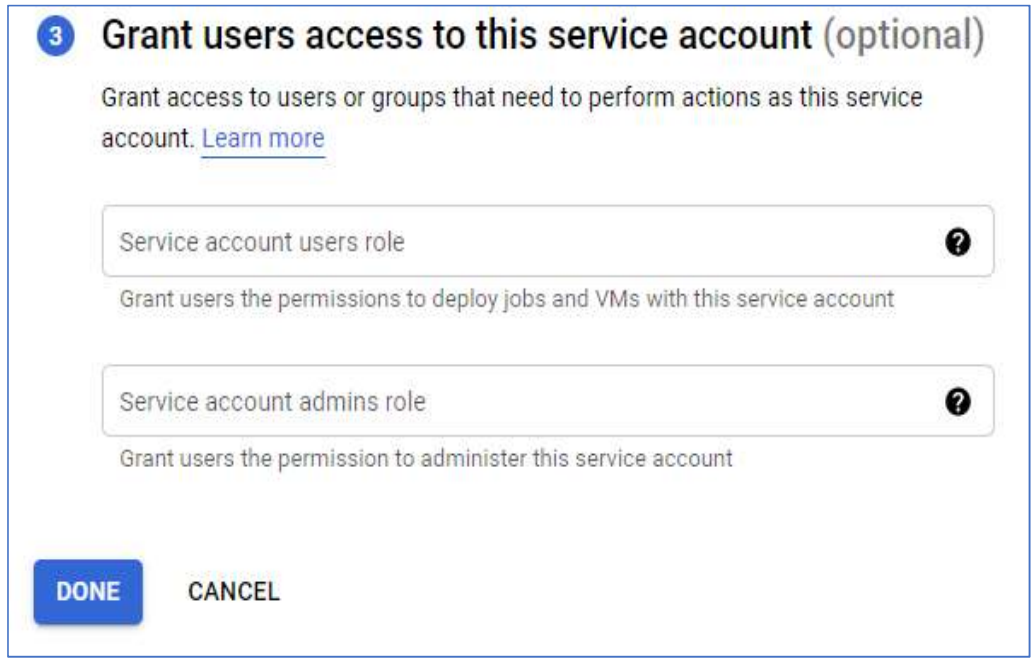


GCP 서비스 계정 키 설정

- Grant this service account access to project에서 Role을 아래와 같이 Owner로 선택하고 CONTINUE를 누른다

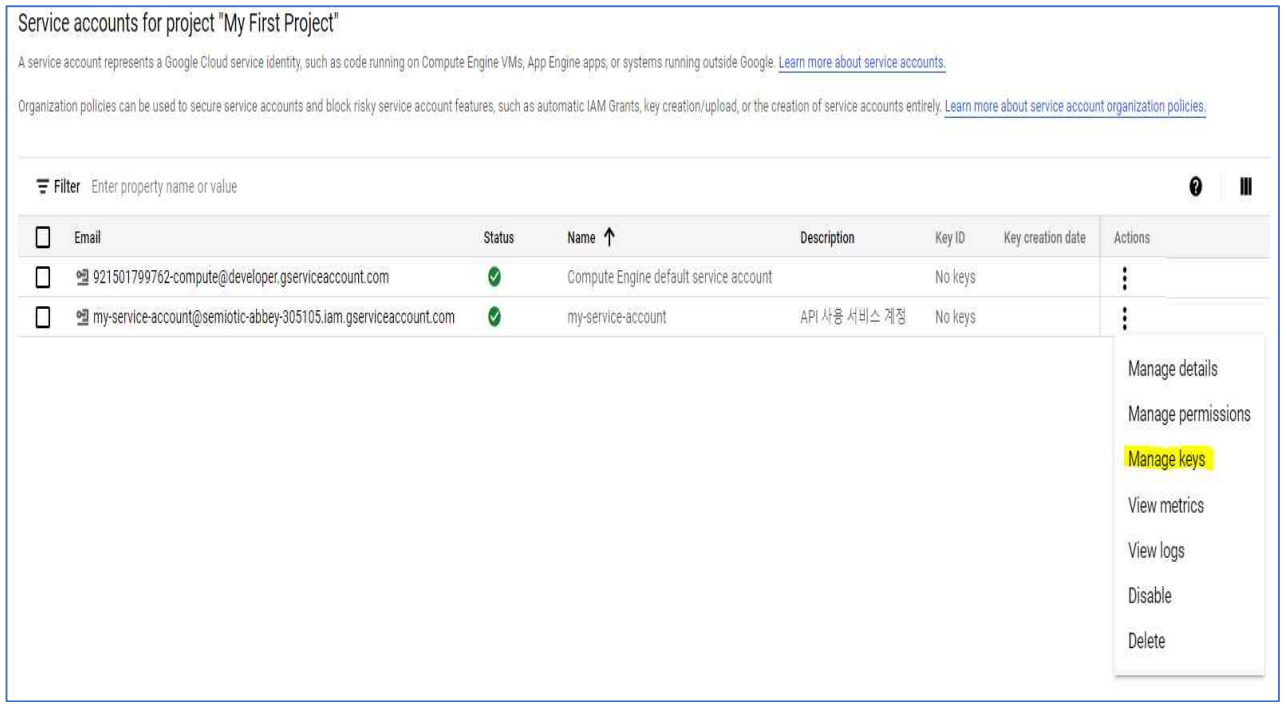


- 아래 메뉴에서는 입력하지 말고 그대로 DONE을 눌러준다

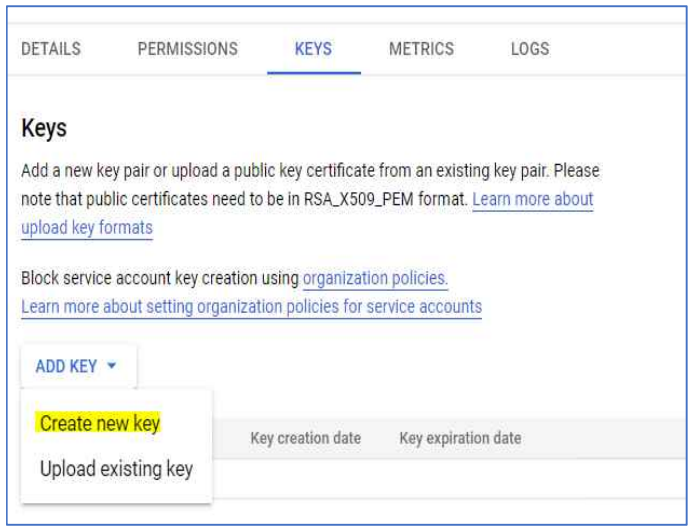


GCP 서비스 계정 키 설정

- 아래 목록에서 새로 생성된 두번째 서비스 계정의 Actions를 누르고 Manage keys를 누른다

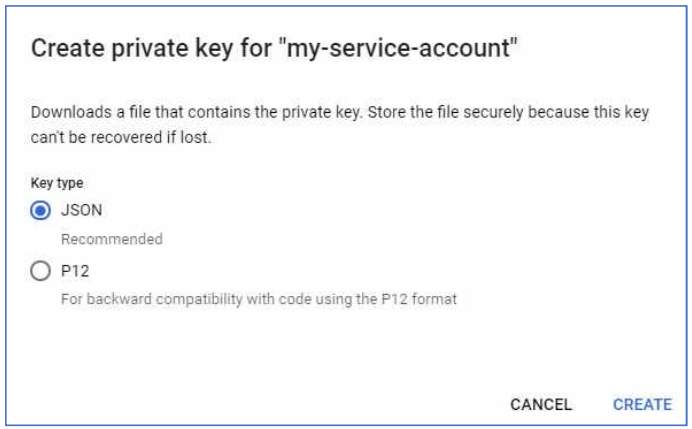


- 다음 아래 메뉴에서는 ADD KEY 에서 Create new key를 누른다

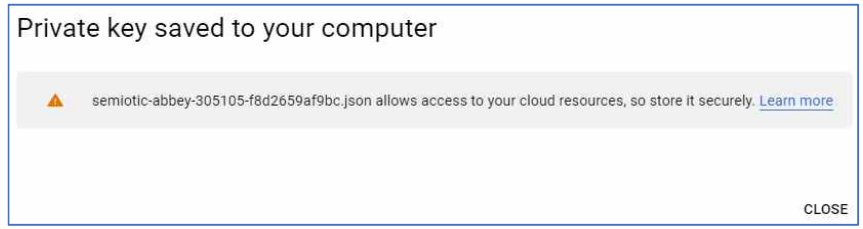


GCP 서비스 계정 키 설정

- 아래 창이 뜨면 JSON으로 선택하고 하단의 CREATE을 누른다



- 아래 창이 뜨면 하단의 CLOSE를 눌러준다

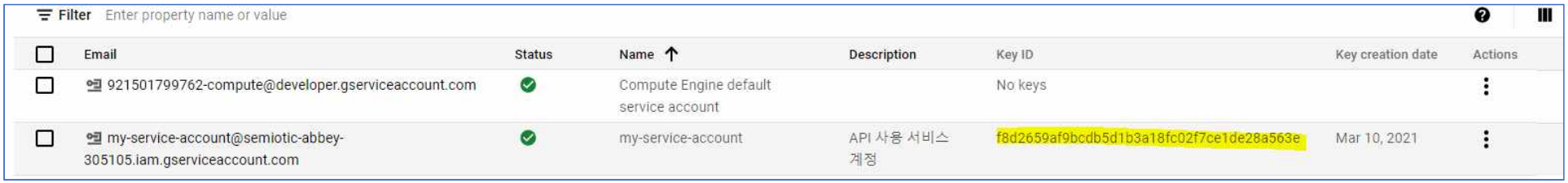


- 키생성이 완료 되면 아래와 같이 목록이 나타난다



Type	Status	Key	Key creation date	Key expiration date
	Active	f8d2659af9bcdb5d1b3a18fc02f7ce1de28a563e	Mar 10, 2021	Jan 1, 10000

- Service Accounts메뉴로 돌아가보면 아래와 같이 서비스 계정에 키가 추가 되었음을 확인할 수 있다

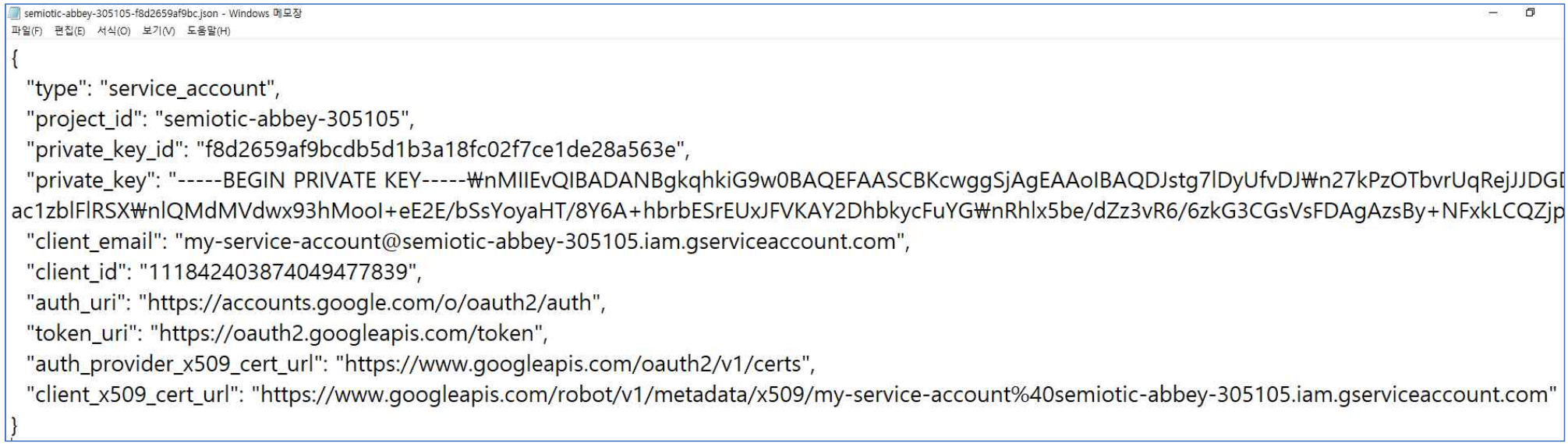


Filter Enter property name or value

Email	Status	Name	Description	Key ID	Key creation date	Actions
921501799762-compute@developer.gserviceaccount.com	✓	Compute Engine default service account		No keys		
my-service-account@semiotic-abbey-305105.iam.gserviceaccount.com	✓	my-service-account	API 사용 서비스 계정	f8d2659af9bcdb5d1b3a18fc02f7ce1de28a563e	Mar 10, 2021	

GCP 서비스 계정 키 설정

- Windows 의 다운로드 폴더에 JSON 형태의 키파일이 다운로드 되어 있을 것이다
  (예시 semiotic-abbey-305105-f8d2659af9bc.json)
- 메모장으로 열어 내용을 확인해볼 수 있다
- 이 파일을 C:₩KEY₩ 디렉토리에 복사해 놓는다



GCP 서비스 계정 키 설정

- Windows 의 "C:₩Program Files (x86)₩Google₩Cloud SDK₩cloud_env.bat" 파일을 메모장으로 열어 아래 빨간색 부분을 추가한다음 저장한다(파일명은 각자 다르다)

```
ECHO OFF
CLS
SET PATH=C:₩Program Files (x86)₩Google₩Cloud SDK₩google-cloud-sdk₩bin;%PATH%;
cd C:₩Program Files (x86)₩Google₩Cloud SDK
SET GOOGLE_APPLICATION_CREDENTIALS=C:₩KEY₩semiotic-abbey-305105-f8d2659af9bc.json
ECHO Welcome to the Google Cloud SDK! Run "gcloud -h" to get the list of available commands.
ECHO GOOGLE_APPLICATION_CREDENTIALS
ECHO ---
ECHO ON
```

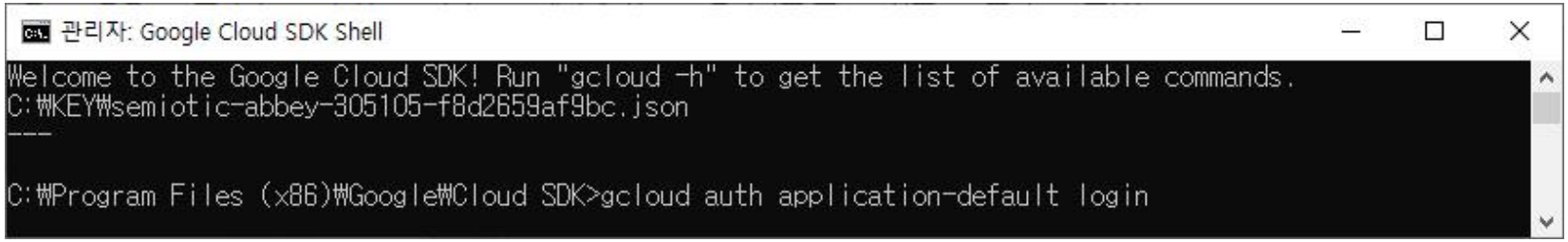
SDK설치 시 기본 credential 파일은 아래 경로에 생성되어 있다
C:₩Users₩storm₩AppData₩Roaming₩gcloud₩application_default_credentials.json

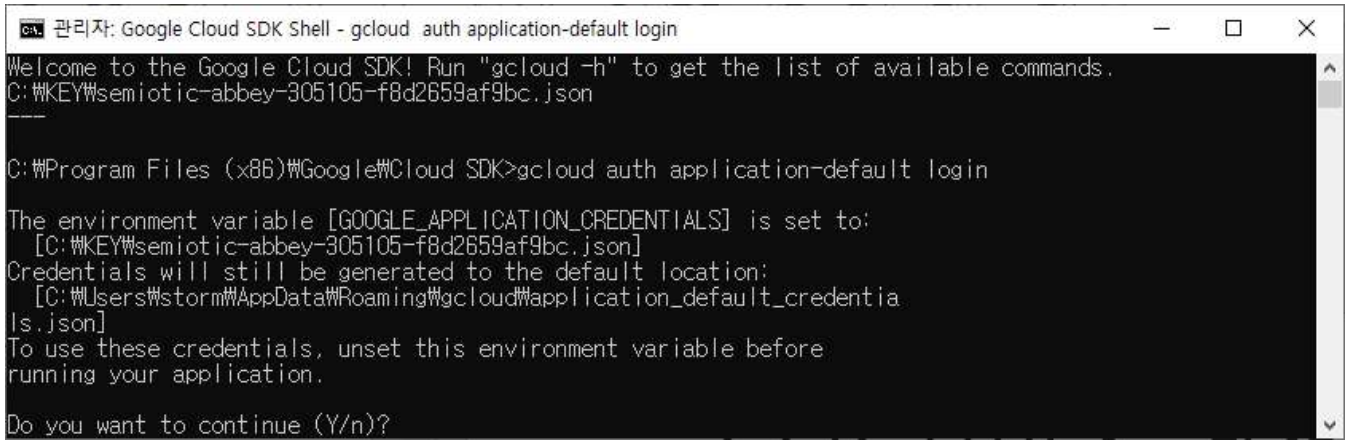
GCP 서비스 계정 키 설정



- Windows 의 바탕화면에서 Google Cloud SDK Shell을 실행시킨다
- 아래와 같이 gcloud 명령을 입력한 다음 'Y'를 입력해준다

  "gcloud auth application-default login"

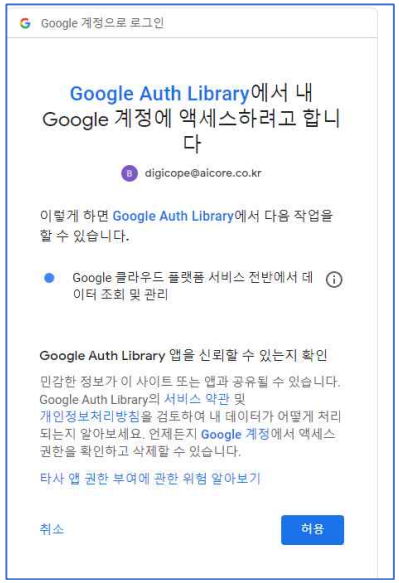


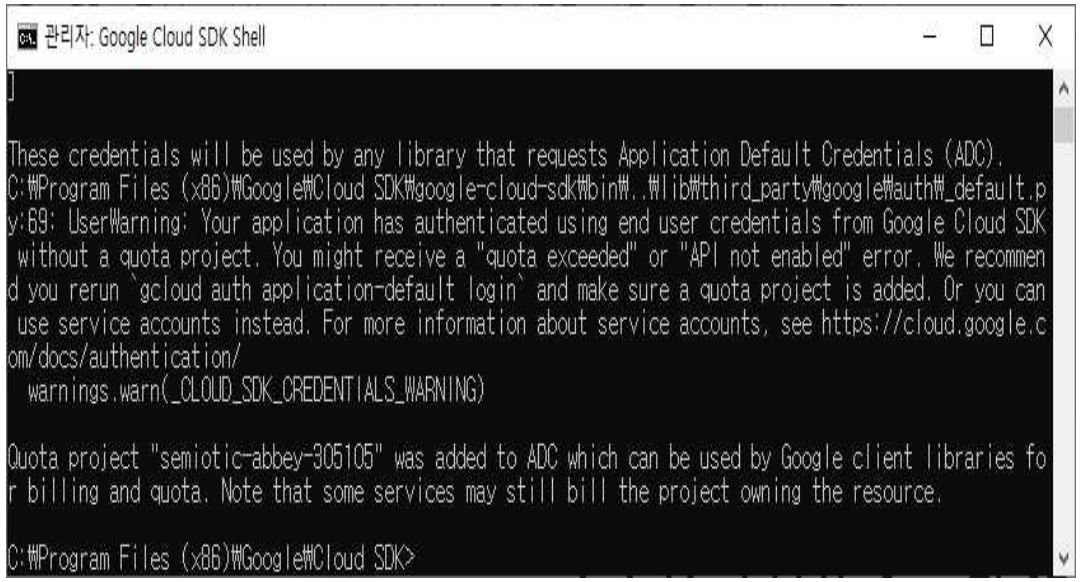


GCP 서비스 계정 키 설정

- 웹브라우저에서 구글 계정을 선택해주고 다음에 액세스를 허용해주면 Shell에서 아래와 같이 나타난다

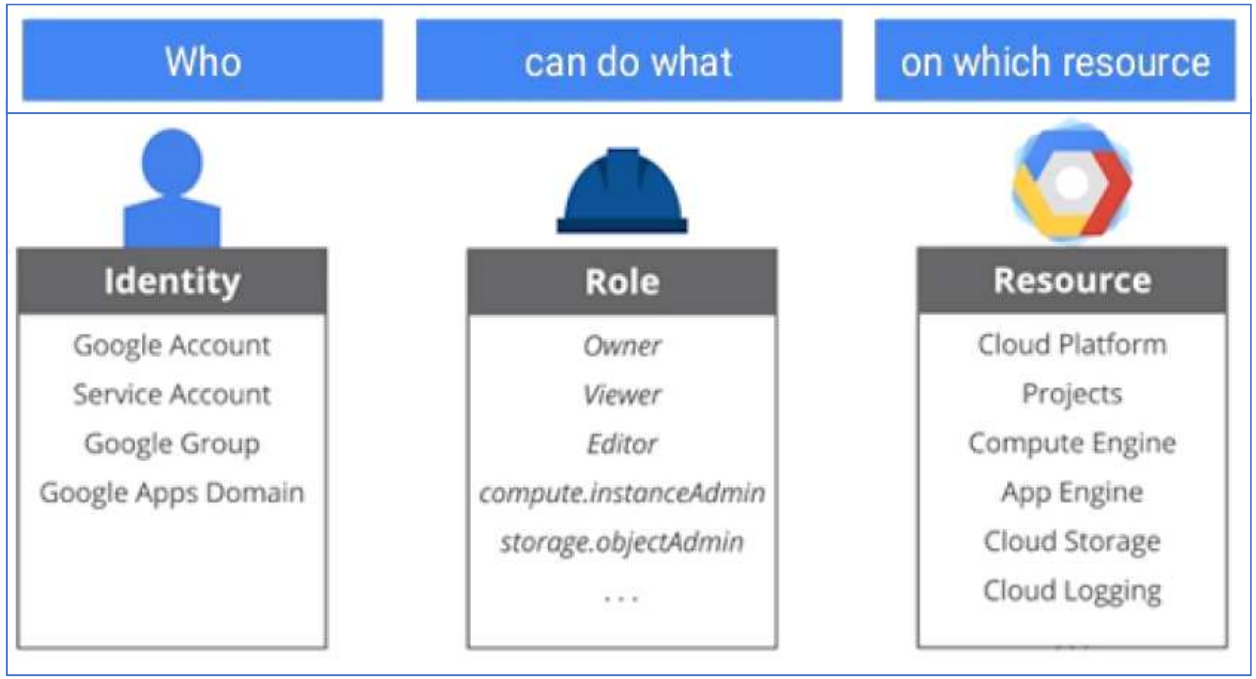






IAM (Identity and Access Management)

- GCP 서비스의 리소스에 대한 접근 권한을 제어 할 수 있다



Who = Member = Identity

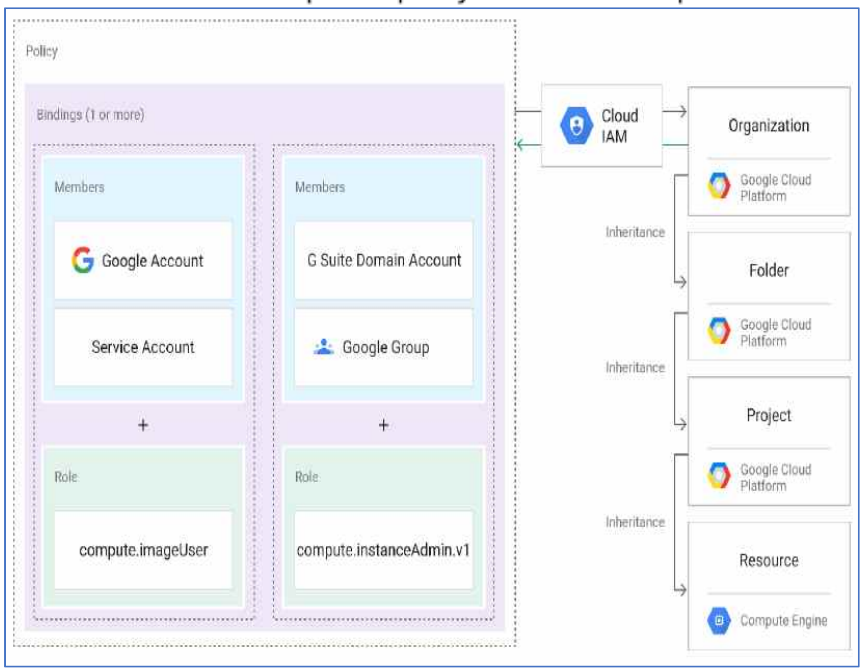
- Person or a service account
- Identity = email address
  - Log in with email address (joe@companyname.com)
- People authenticate via Google account (in form of email address)
- Service account = application/server account
  - Not associated with a person

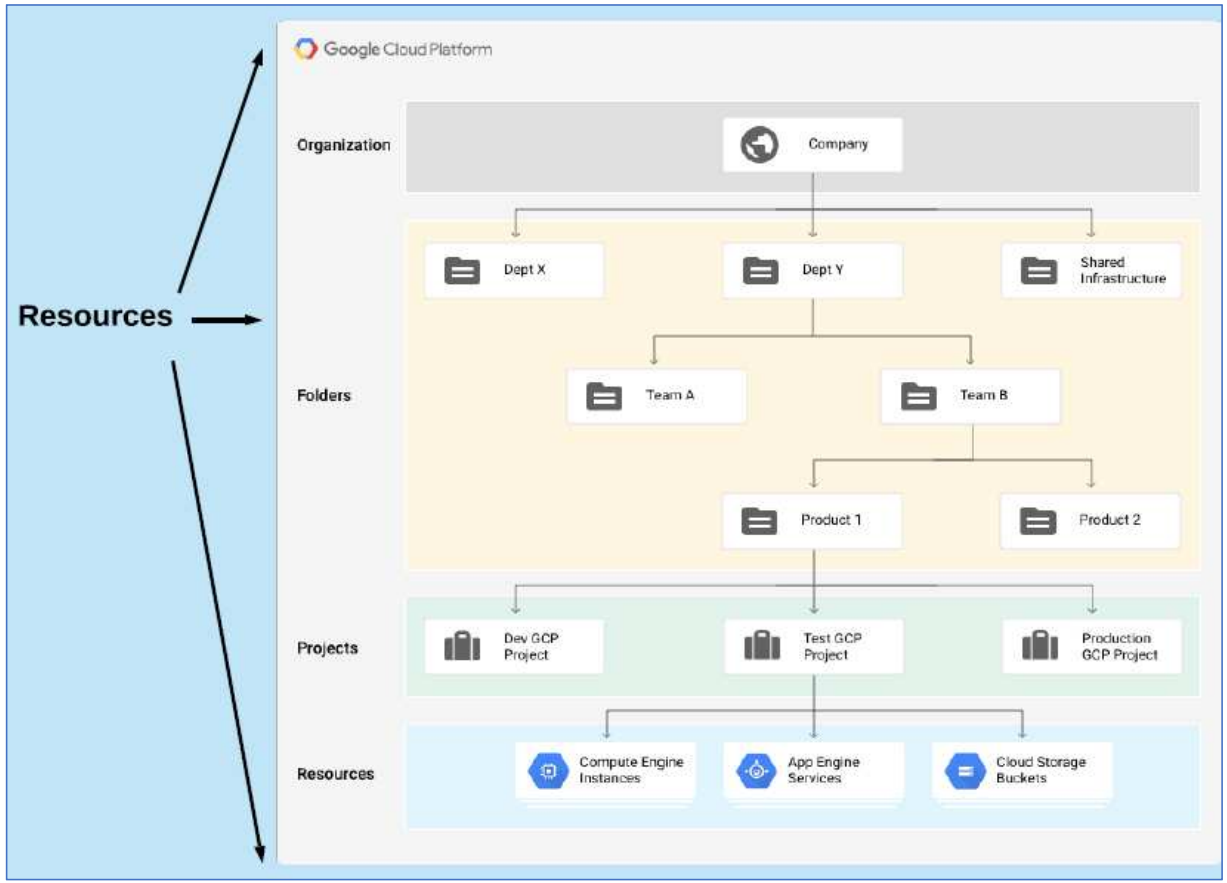
- Role(역할)
- Owner(소유자) : 멤버 초대 및 삭제, 프로젝트 삭제
- Editor(편집자) : 리소스 설정 변경, 서비스 설정
- Viewer(뷰어) : 읽기 전용

IAM 정책

Policy Inheritance (or Hierarchy)

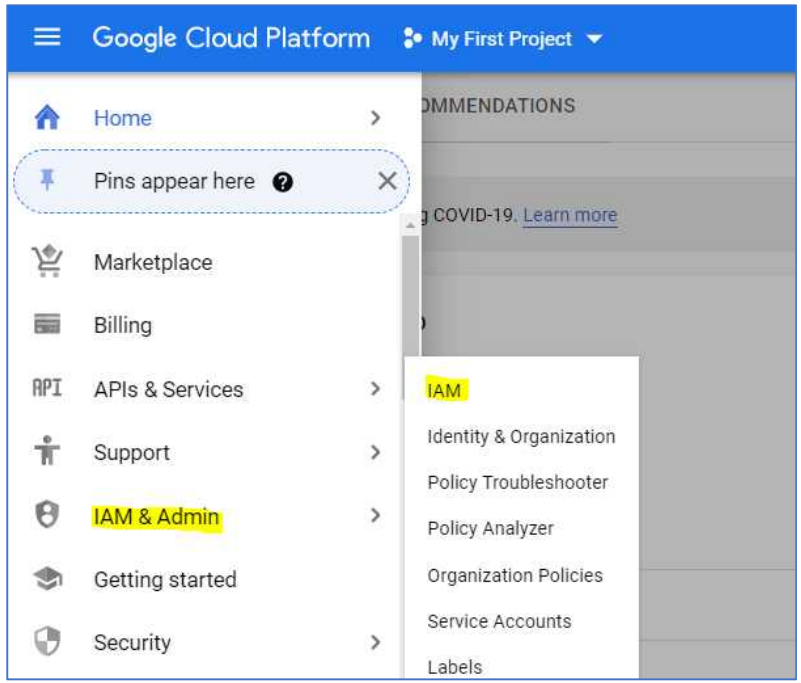
- Roles are enforced from the top down
- Organization > Folders > Project > Service/Resource
- Each child has exactly one parent
- Inherits roles from parent
- Permissive parent policy overrules child policies



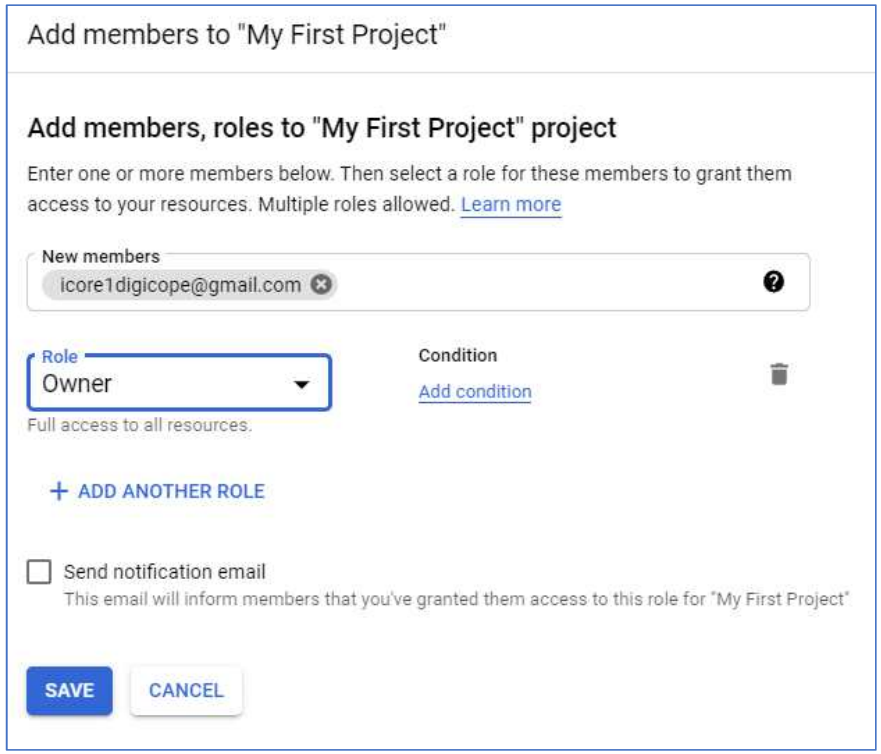


IAM : 다른 계정에 권한 부여하기(Lab)

- GCP 콘솔 메뉴에서 IAM & Admin 아래 IAM을 선택한다

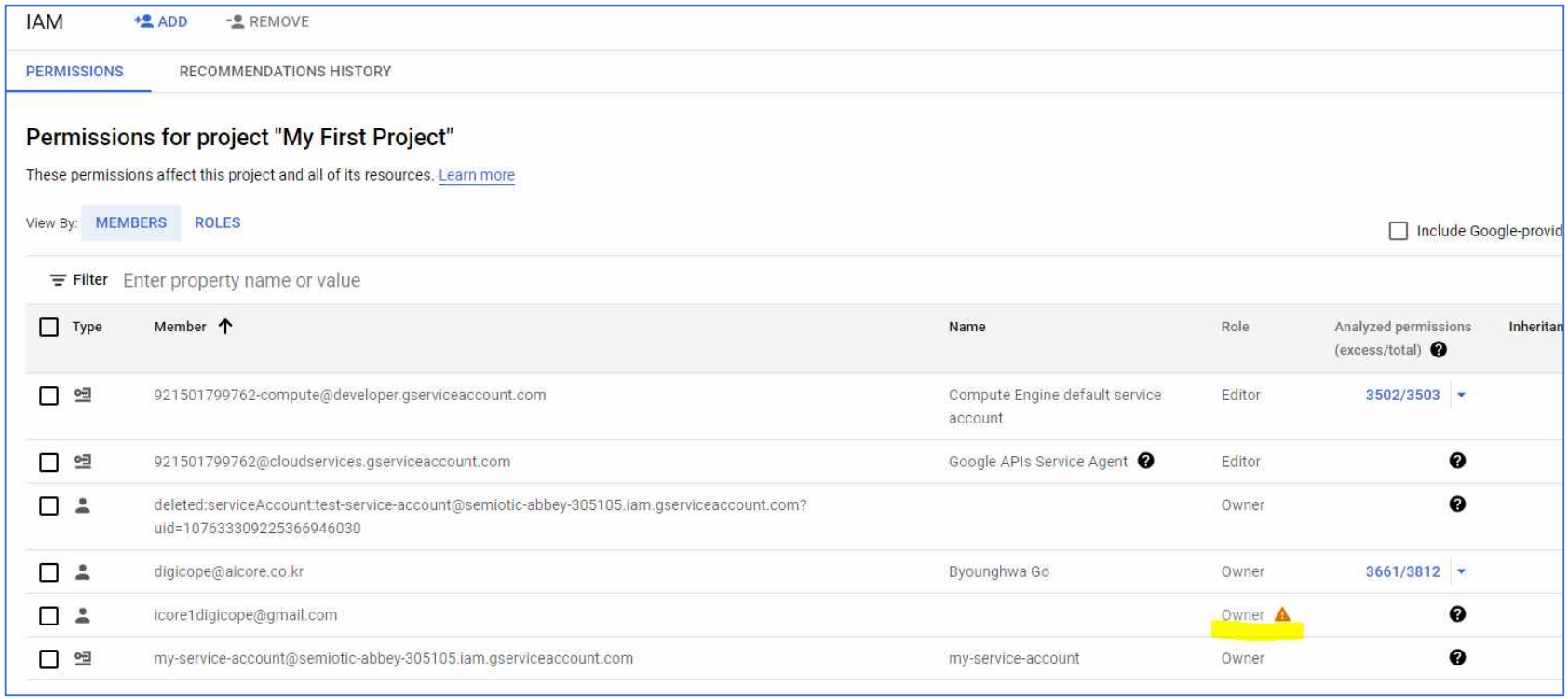


- 상단의 ADD를 누르고 추가할 계정을 입력하고 Role을 Project -> Owner 로 선택한 다음 SAVE 버튼을 누른다



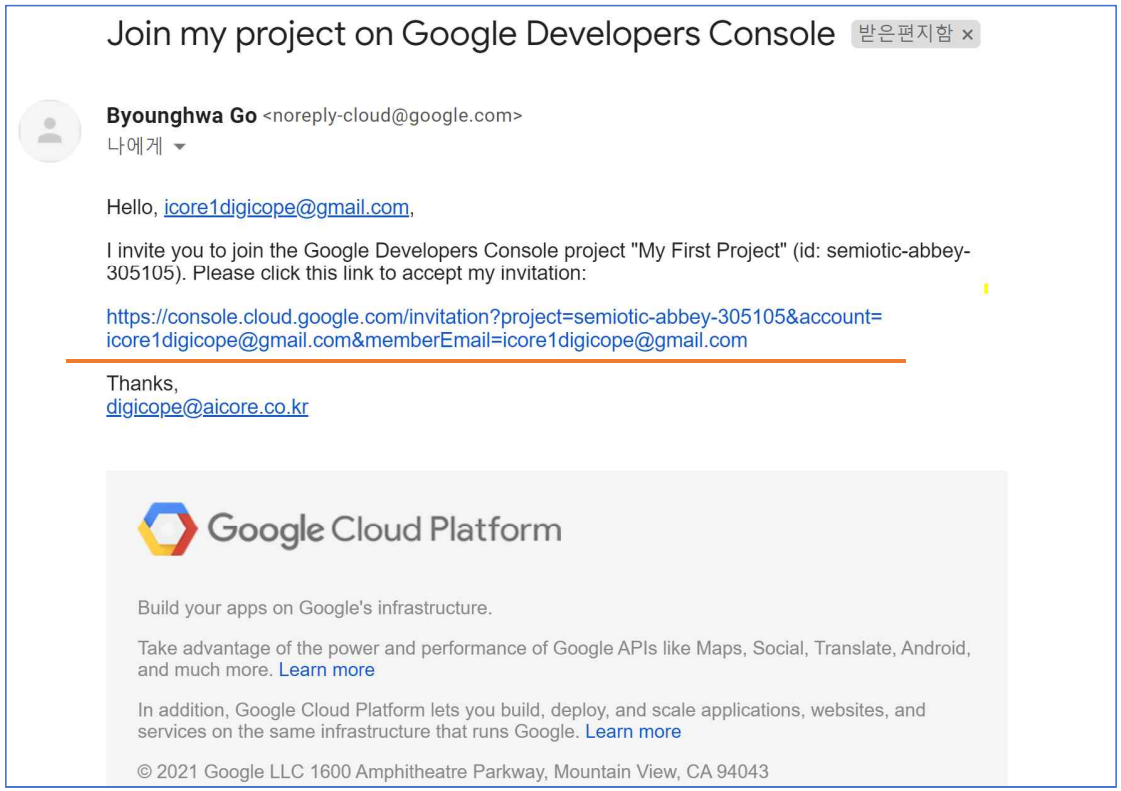
IAM : 다른 계정에 권한 부여하기(Lab)

- 권한을 받은 계정에서 '허용 ' 을 하지 않은 상태라서 아래와 같이 Owner 옆에 경고가 표시된다

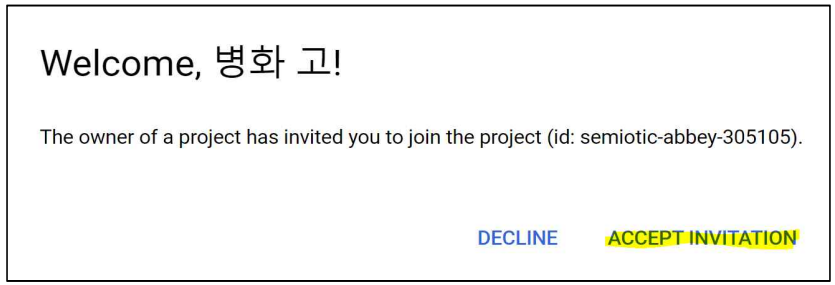


IAM : 다른 계정에 권한 부여하기(Lab)

- 권한을 받을 계정 이메일에서 아래 밑줄 친 링크 부분을 클릭해준다



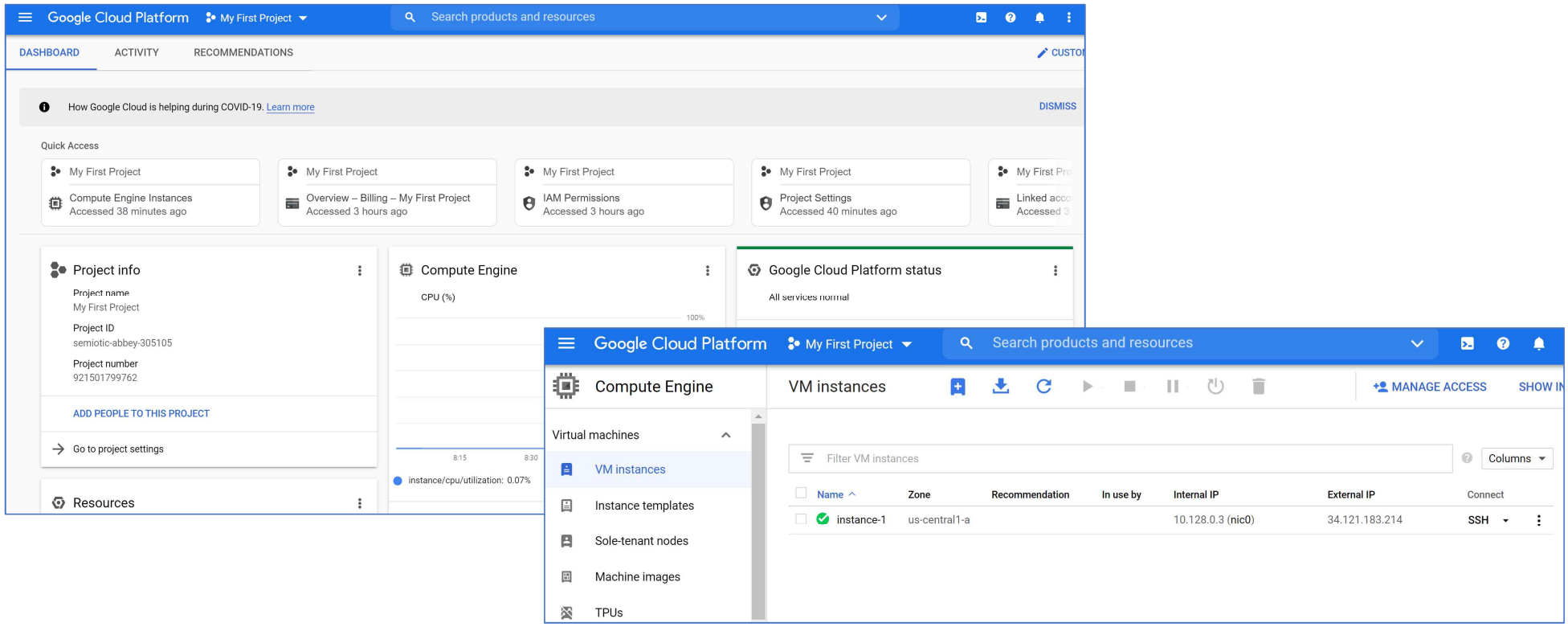
- 아래 창이 뜨면 ACCEPT INVITATION을 누르면 계정에 권한을 부여 받게 된다



- 처음 계정 권한을 부여한 콘솔에서 F5를 눌러 브라우저 화면을 갱신하면 경고 표시가 사라져 보일 것이다

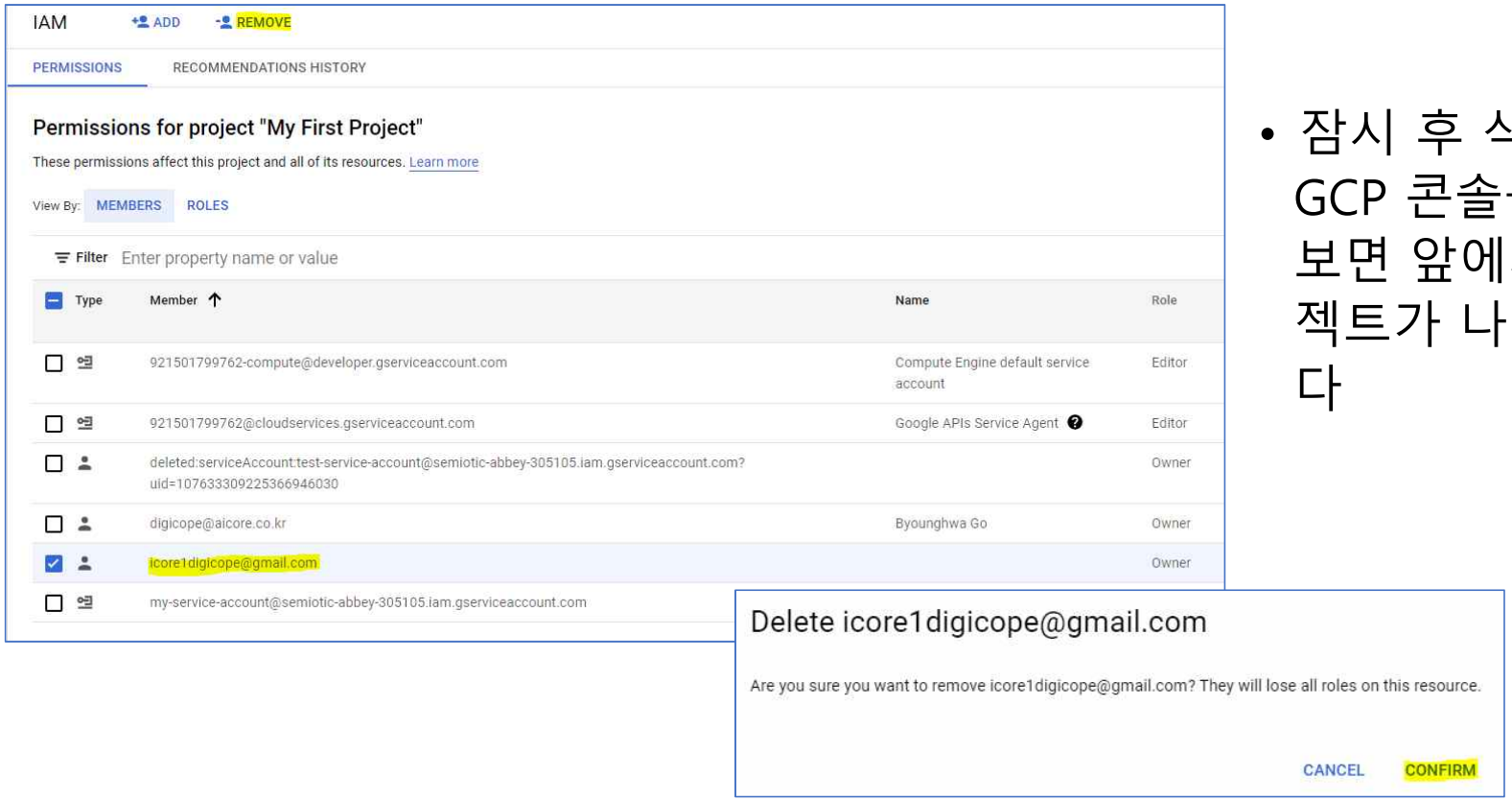
IAM : 다른 계정에 권한 부여하기(Lab)

- 권한을 받은 계정의 콘솔에서 프로젝트가 보여진다
- IAM이나 Compute Engine의 VM instances 도 동일하게 보여진다



IAM : 다른 계정의 권한 삭제(Lab)

- GCP 콘솔 메뉴에서 IAM & Admin 아래 IAM을 선택하고 해당 계정을 선택하고 상단의 REMOVE를 누른 다음 CONFIRM을 눌러 삭제를 완료한다



- 잠시 후 삭제된 계정의 GCP 콘솔을 다시 접속해 보면 앞에서 보여진 프로젝트가 나타나지 않게 된다

IAM : gcloud 사용 계정에 권한 부여하기(Lab)

- Windows 에서 Google Cloud SDK Shell을 실행시키고 아래와 같이 gcloud 명령을 입력한다 (semiotic-abbey-305105는 프로젝트 ID이다)
- gcloud projects get-iam-policy semiotic-abbey-305105 > iam.yaml




```
관리자: Google Cloud SDK Shell
C:₩Program Files (x86)₩Google₩Cloud SDK>gcloud projects get-iam-policy  semiotic-abbey-305105 > iam.yaml

C:₩Program Files (x86)₩Google₩Cloud SDK>dir/w
 C 드라이브의 볼륨: Windows
 볼륨 일련 번호: 50A6-51F3

 C:₩Program Files (x86)₩Google₩Cloud SDK 디렉터리

[.]                  [..]                 cloud_env.bat        cloud_platform_logo.ico
[google-cloud-sdk]   iam.yaml             install_mode         uninstaller.exe
               5개 파일             306,724 바이트
               3개 디렉터리  76,428,742,656 바이트 남음

C:₩Program Files (x86)₩Google₩Cloud SDK>type iam.yaml
bindings:
- members:
  - serviceAccount:service-921501799762@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:921501799762-compute@developer.gserviceaccount.com
  - serviceAccount:921501799762@cloudservices.gserviceaccount.com
  role: roles/editor
- members:
  - deleted:serviceAccount:test-service-account@semiotic-abbey-305105.iam.gserviceaccount.com?uid=107633309225366946030
  - serviceAccount:my-service-account@semiotic-abbey-305105.iam.gserviceaccount.com
  - user:digicope@aicore.co.kr
  role: roles/owner
etag: BwW9LdOLnzI=
version: 1

C:₩Program Files (x86)₩Google₩Cloud SDK>notepad iam.yaml

C:₩Program Files (x86)₩Google₩Cloud SDK>
```

- dir/w를 사용하여 생성된 iam.yaml 파일을 확인한다
- type iam.yaml 를 사용하여 파일 내용을 확인한다
- notepad iam.yaml을 입력하여 편집을 시작한다

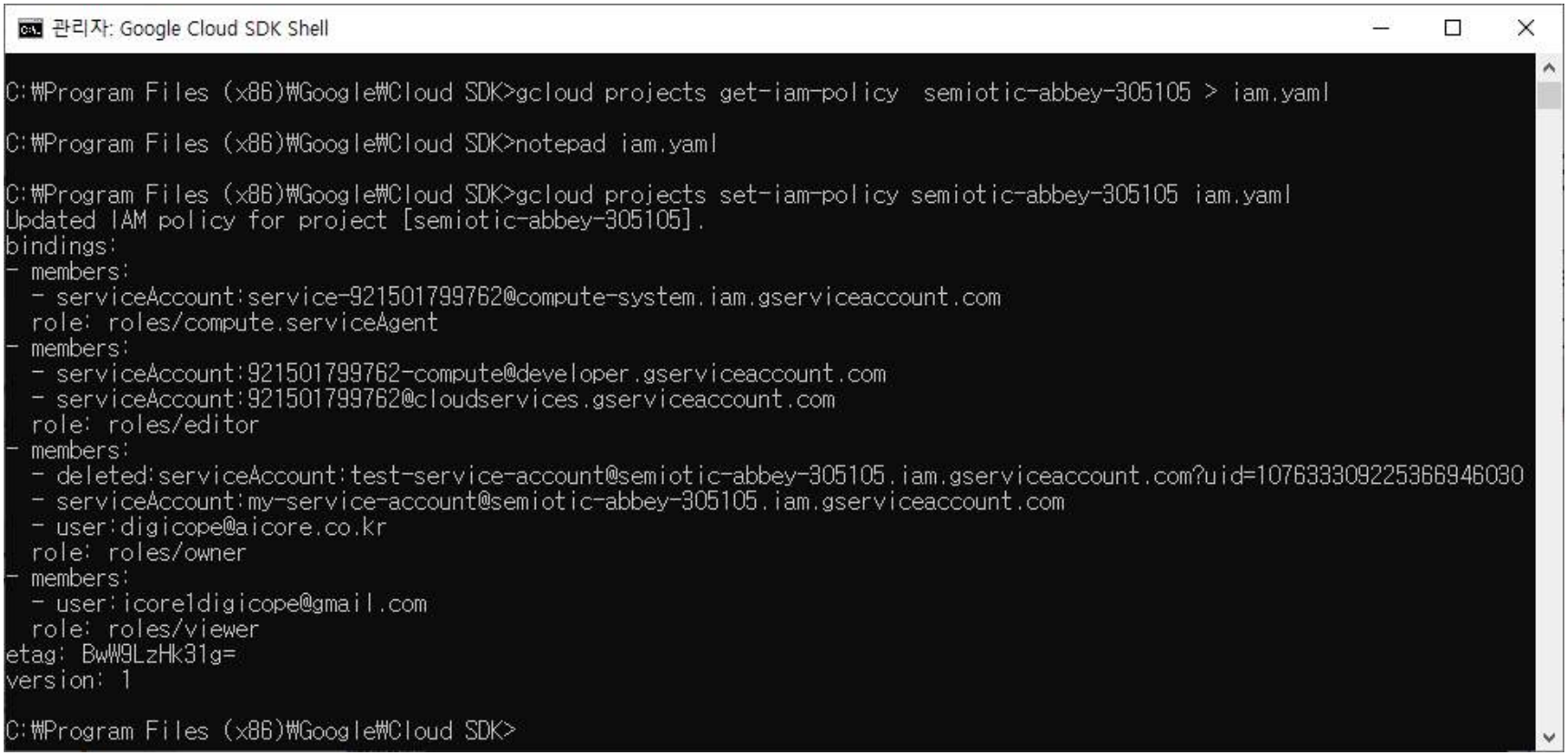
IAM : gcloud 사용 계정에 권한 부여하기(Lab)

- 노트패드에서 아래 빨간색 부분을 추가하고(사용 계정과 role 설정) 저장한다

```
bindings:
- members:
  - serviceAccount:service-921501799762@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
    <중간 생략>
 - serviceAccount:my-service-account@semiotic-abbey-305105.iam.gserviceaccount.com
 - user:digicope@aicore.co.kr
 role: roles/owner
- members:
  - user:icore1digicope@gmail.com
  role: roles/viewer
etag: BwW9Ld0Lnzl=
version: 1
```

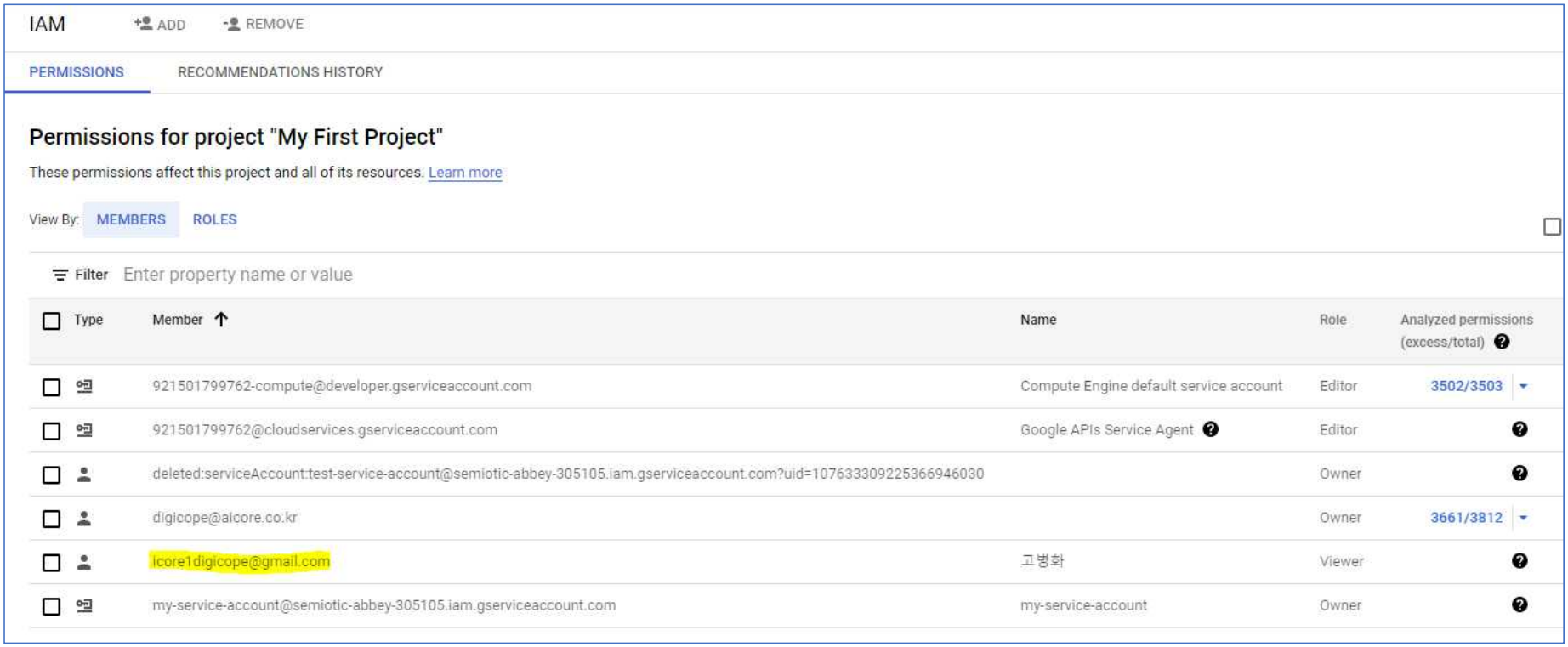
IAM : gcloud 사용 계정에 권한 부여하기(Lab)

- Shell에서 아래와 같이 명령을 입력하여 변경된 내용을 적용시켜준다(viewer 만 가능)
- gcloud projects set-iam-policy semiotic-abbey-305105 iam.yaml



IAM : gcloud 사용 계정에 권한 부여하기(Lab)

- 원본 계정과 권한 부여 받은 계정 모두 GCP 콘솔에서 생성된 결과를 아래와 같이 확인 해 볼 수 있다

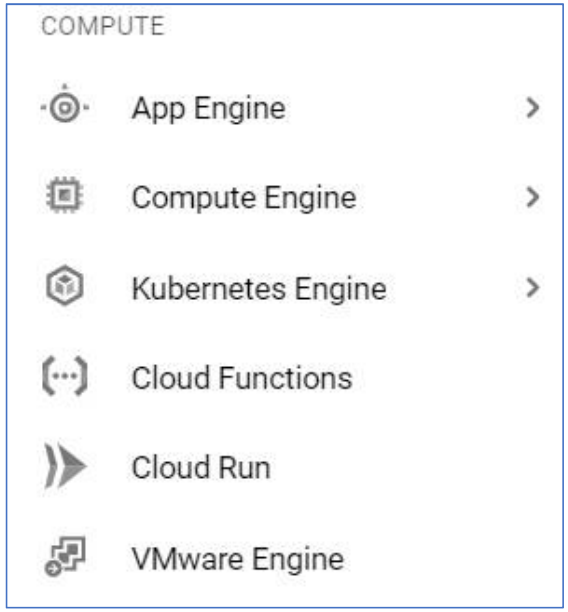


Compute Engine



Compute Engine : GCP 제공 가상머신 서비스

- 미리 정의된 리눅스 및 윈도우 서버나 사용자 이미지나 비공개 커스텀 이미지도 사용 가능하다



- VM's are the heart and soul of GCP
- GCE, GKE, and GAE all run on VM's
- Substantial portion of exam will be VM focused
- Start with single VM's
- Move up to 'force multipliers' – the true 'magic' of cloud computing
  - Automation
  - Auto-scaling
  - Managed instance groups
  - Load balancers

Compute Engine : Machine types

- VM instance에서는 가상화 된 리소스로 vCPU, system memory를 선택하여 사용할 수 있다

The following table provides machine type recommendations for different workloads.

E2 General purpose	N2, N2D, N1 General purpose	M2, M1 Memory-Optimized	C2 Compute-Optimized	A2 Accelerator-Optimized
Day-to-day computing at a lower cost	Balanced price/performance across a wide range of VM shapes	Ultra high-memory workloads	Ultra high performance for compute-intensive workloads	Optimized for high performance computing workloads
• Web serving • App serving • Back office applications • Small-medium databases • Microservices • Virtual desktops • Development environments	• Web serving • App serving • Back office applications • Medium-large databases • Cache • Media/streaming	• Large in-memory databases like SAP HANA • In-memory analytics	• HPC • Electronic Design Automation (EDA) • Gaming • Single-threaded applications	• High performance computing (HPC) • Machine learning (ML) • Massive parallelized computation

Compute Engine : Live Migration

- VM instance의 종료 및 재부팅 없이 실시간으로 다른 호스트로 이전 시킨다

Live migration keeps your instances running during:

- Regular infrastructure maintenance and upgrades.
- Network and power grid maintenance in the data centers.
- Failed hardware such as memory, CPU, network interface cards, disks, power, and so on. This is done on a best-effort basis; if a hardware fails completely or otherwise prevents live migration, the VM crashes and restarts automatically and a `hostError` is logged.
- Host OS and BIOS upgrades.
- Security-related updates, with the need to respond quickly.
- System configuration changes, including changing the size of the host root partition, for storage of the host image and packages.

Compute Engine : Disks

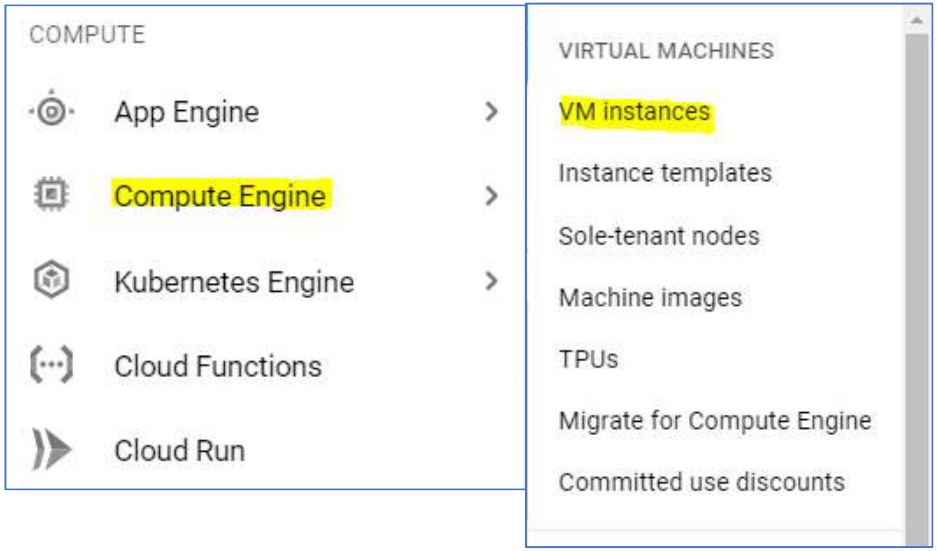
- Boot disk로는 Persistent disk 만 사용가능
- SSD는 375GB로 고정, 8개까지 연결 가능
- GCS는 유연성과 지속성이 높으나 수행 성능은 가장 낮음

- All instances have a single root disk for the OS (persistent disk)
- **Persistent Disk**
  - Most common, default option
  - Not directly attached - network attached storage
  - Standard and SSD variety
- **Local SSD**
  - Directly attached to your VM
- **Cloud Storage buckets**
  - Niche option
  - High collaboration, 'infinite' space
  - Not block storage

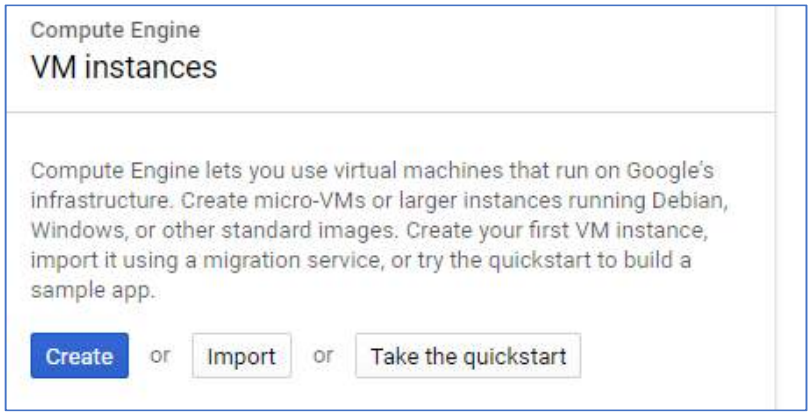


Compute Engine : VM 인스턴스 생성 (Lab)

- GCP 콘솔 메뉴에서 Compute Engine 아래 VM instances를 선택한다

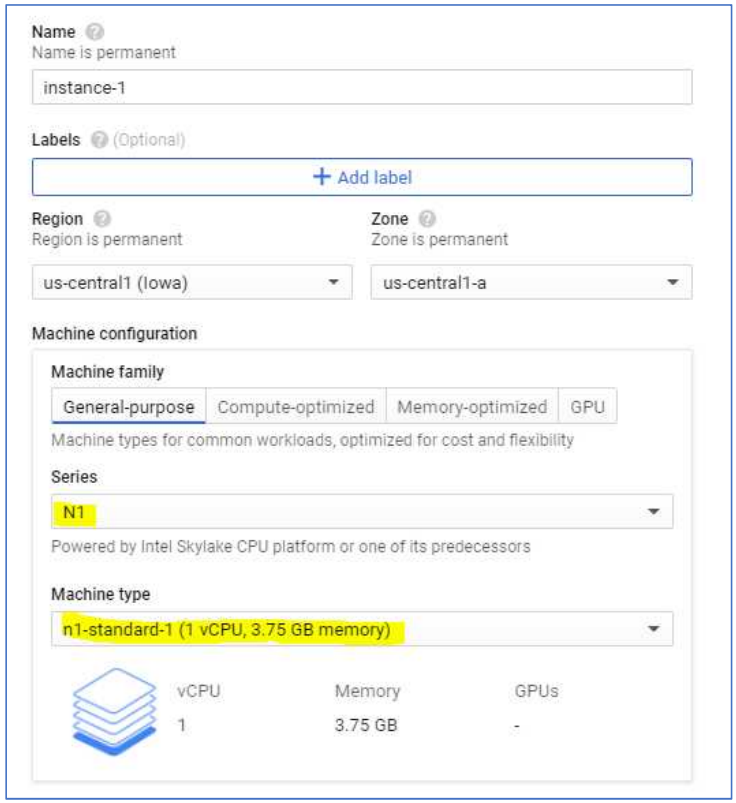


- 아래 화면에서 CREATE 버튼을 누른다

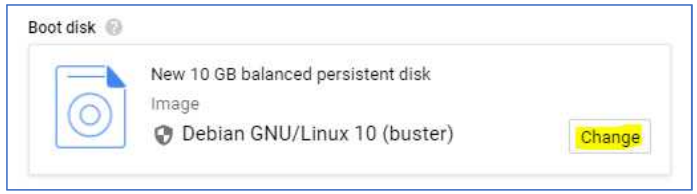


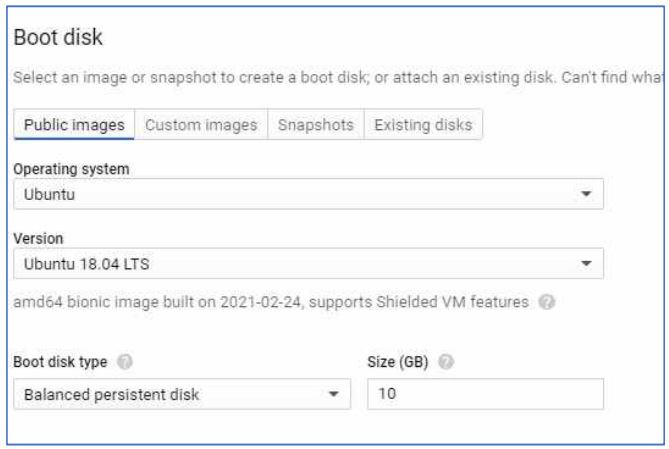
Compute Engine : VM 인스턴스 생성 (Lab)

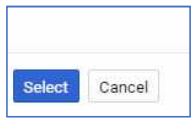
- 아래 화면에서 Name/Region/Zone 등은 그대로 놔두고 Series와 Machine type을 변경해준다 (저렴한 n1-standard-1(1 vCPU,3.75GB memory을 사용)



- 아래 화면에서 Boot disk의 Change 버튼을 누른 다음 Ubutu 18.04LTS를 선택해주고 하단의 Select 버튼을 누른다

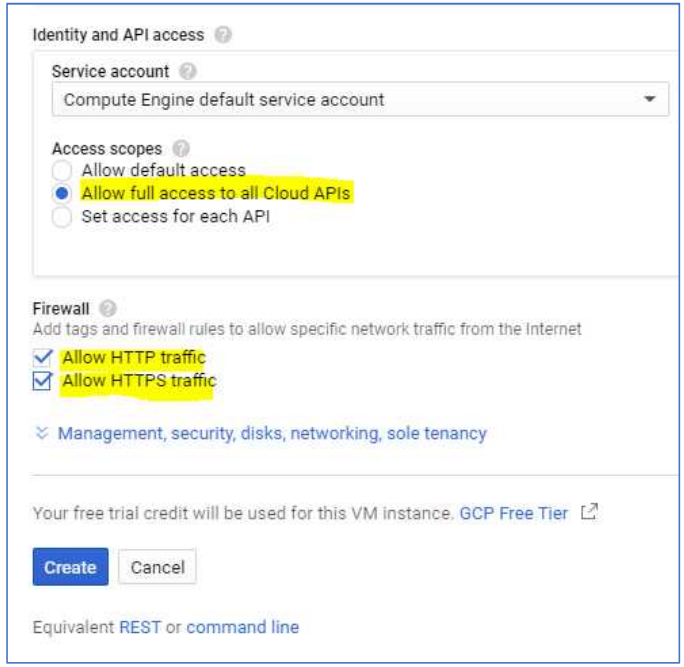




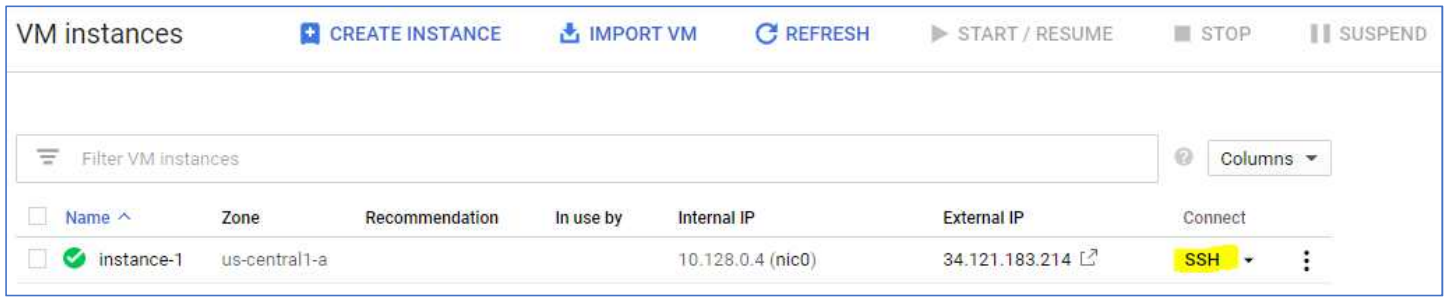


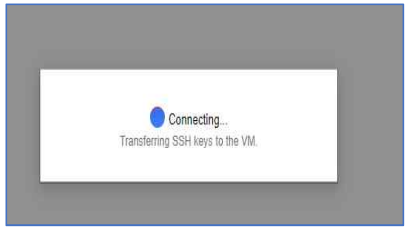
Compute Engine : VM 인스턴스 생성 (Lab)

- 오른쪽 화면에서 Access scopes의
  Allow full access to all Cloud API를 선택하고
  Firewall의 Allow HTTP traffic와 Allow HTTPS traffic를 모두 선택해준 다음Create버튼을 눌러 VM을 생성한다



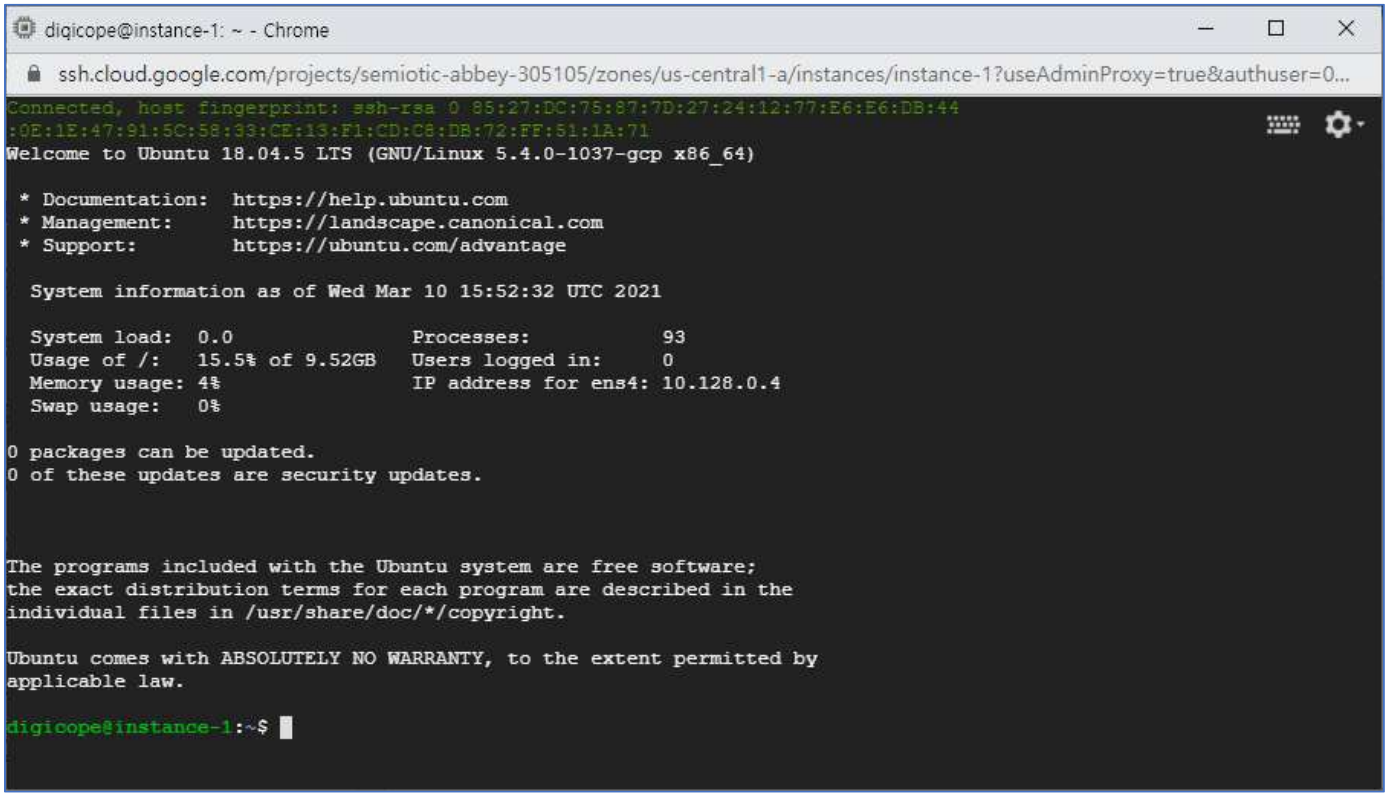
- 아래 와 같이 VM 인스턴스가 생성된 것을 확인 할 수가 있다. 하단의 SSH 버튼을 눌러 실행 시킨다





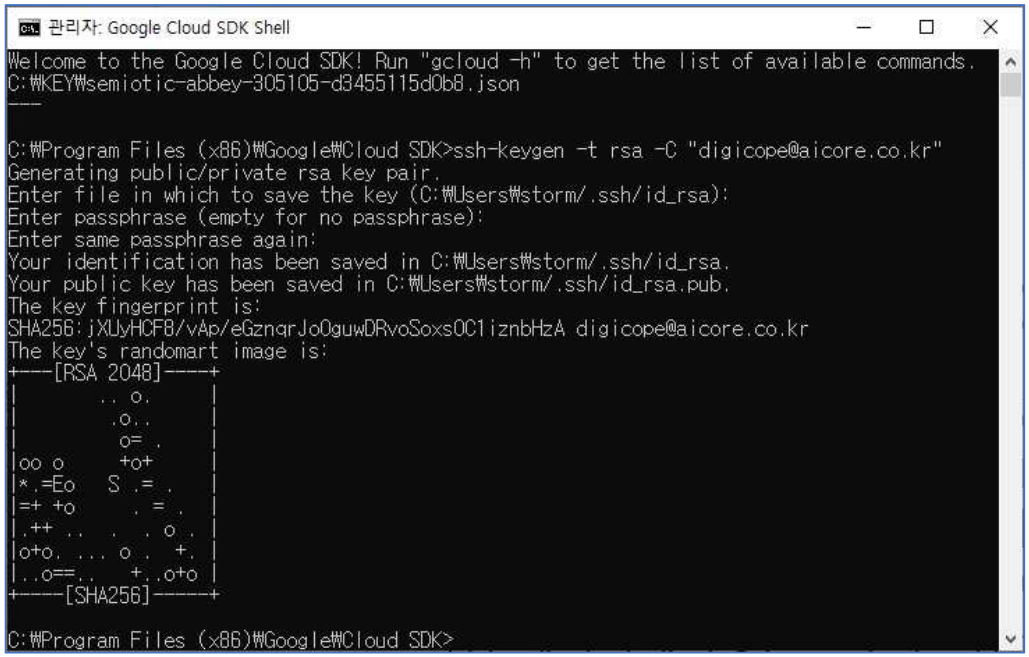
Compute Engine : VM 인스턴스 생성 (Lab)

- SSH 연결이 완료되면 아래와 같이 Ubuntu 터미널을 볼 수가 있다



Compute Engine : SSH로 VM에 접속하기(Lab)

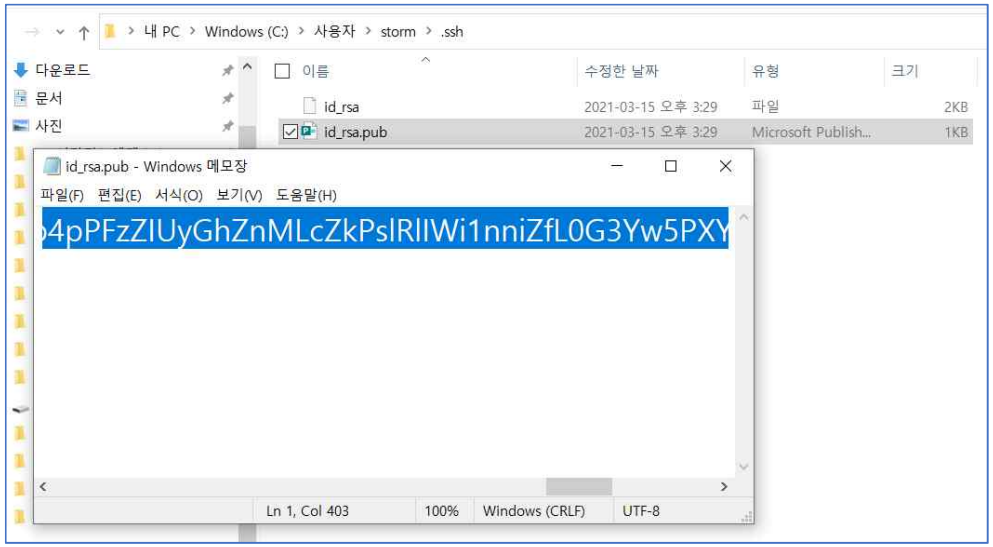
- 로컬 터미널에서 SSH명령어로 VM에 연결이 가능하다
- Windows 에서 Google Cloud SDK Shell을 실행시키고 명령을 입력한다 (마지막인자는 GCP 계정 이메일이다)
- ssh-keygen –t rsa –C “digicope@aicore.co.kr”
- 다음 메시지에서 3번 모두 엔터를 입력하여 넘어간다



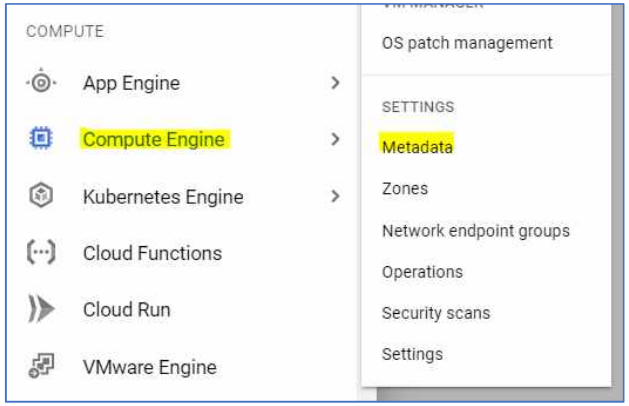
- C:₩Users₩(윈도우 계정명)₩.ssh 폴더 밑에 RSA 키가 만들어 진다

Compute Engine : SSH로 VM에 접속하기(Lab)

- 윈도 파일 탐색기에서 C:₩사용자₩(윈도우 계정명)₩.ssh₩id_rsa.pub 파일을 메모장으로 열어 파일 내용을 전체 선택하고 복사한다

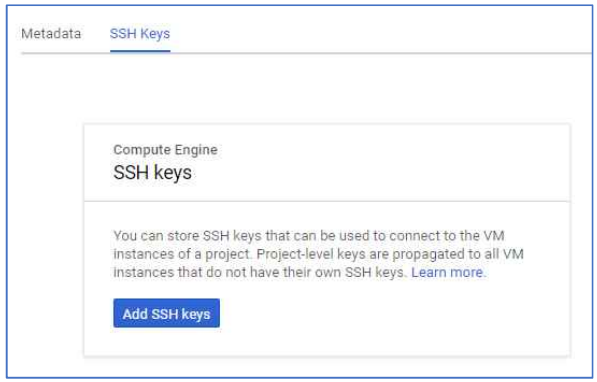


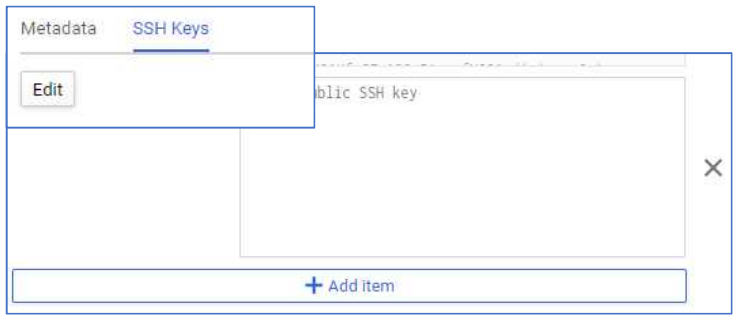
- GCP 콘솔의 메뉴에서 Compute Engine안에서 Metadata를 선택해준다



Compute Engine : SSH로 VM에 접속하기(Lab)

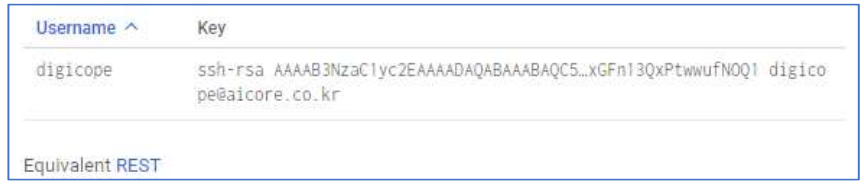
- SSH Keys 탭을 선택하고 Add SSH keys를 누른다
- 이미 다른 키가 만들어져 있는 경우는 Edit 버튼을 누르고 Add item을 누른다





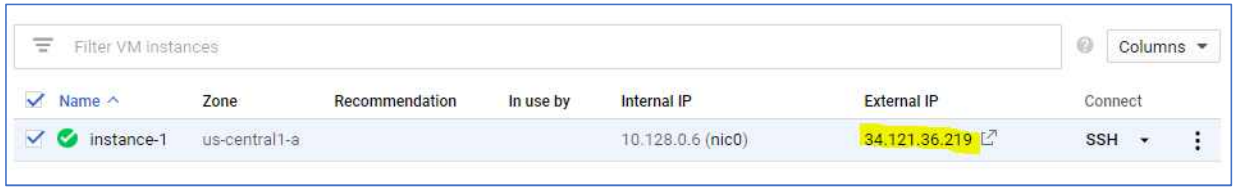
- 메모장에서 복사한 키 내용을 붙여 넣고 Save 버튼을 눌러 저장하면 목록에서 아래와 같이 등록된 SSH 키를 확인할 수 있을 것이다



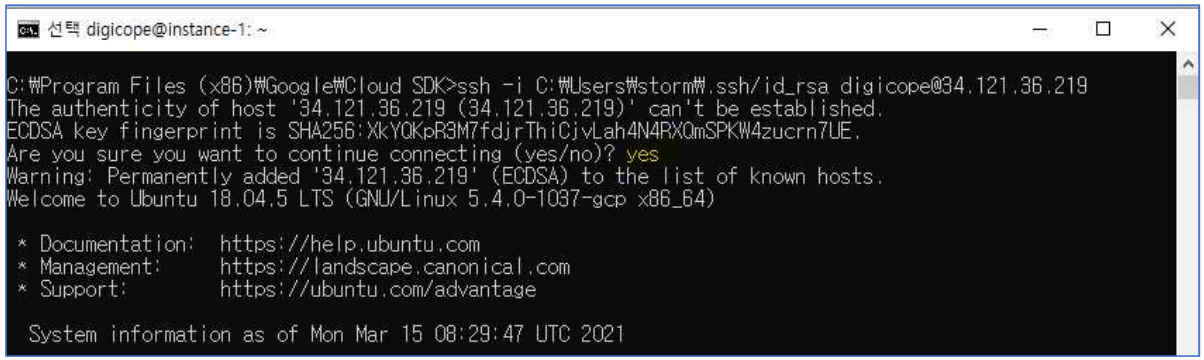


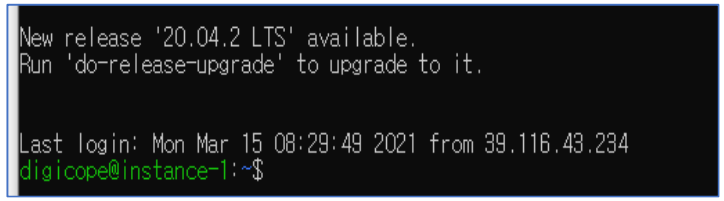
Compute Engine : SSH로 VM에 접속하기(Lab)

- Compute Engine → VM 인스턴스를 눌러서 External IP를 복사한다



- 윈도우 터미널에서 아래 명령어를 입력하여 접속을 시도한다(정확한 계정 ID와 외부 IP주소 입력)
- ssh -i C:₩Users₩storm₩.ssh/id_rsa digicope@34.121.36.219
- 접속 시 아래 메시지 출력 시 yes를 입력하면 연결이 완료 된다

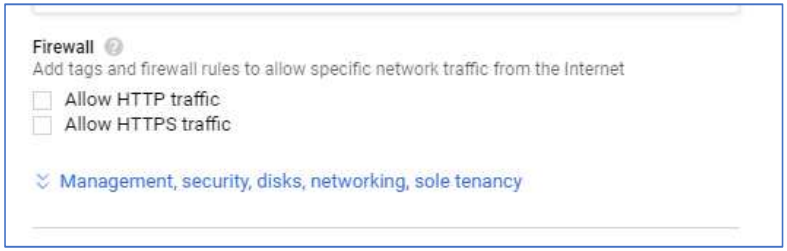




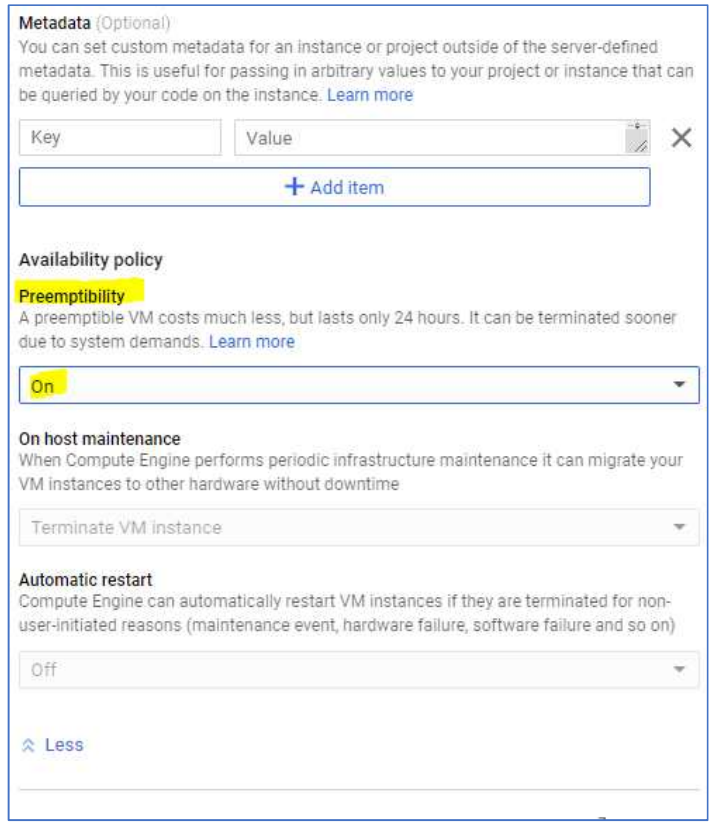
Compute Engine : 선점형 인스턴스

- 선점형 인스턴스는 일반 인스턴스 보다 비용이 저렴하게 이용할 수 있다
- 사용도중에 임의로 종료될 수 있다
- 항상 24시간 후에 자동 종료되므로 종료 시 수행할 스크립트를 만들어 놓는게 좋다

- Compute Engine → VM Instances에서 CREATE INSTANCE를 선택하고 아래 부분을 찾아 Management 왼쪽을 클릭해준다



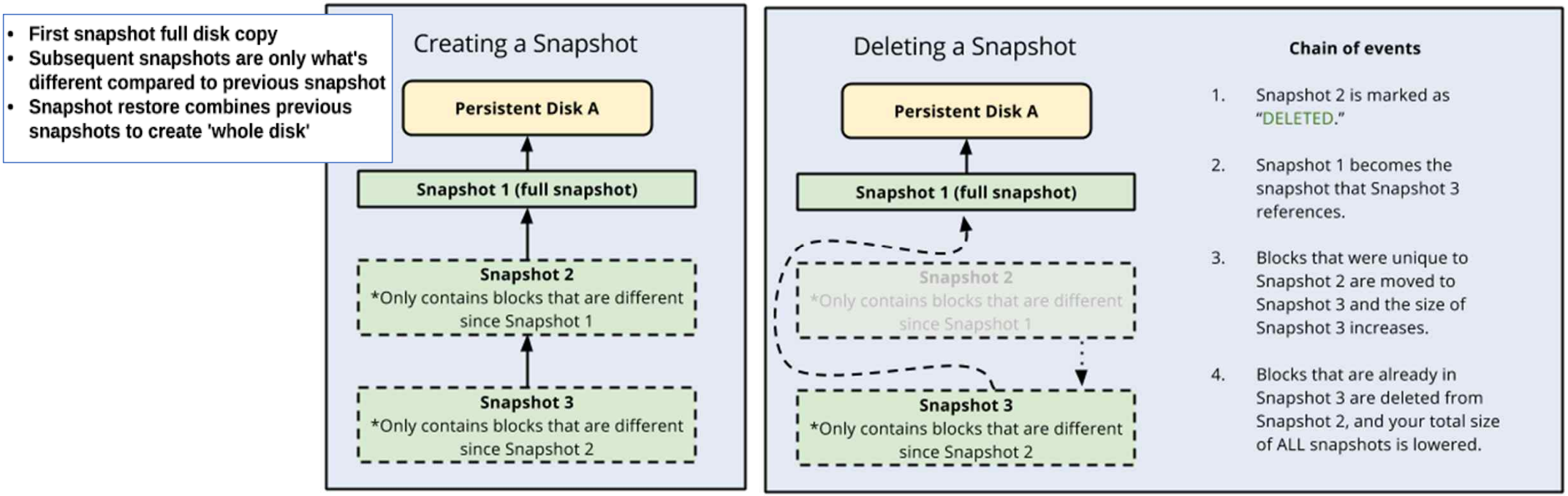
- Preemptibility 부분을 On으로 변경 하여 인스턴스를 만들면 선점형 인스턴스가 만들어진다



Compute Engine : Snapshot(스냅샷)

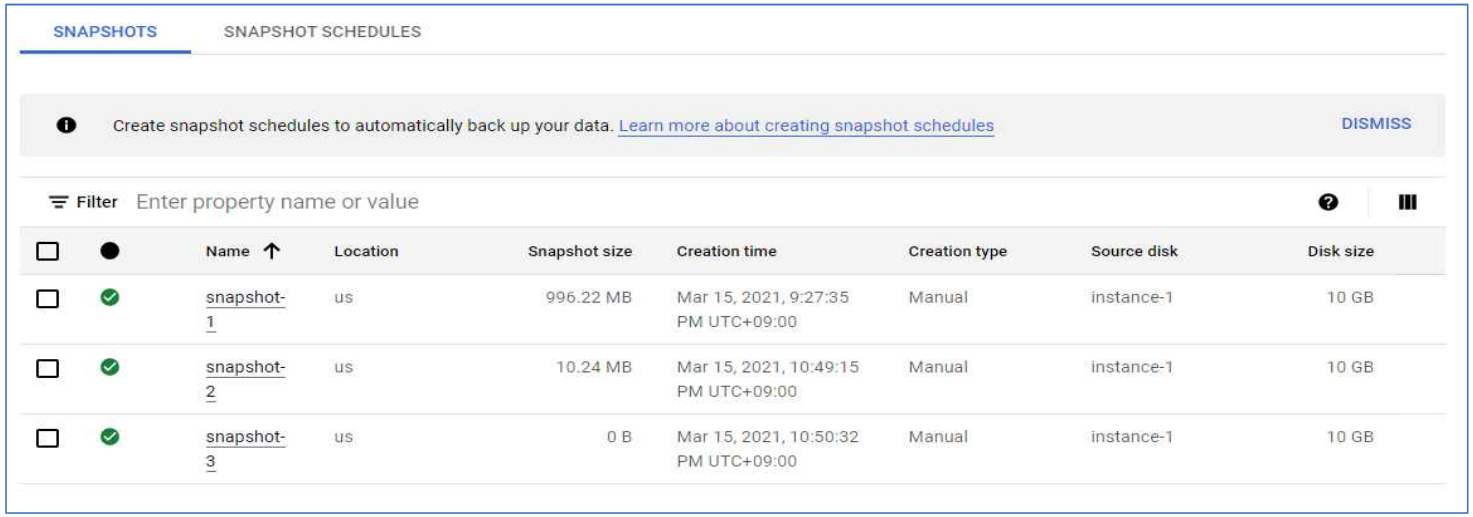
https://cloud.google.com/compute/docs/disks/snapshots

- 스냅샷은 이미 생성된 인스턴스/디스크를 백업
- 인스턴스 실행 중에도 백업 가능, 백업 시점으로 복구 가능



Compute Engine : Snapshot(스냅샷)

- 동일한 VM인스턴스를 사용하여 3개의 스냅샷 생성



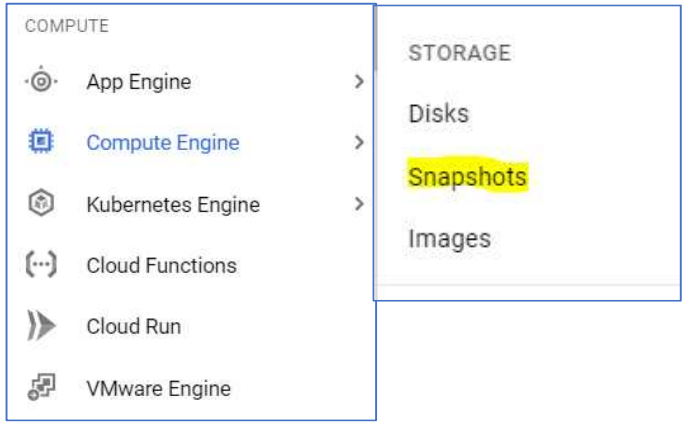
Name	Location	Snapshot size	Creation time	Creation type	Source disk	Disk size
snapshot-1	us	996.22 MB	Mar 15, 2021, 9:27:35 PM UTC+09:00	Manual	instance-1	10 GB
snapshot-2	us	10.24 MB	Mar 15, 2021, 10:49:15 PM UTC+09:00	Manual	instance-1	10 GB
snapshot-3	us	0 B	Mar 15, 2021, 10:50:32 PM UTC+09:00	Manual	instance-1	10 GB

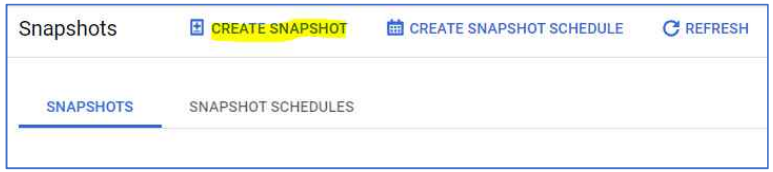
- 2번 스냅샷을 삭제 : snapshot-3가 snapshot-2의 정보를 갖는다

Name	Location	Snapshot size	Creation time	Creation type	Source disk	Disk size
snapshot-1	us	996.22 MB	Mar 15, 2021, 9:27:35 PM UTC+09:00	Manual	instance-1	10 GB
snapshot-3	us	10.24 MB	Mar 15, 2021, 10:50:32 PM UTC+09:00	Manual	instance-1	10 GB

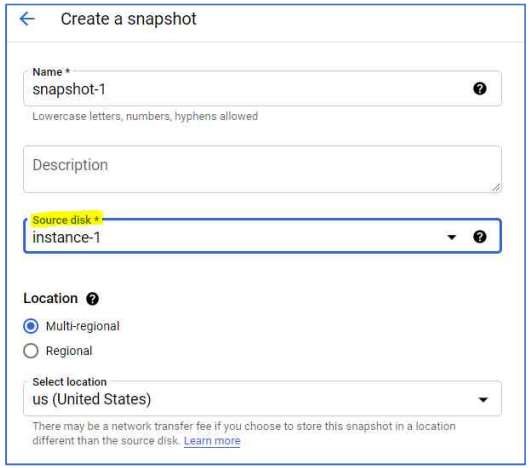
Compute Engine : 인스턴스 스냅샷 만들기(Lab)

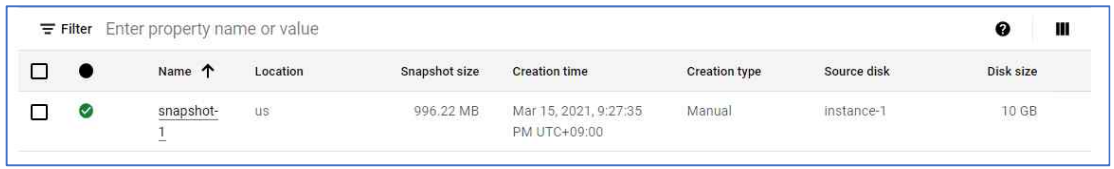
- Compute Engine → Snaspshots 을 선택하고 CREATE SNAPSHOT을 누른다





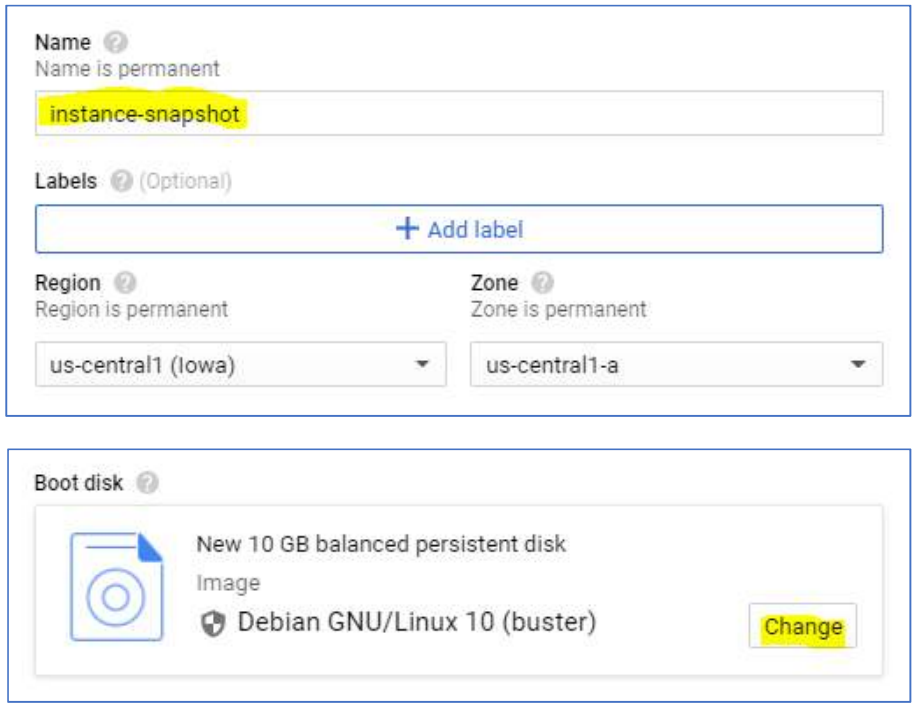
- Source disk를 선택하여 이미 만들어 놓은 VM instance를 선택해준 다음 나머지는 그대로 두고 하단의 CREATE 버튼을 누르면 스냅샷이 만들어진다



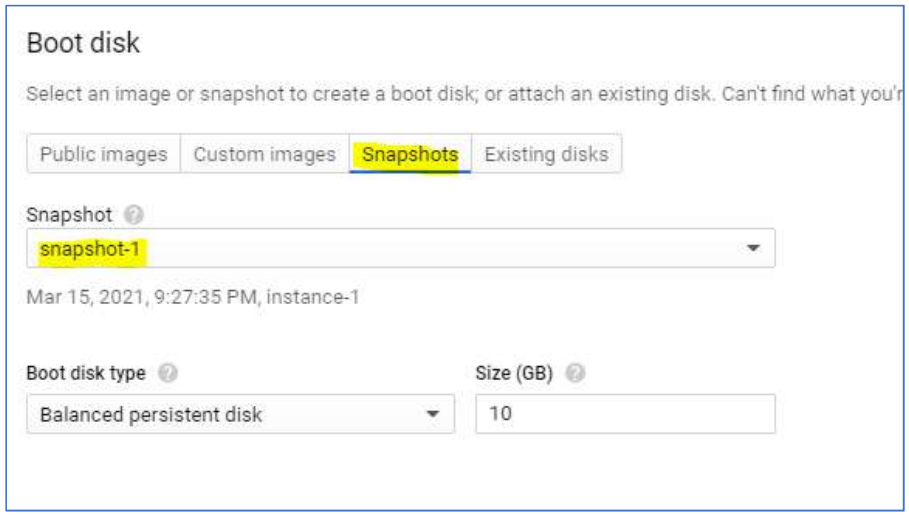


Compute Engine : 스냅샷으로 인스턴스 만들기(Lab)

- Compute Engine → VM instances를 선택한 다음 CREATE INSTANCE 버튼을 누른다
- Name을 instance-snapshot 으로 변경해주고
- 아래 Boot disk에서 Change 버튼을 누른다

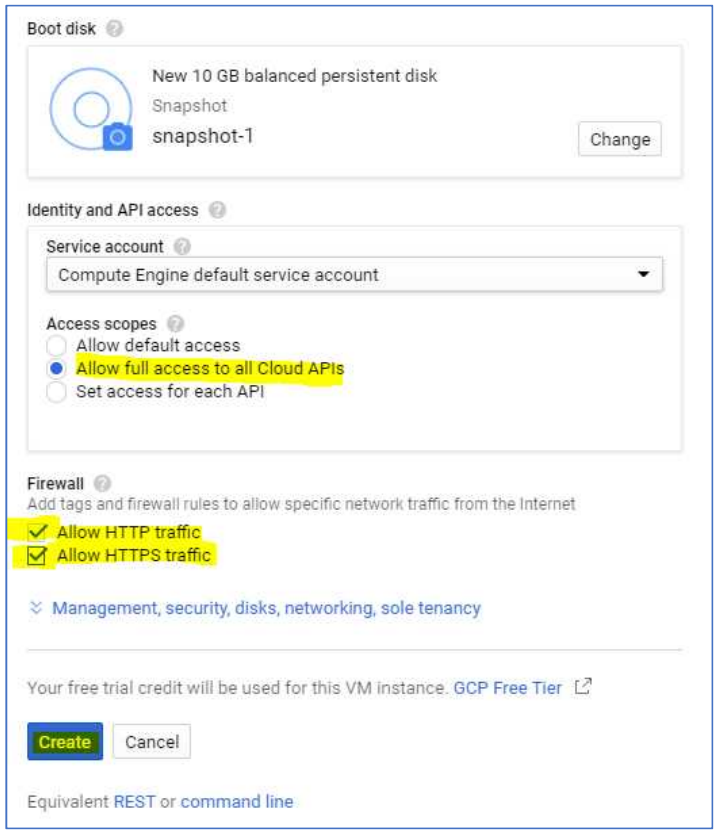


- Snapshots 탭을 선택해주고 Snapshot을 이미 만들어 놓은 snapshot-1 으로 선택해준 다음 하단의 SELECT 버튼을 누른다

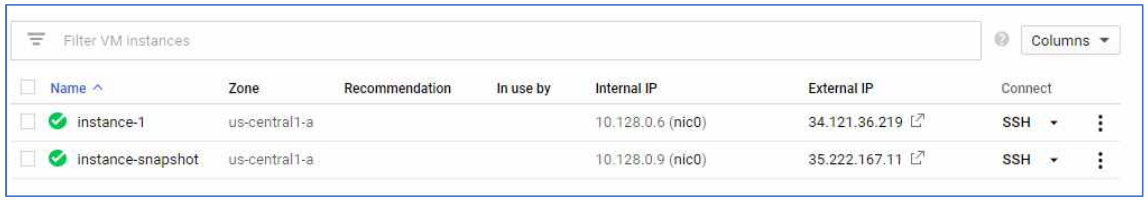


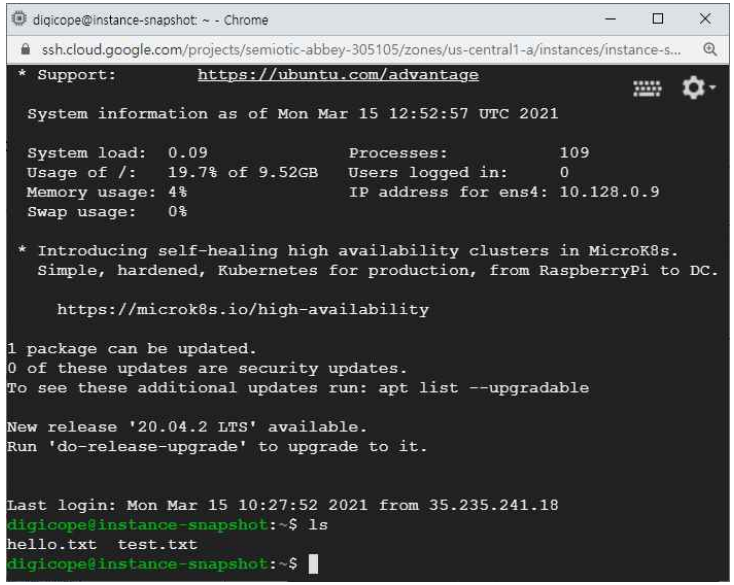
Compute Engine : 스냅샷으로 인스턴스 만들기(Lab)

- Access scopes와 Firewall을 아래와 같이 선택해주고 Create 버튼을 누른다



- 인스턴스 생성이 완료되면 앞과 같이 SSH로 접속하여 연결해 본다(instance-1과 동일한 인스턴스임)





Compute Engine : Image

https://cloud.google.com/compute/docs/images

- 인스턴스의 부트 디스크 생성을 위한 image를 사용할 수 있다

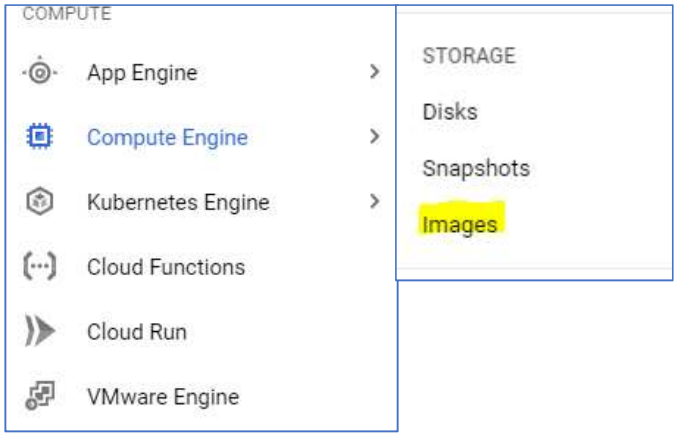
- Necessary to create boot disks for GCE instances
- Unlike disks, not limited to zone - much more accessible
- Public and Custom images
- Public - provided and maintained by Google/community/vendors
  - All projects have access
- Custom - Customized boot disks you create
  - Only available to your project (and other sources you control)

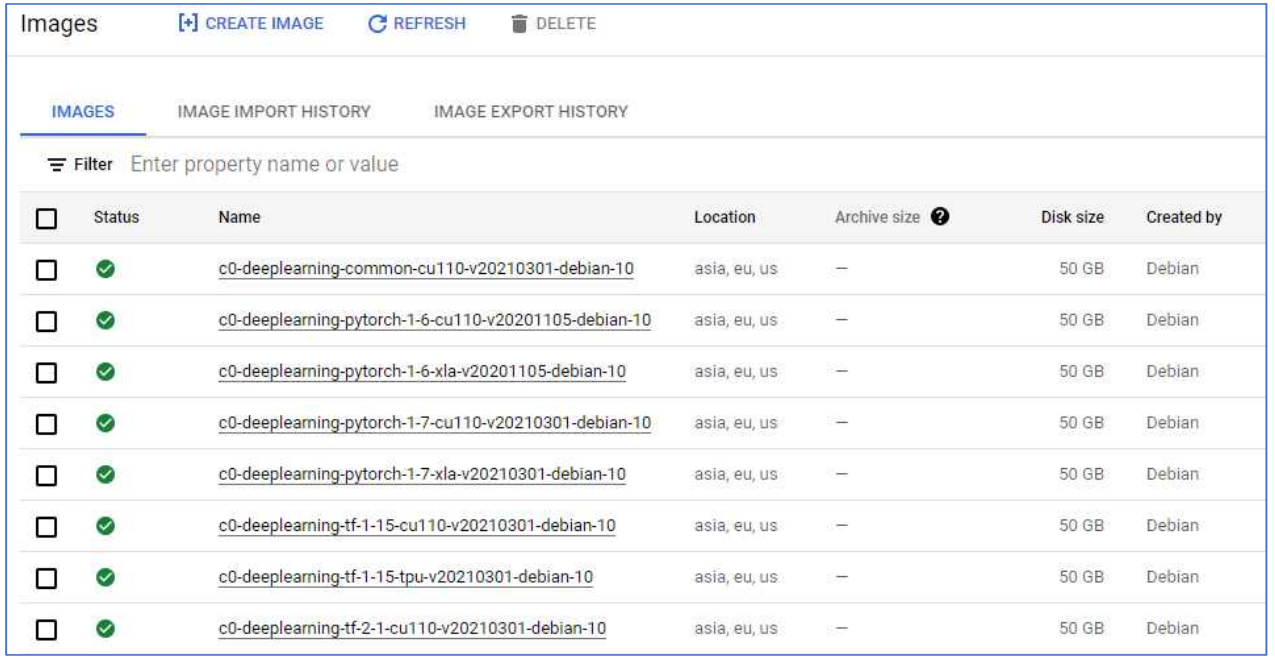
Differences between Images and Snapshots

- Images
  - Purpose - create new instances, configure instance templates
  - Recommended to shut down instances before creating new image
  - Access across projects (even different organizations)
- Snapshots
  - Purpose - periodic incremental backup of existing disk/instance
  - Can create while running
  - Cross-project access in 'transition'

Compute Engine : 커스텀 이미지 만들기(Lab)

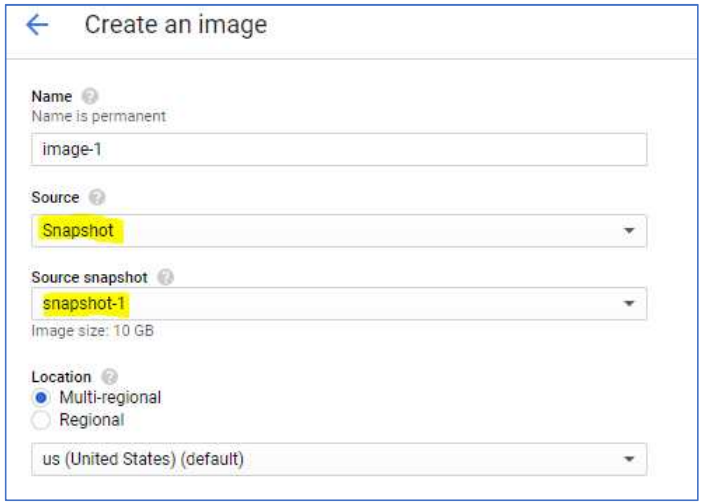
- Compute Engine → Image를 선택한 다음 상단의 [+]CREATE IMAGE 버튼을 누른다





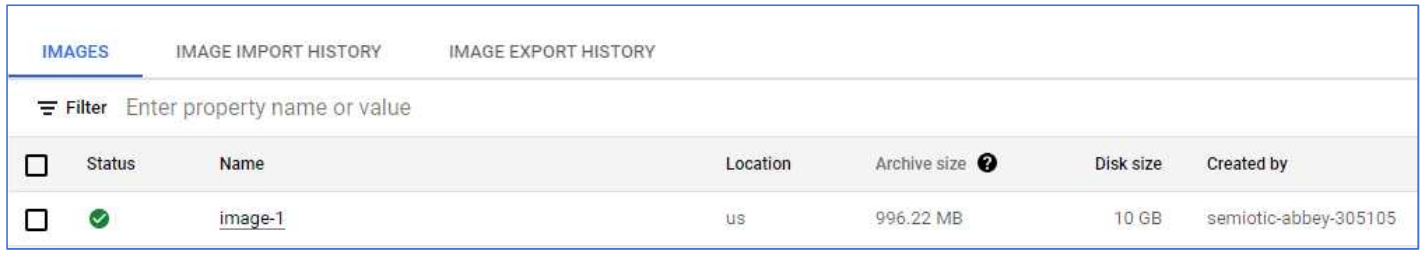
Compute Engine : 커스텀 이미지 만들기(Lab)

- Source를 Snapshot으로 변경하고 Source snapshot을 snapshot-1으로 변경하고 하단의 Create 버튼을 눌러준다



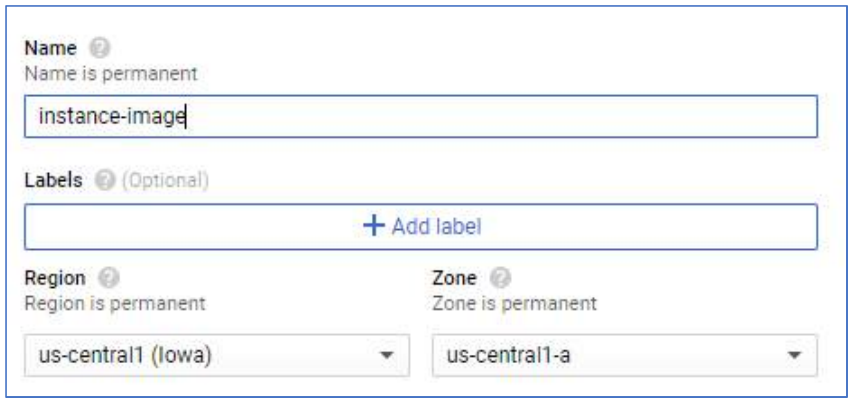
- 이미지 생성이 완료 되면 아래와 같이 목록에서 확인할 수 있다
- 이미지를 사용하여 인스턴스나 인스턴스 템플릿을 만들 수 있다

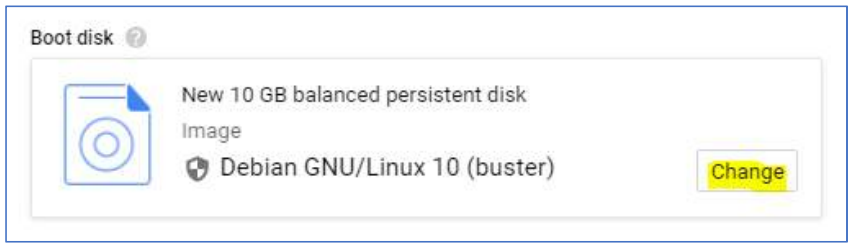




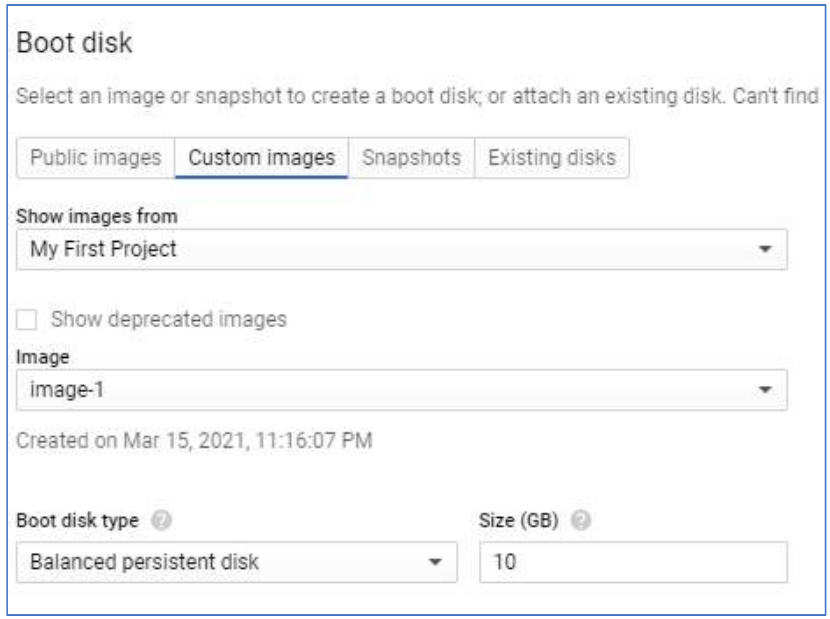
Compute Engine : 이미지로 인스턴스 만들기(Lab)

- Compute Engine → VM instances를 선택한 다음 CREATE INSTANCE 버튼을 누른다
- Name을 instance-image로 변경해주고
- 아래 Boot disk에서 Change 버튼을 누른다



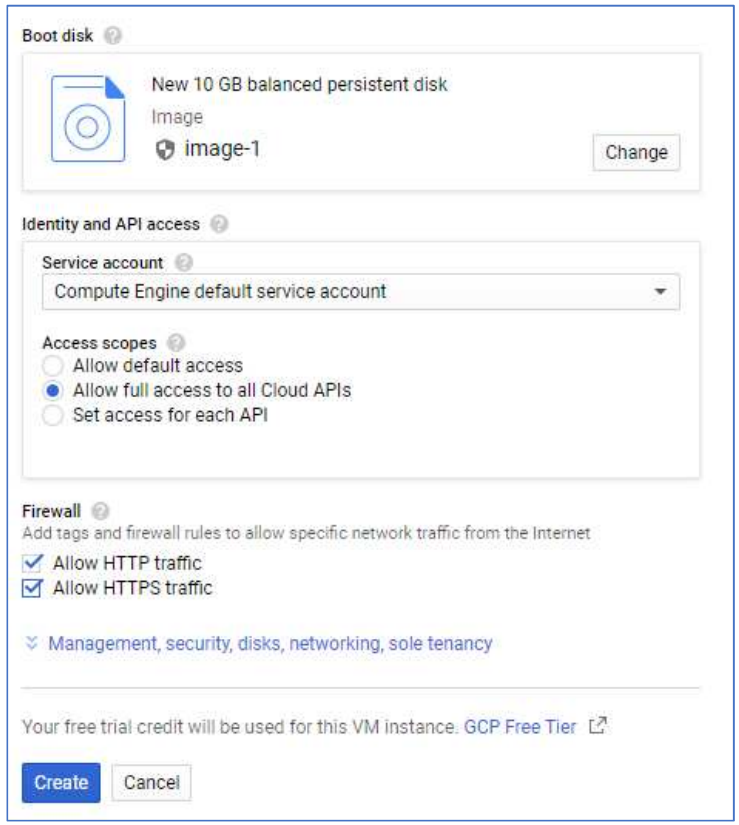


- Custom image 탭을 선택해주고 Image를 프로젝트에서 이미 만들어 놓은 image-1 으로 선택해준 다음 하단의 SELECT 버튼을 누른다



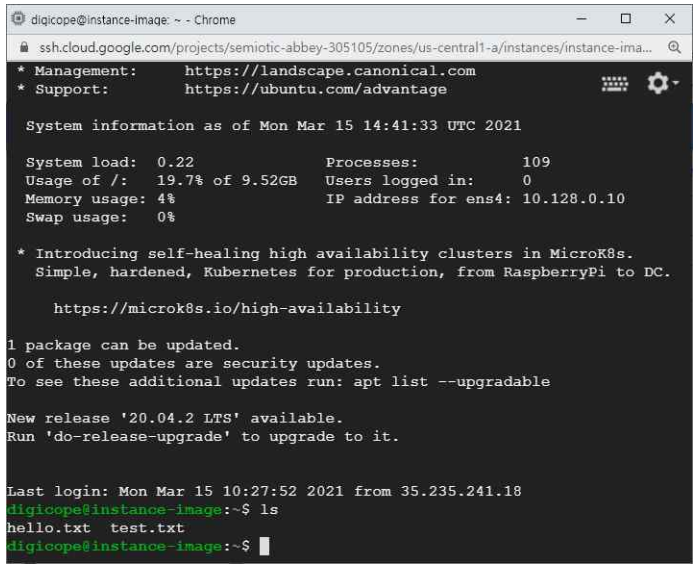
Compute Engine : 이미지로 인스턴스 만들기(Lab)

- Access scopes와 Firewall을 아래와 같이 선택해주고 Create 버튼을 누른다



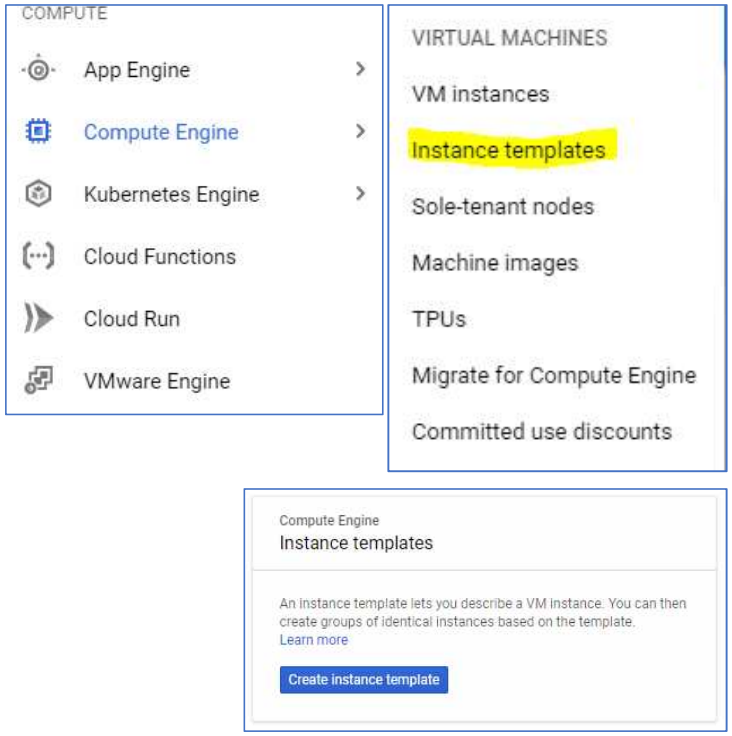
- 인스턴스 생성이 완료되면 앞과 같이 SSH로 접속하여 연결해 본다



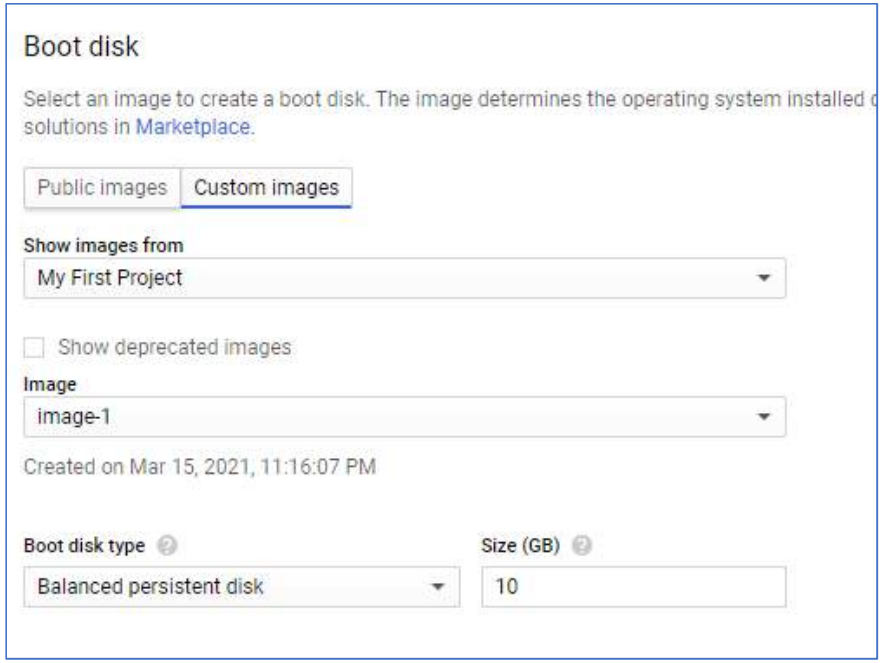


Compute Engine : 인스턴스 템플릿 만들기(Lab)

- Compute Engine → Instance templates를 선택한 다음 Create instance template 버튼을 누른다

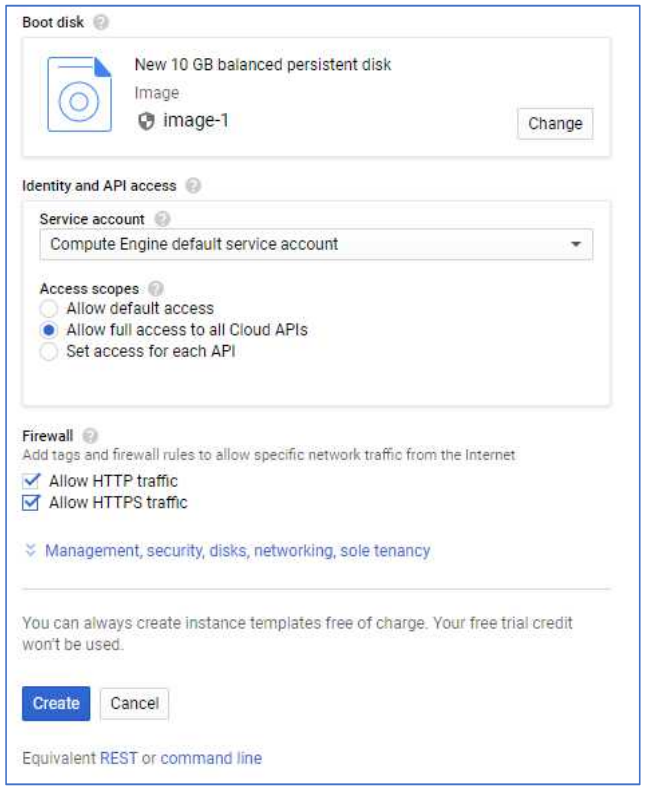


- Name을 instance-template-1로 그대로 두고 Boot disk에서 Change 버튼을 누른다
- Custom images 탭을 선택해주고 Image 를 이미 만들어 놓은 image-1 으로 선택해준 다음 하단의 SELECT 버튼을 누른다

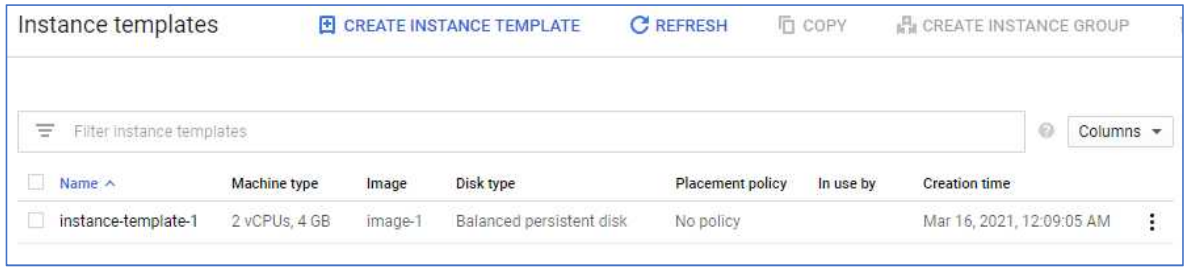


Compute Engine : 인스턴스 템플릿 만들기(Lab)

- Access scopes와 Firewall을 아래와 같이 선택해주고 Create 버튼을 누른다



- 인스턴스 템플릿 생성이 완료되면 아래와 같이 목록에서 확인 할 수 있다 (뒤에서 인스턴스 템플릿을 이용하여 인스턴스 그룹을 선택할 수 있다)



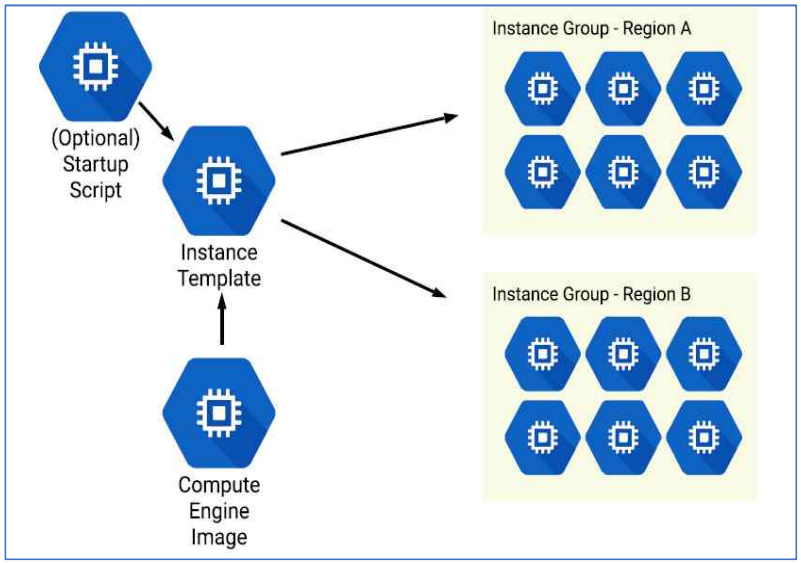
Compute Engine : 인스턴스 그룹과 오토 스케일링

https://cloud.google.com/compute/docs/instance-groups

- 인스턴스 그룹은 단일 항목으로 관리 할 수 있는 가상 머신 (VM) 인스턴스 모음
- Compute Engine은 관리 형 및 비 관리 형의 두 가지 VM 인스턴스 그룹을 제공한다
- 관리 형 인스턴스 그룹 : 동일한 여러 VM에서 앱을 운영 할 수 있다. 자동 확장, 자동 복구, 지역 (다중 영역) 배포 및 자동 업데이트를 포함한 자동화 된 MIG 서비스를 활용하여 워크로드를 확장 가능하고 가용성이 높게 만들 수 있다.
- 비 관리 형 인스턴스 그룹 : 직접 관리하는 VM 집합에 부하를 분산 할 수 있다

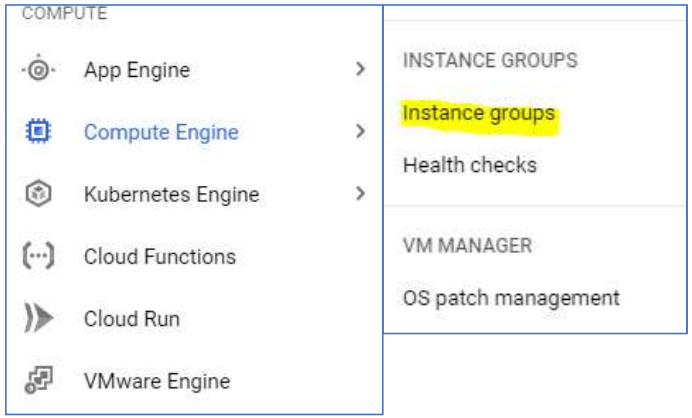
How it works

- Short version (managed instance groups):
  a. Create instance template
  b. Create instance group from instance template
- Instance Template
  - Defines group configuration
  - Machine type, zone, image, scripts
  - Re-usable for multiple group configurations
  - Instance template = Global – not region bound
  - Can specify zonal resources (i.e. read only disk), which effectively binds it
- From template – create managed instance group
  - Instance group = Regional, can use more than one zone in single group



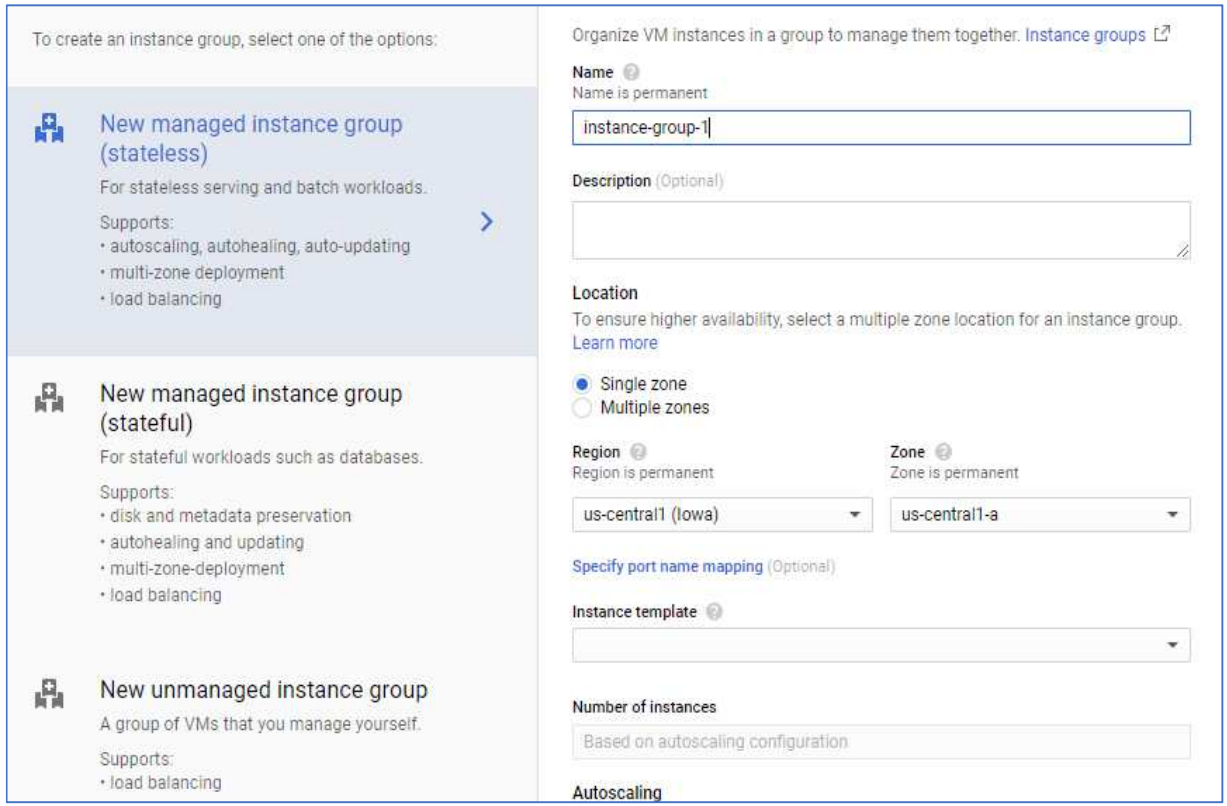
Compute Engine : 인스턴스 그룹 만들기(Lab)

- Compute Engine → Instance groups를 선택한 다음 CREATE INSTANCE GROUP 버튼을 누른다



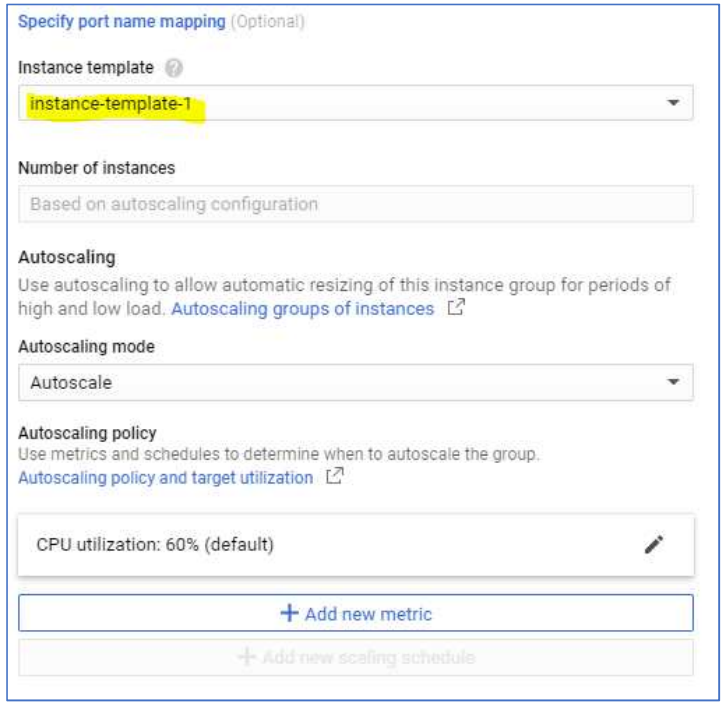


- New managed instance group(stateless)를 선택 하고 Name은 instance-group-1으로 한다(기본값으로 사용)

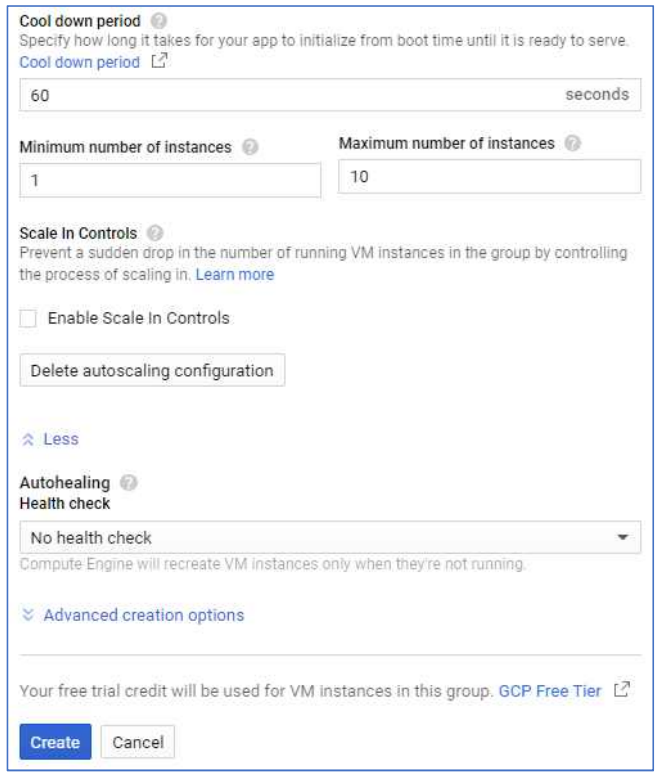


Compute Engine : 인스턴스 그룹 만들기(Lab)

- Instance template 을 미리 만들어 놓은 instance-template-1으로 선택한다 버튼을 누른다

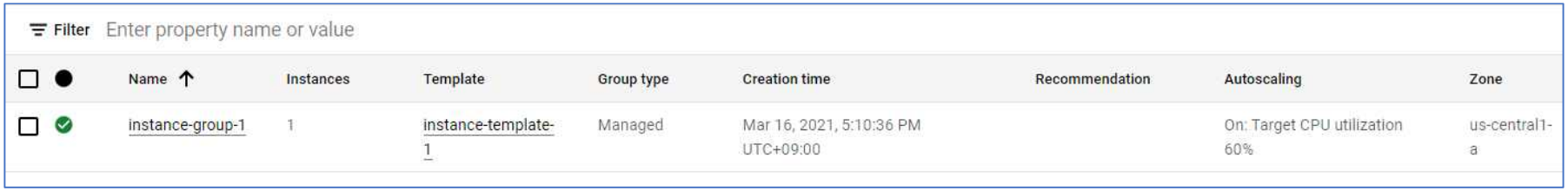


- 아래와 같이 기본 설정으로 놔두고 Create 버튼을 눌러 인스턴스 그룹 작성을 완료한다



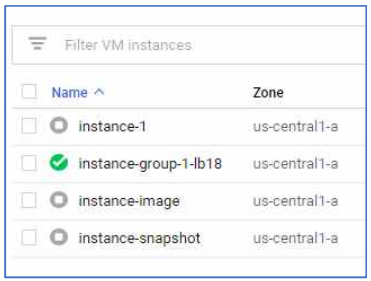
Compute Engine : 인스턴스 그룹 만들기(Lab)

- 인스턴스 그룹이 만들어지면 아래와 같이 목록에 나타난다



- Compute Engine아래 VM Instance 에서도 아래와 같이 추가 된 것을 확인할 수 있다
- 오토 스케일링 : 부하가 늘어나면 자동으로 인스턴스의 수가 늘어 난다





VPC
(Virtual Private Cloud)



VPC (Virtual Private Cloud)

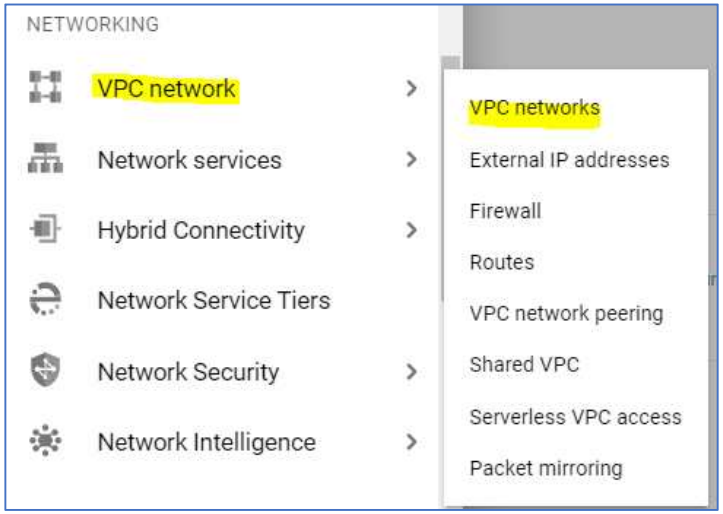
- VPC는 GCP 리소스를 위한 관리형 네트워크 기능을 제공

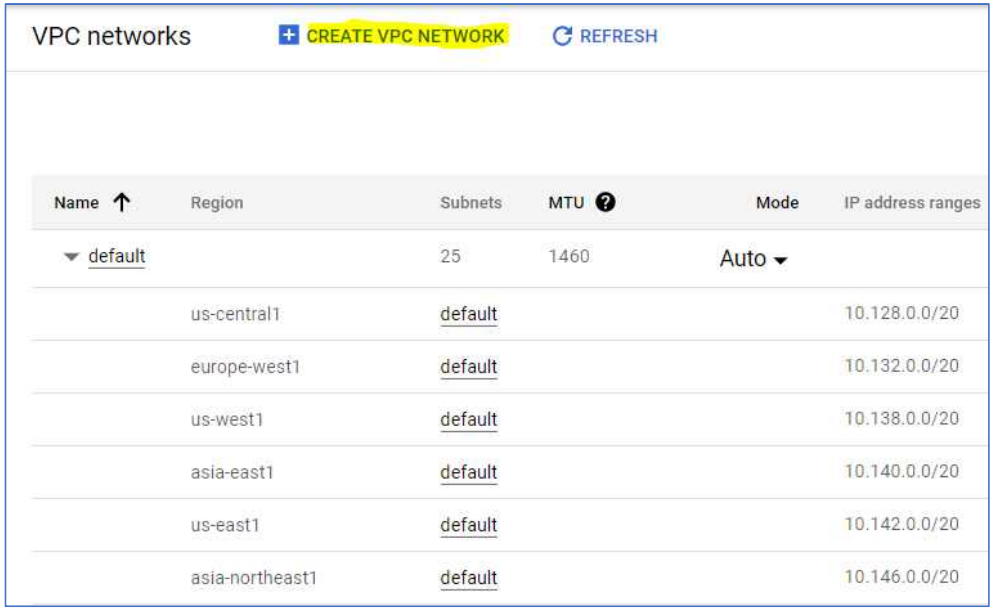
What is a Virtual Private Cloud (VPC)?: "The basics"

- Software-Defined Network (SDN)
  - Virtual version of traditional physical networks
- Terms 'VPC' and 'network' are interchangeable
- Central foundation of all other networking functions on GCP
- **Global** (multi-regional) communications space, private communication among resources
- Create subnetworks from single VPC
  - Subnets are **regional**. Can have single subnet span multiple zones.
- Project based, but can share between projects with Shared VPC
- Traditional networking concepts apply
  - Firewalls, routes, load balancing, DNS, etc.
- Traffic to/from instances controlled with **firewall rules**.
- VPC instances can have both **external** and **internal** IP addresses
  - Assign to a network and subnet (if not default)
- Hybrid networking with on-premises networks with interconnect options
- Can configure private (internal-only) access to other GCP resources
- Incoming (**ingress**) traffic is free. Outgoing (**egress**) traffic has a cost

VPC : VPC 네트워크 구현(Lab)

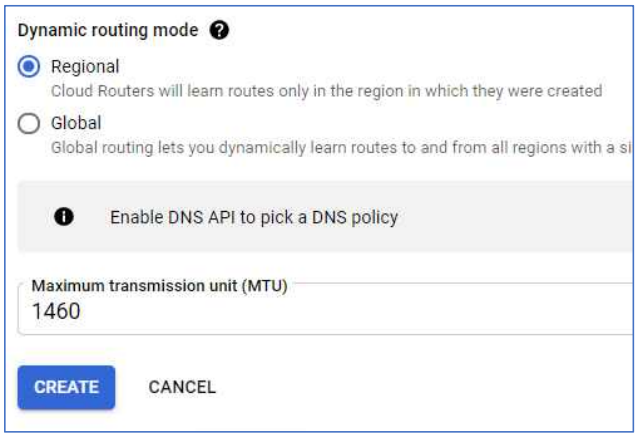
- GCP의 기본(default) 네트워크이외의 자신만의 설정을 가진 네트워크를 구현한다
- GCP Console로 가서 VPC network안의 VPC networks를 선택하고 다음 화면에서 상단의 CREATE VPC NETWORK를 누른다

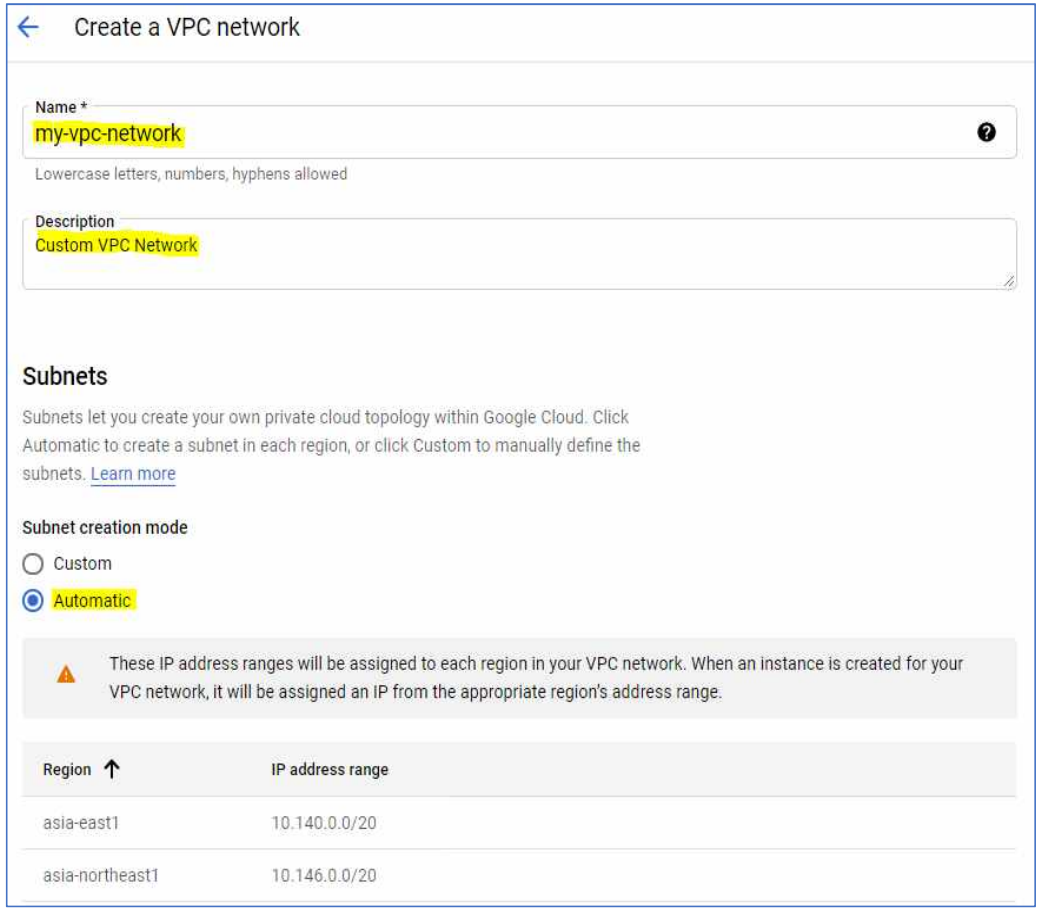




VPC : VPC 네트워크 구현(Lab)

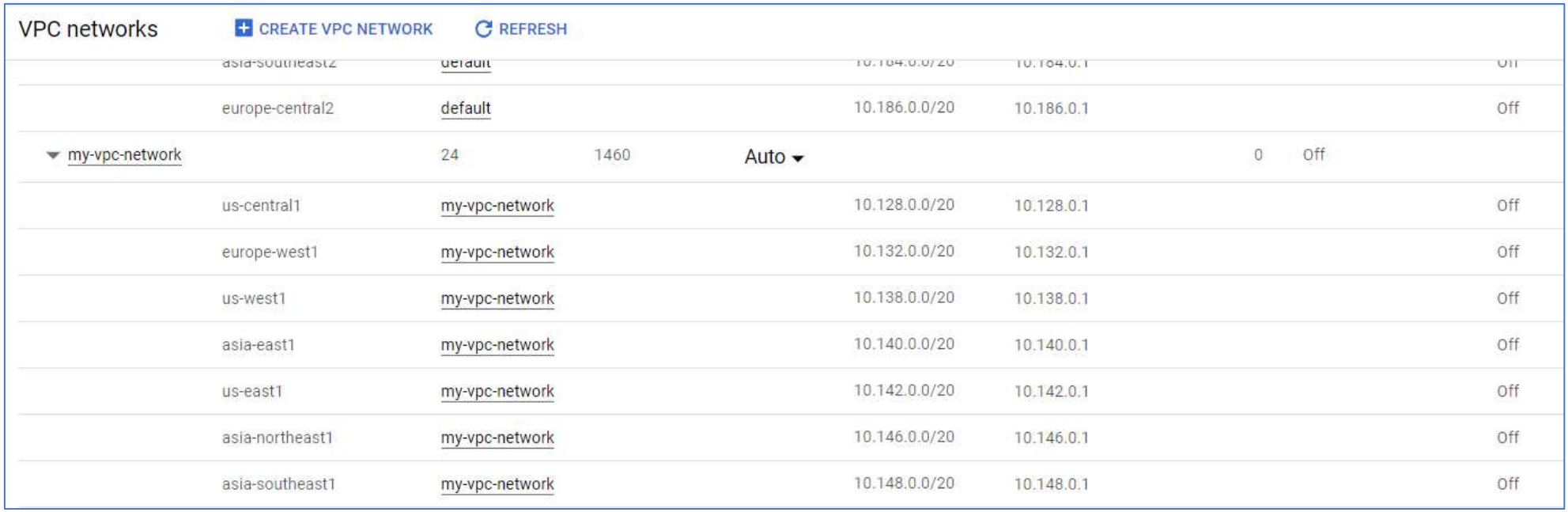
- Name 을 my-vpc-network 으로 입력하고 Description은 Custom VPC Network로 입력한다
- Subnet creation mode는 Automatic으로 선택하여 준다
- 하단의 CREATE버튼을 눌러준다





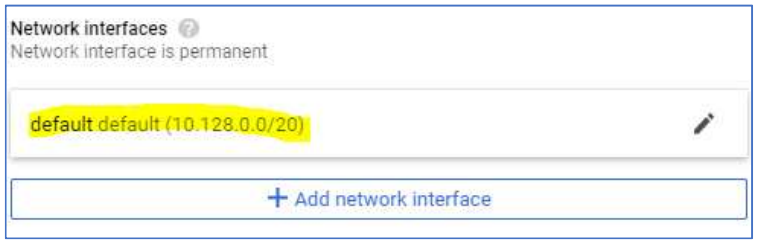
VPC : VPC 네트워크 구현(Lab)

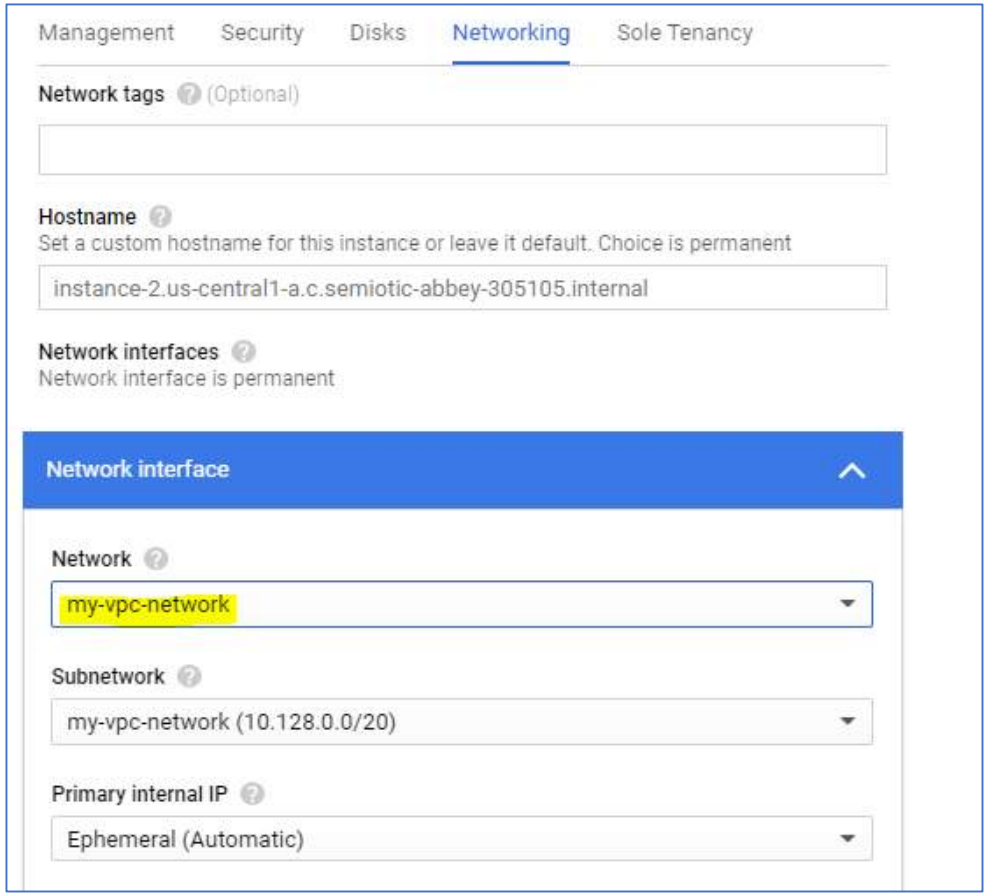
- VPC 네트워크가 생성 완료되면 상단의 default 네트워크 이외에 아래와 같이 새로운 네트워크가 추가된 것을 확인할 수 있다



VPC : VPC 네트워크 구현(Lab)

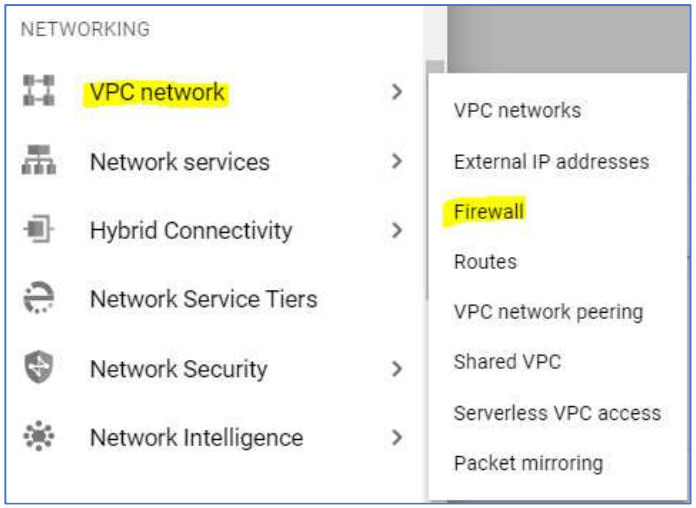
- Compute Engine → VM instances 메뉴를 선택하고 상단의 CREATE INSTANCE를 클릭해준다.
  맨 아래의 Management, security, disks, networking, sole tenancy를 클릭하고 Networking 탭을 선택하고 Network interfaces 부분을 선택하면 왼쪽과 같이 default 이외에 새로 추가한 네트워크를 볼 수가 있다

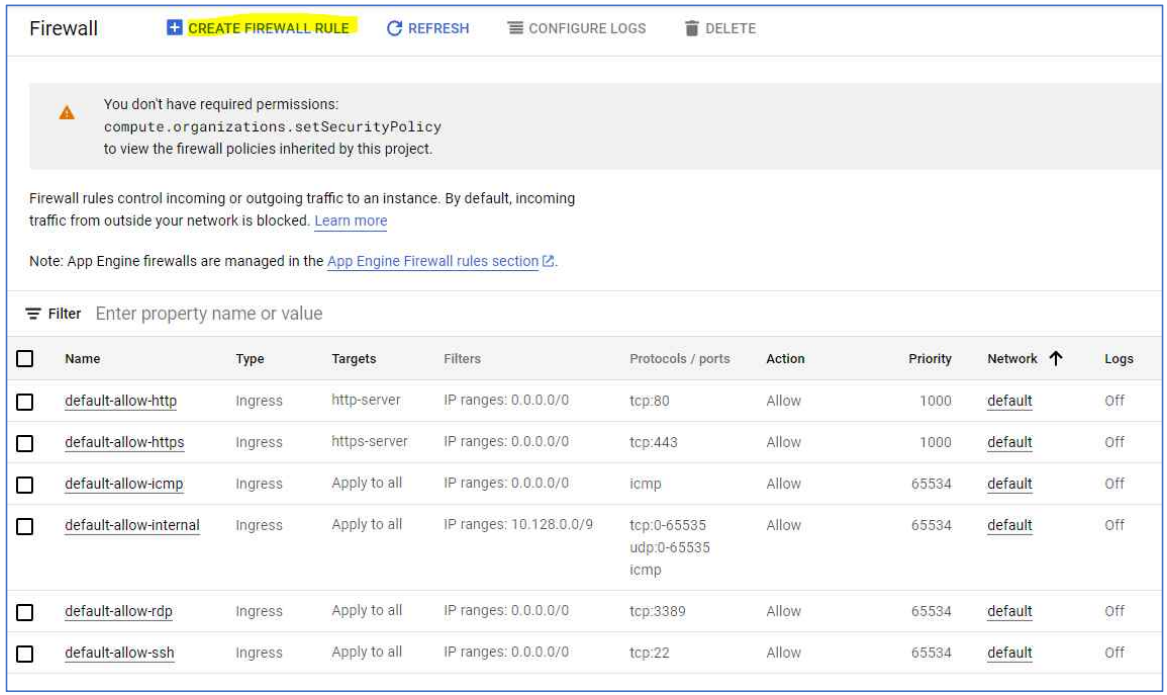




VPC : 방화벽 규칙 만들기(Lab)

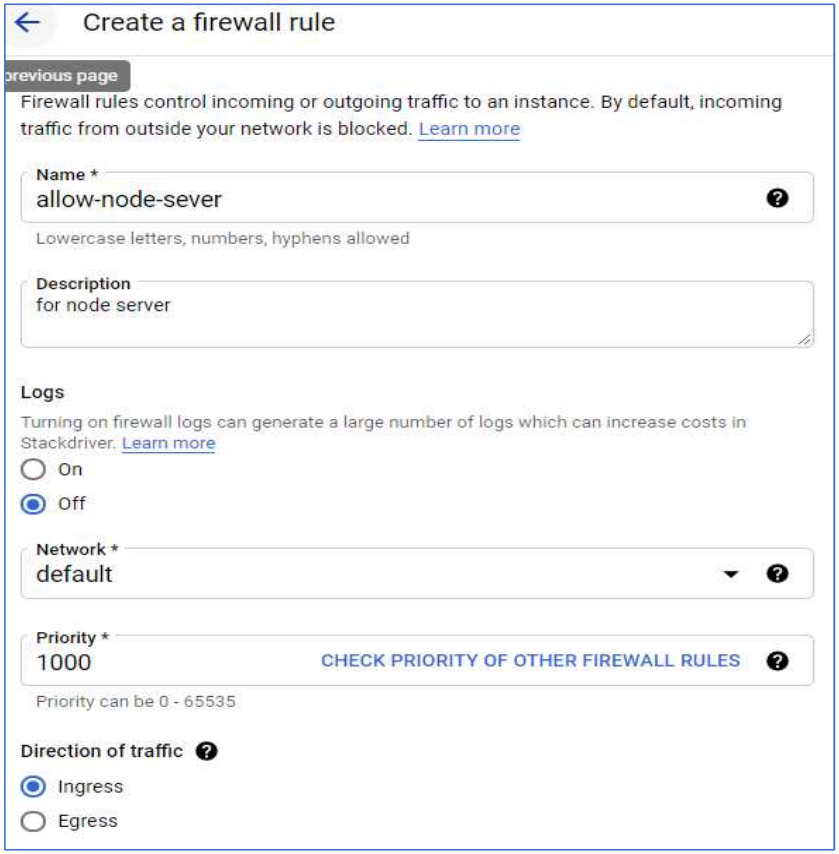
- GCP Console로 가서 VPC network안의 Firewall을 선택하고 다음 화면에서 상단의 CREATE FIREWALL RULE을 누른다

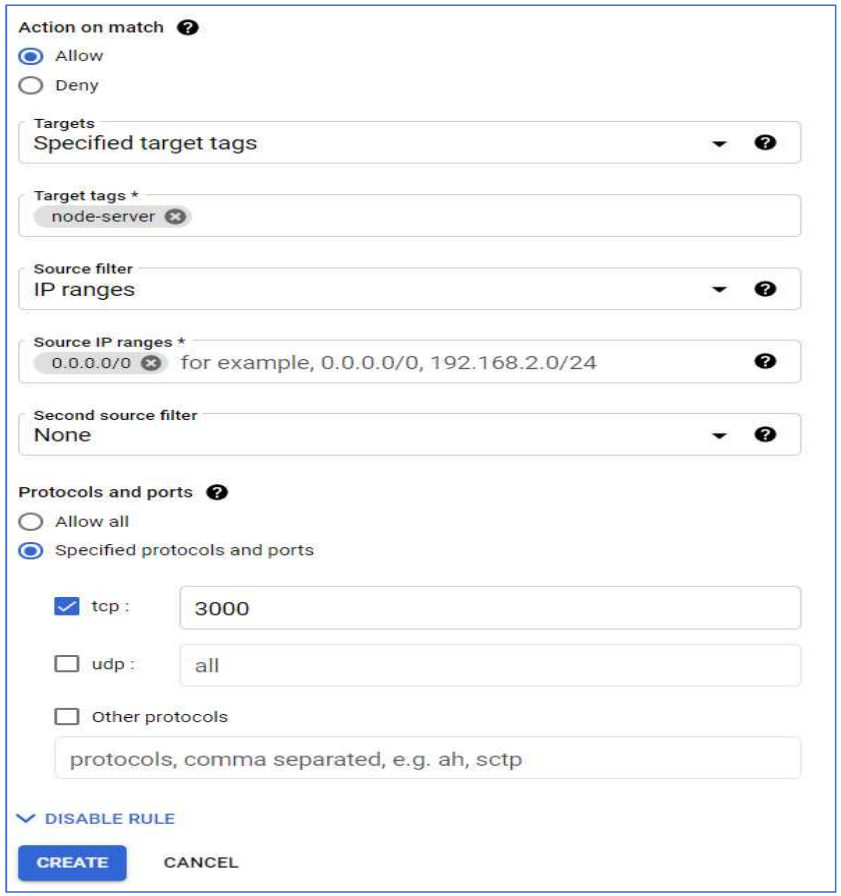




VPC : 방화벽 규칙 만들기(Lab)

- 아래와 같이 선택하고 CREATE 버튼을 누른다





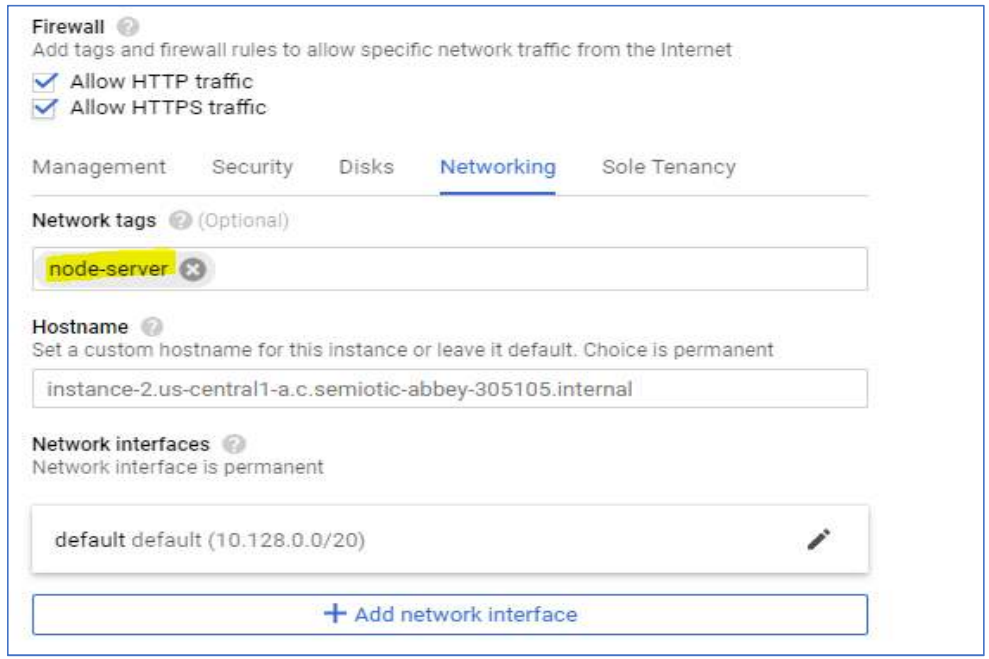
VPC : 방화벽 규칙 만들기(Lab)

- 아래와 같이 방화벽 규칙이 생성된 것을 확인할 수 있다

Filter Enter property name or value

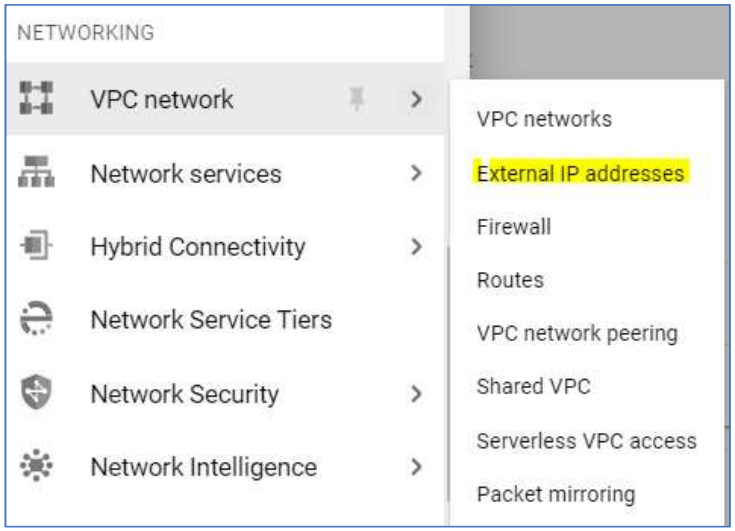
Name	Type	Targets	Filters	Protocols / ports	Action	Priority	Network	Logs
allow-node-sever	Ingress	node-server	IP ranges: 0.0.0.0/0	tcp:3000	Allow	1000	default	Off
default-allow-http	Ingress	http-server	IP ranges: 0.0.0.0/0	tcp:80	Allow	1000	default	Off

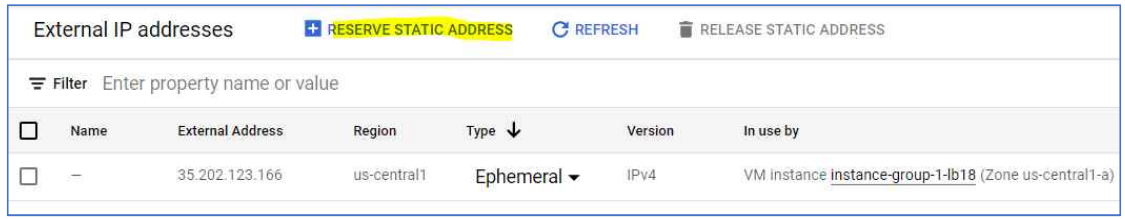
- Compute Engine → VM instances 메뉴를 선택하고 상단의 CREATE INSTANCE를 클릭해준다.
  맨 아래의 Management, security, disks, networking, sole tenancy를 클릭하고 Networking 탭을 선택하고 Network tags에 왼쪽과 같이 태그를 입력하고 CREATE 버튼을 눌러 방화벽 규칙이 적용된 인스턴스를 만들 수 있다



VPC : 외부 고정 IP 만들기(Lab)

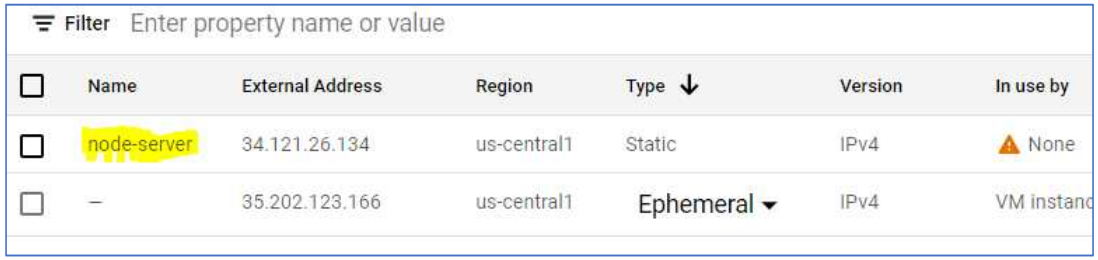
- GCP Console로 가서 VPC network안의 External IP addresses를 선택하고 다음 화면에서 상단의 RESERVE STATIC ADDRESS를 누른다

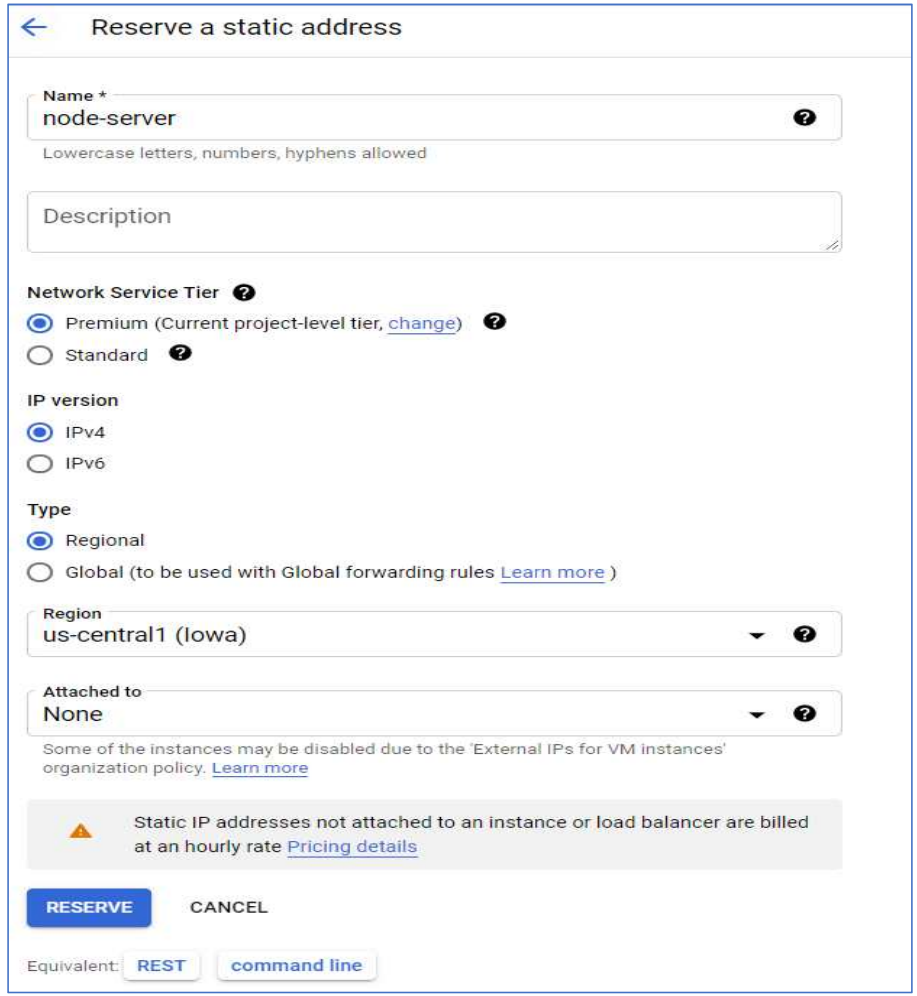




VPC : 외부 고정 IP 만들기(Lab)

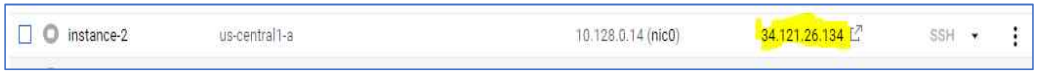
- Name을 node-server로 입력하고 기본 설정 그대로 하단의 RESERVE 버튼을 누른다
- 아래와 같이 임의의 외부 고정 IP 주소가 만들어 진다

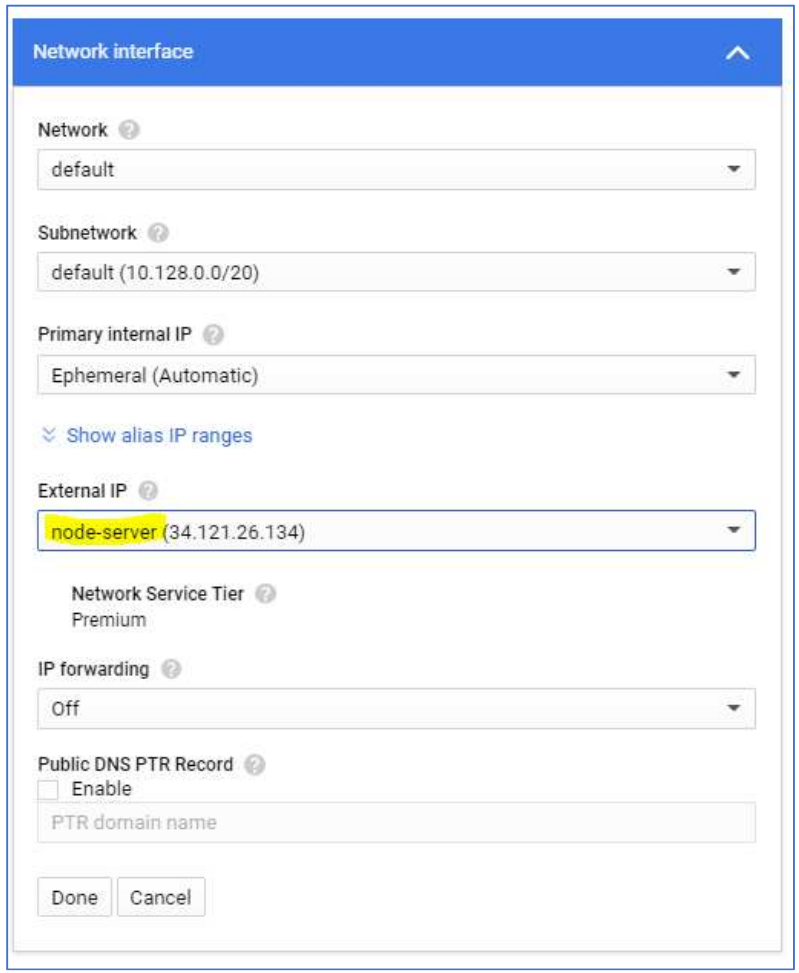




VPC : 외부 고정 IP 만들기(Lab)

- Compute Engine → VM instances 메뉴를 선택하고 상단의 CREATE INSTANCE를 클릭해준다.
  맨 아래의 Management, security, disks, networking, sole tenancy를 클릭하고 Networking 탭을 선택하고 External IP에 왼쪽과 같이 node-server를 선택하고 Done 버튼을 누르고 하단의 Create 버튼을 눌러 인스턴스를 만든다
  이 인스턴스는 VM을 중지(STOP)시키고 다시 시작해도 IP가 변하지 않게 된다





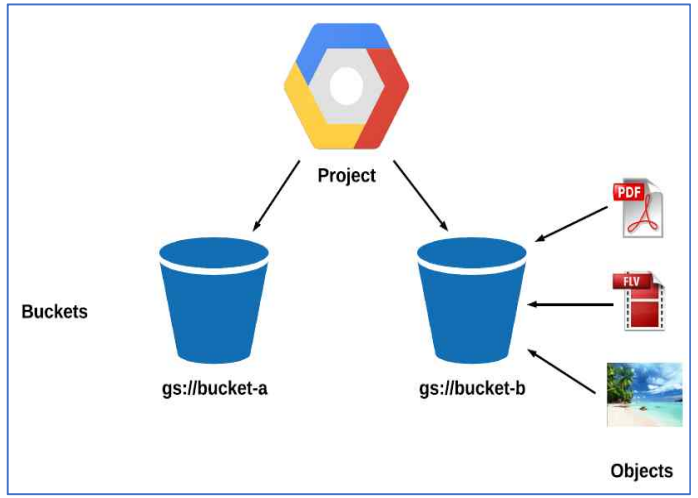
GCS
(Google Cloud Storage)



GCS : Google Cloud Storage

- GCS : GCP 제공 객체 Repository

- Storage for unstructured data
  - What can a bucket hold? Everything!
  - Pictures, videos, files, scripts, etc.
  - Even databases for transfer/backup (or BigQuery read)
- Virtually infinite size
- Pay-per-use, not allocation (elastic)
- Primary unit is a bucket
  - Access managed via IAM
  - Can be arranged in file/folder format for organizational purposes
- Objects go inside of the bucket
  - Objects = files
  - Inherit bucket permissions and storage class
  - **Note**: Folders are also considered objects
- GCS is not a file system/block storage
  - Technically an object storage system
  - Different write/performance characteristics
  - Block storage = persistent disk (useful for SANs)



GCS : Storage Classes

- GCS 는 등급별 가격이 다르다

Multi-Regional (standard)	Span multiple regions Hot data from different geographical locations No retrieval costs 99.95% availability SLA	$0.026/GB/mo
Regional (standard)	Single region Hot data in single geographical location Highest regional performance No retrieval costs 99.9% availability SLA	$0.02/GB/mo
Nearline	Regional or multi-regional Archive data accessed once per month Lower monthly cost with some retrieval costs Minimum 30 day duration Ideal for backups 99.9% SLA multi-regional, 99% regional	$0.01/GB/mo $0.01 per GB retrieval
Coldline	Regional or multi-regional Archive data accessed once per year Even lower monthly costs with higher retrieval costs Minimum 90 day duration Ideal for 'cold' storage - legal compliance, disaster recovery 99.9% SLA multi-regional, 99% regional	$0.007/GB/mo $0.05 per GB retrieval

GCS : gsutil 명령어

- gsutil은 CLI(command line interface) 환경에서 GCS에 접근하는 명령

What is gsutil?

- Command line format for working with Cloud Storage buckets/objects
- Different format compared to gcloud commands
- Useful for:
  - Creating and deleting buckets
  - Uploading, downloading, and deleting objects
  - Listing buckets and objects
  - Moving, copying, and renaming objects
  - Editing object and bucket ACLs

Accessing buckets

Bucket syntax = `gs://<BUCKET_NAME>/<OBJECT_NAME>`

gsutil Command Structure

- `gsutil <command> <options> <target>`
- `gsutil <command> <options> <source> <target>`
- Very similar to Linux commands
  - `cp`, `mv`, `rm`, `ls`; same for both Linux and `gsutil`
  - Almost identical structure
- Example: `gsutil cp file1.txt gs://mybucket`
  - `gsutil ls -a gs://mybucket`
- Top-level command options (before the command)
  - `gsutil -m cp -r gs://<BUCKET> <target>`

GCS : gsutil 명령어

https://cloud.google.com/storage/docs/gsutil

- gsutil help 명령으로 gsutil 명령어 리스트를 확인할 수 있다

```
C:₩Program Files (x86)₩Google₩Cloud SDK>gsutil help


Updates are available for some Cloud SDK components.  To install them,
please run:
  $ gcloud components update

Usage: gsutil [-D] [-DD] [-h header]... [-i service_account] [-m] [-o] [-q] [-u user_project] [command [opts...] args...]
Available commands:
  acl               Get, set, or change bucket and/or object ACLs
  bucketpolicyonly  Configure uniform bucket-level access
  cat               Concatenate object content to stdout
  compose           Concatenate a sequence of objects into a new composite object.
  config            Obtain credentials and create configuration file
  cors              Get or set a CORS JSON document for one or more buckets
  cp                Copy files and objects
  defacl            Get, set, or change default ACL on buckets
  defstorageclass   Get or set the default storage class on buckets
  du                Display object size usage
  hash              Calculate file hashes
  help              Get help about commands and topics
  hmac              CRUD operations on service account HMAC keys.
  iam               Get, set, or change bucket and/or object IAM permissions.
  kms               Configure Cloud KMS encryption
  label             Get, set, or change the label configuration of a bucket.
  lifecycle         Get or set lifecycle configuration for a bucket
  logging           Configure or retrieve logging on buckets
  ls                List providers, buckets, or objects
  mb                Make buckets
  mv                Move/rename objects
  notification      Configure object change notification
  perfdiag          Run performance diagnostic
  rb                Remove buckets
  requesterpays     Enable or disable requester pays for one or more buckets
  retention         Provides utilities to interact with Retention Policy feature.
  rewrite           Rewrite objects
  rm                Remove objects
  rsync             Synchronize content of two buckets/directories
  setmeta           Set metadata on already uploaded objects
  signurl           Create a signed url
  stat              Display object status
  test              Run gsutil unit/integration tests (for developers)
  ubla              Configure Uniform bucket-level access
  update            Update to the latest gsutil release
  version           Print version info about gsutil
  versioning        Enable or suspend versioning for one or more buckets
  web               Set a main page and/or error page for one or more buckets
```

GCS : 버킷 만들기 실습

- 강사 안내 참조

Stackdriver



Stackdriver : 모니터링과 로깅 서비스

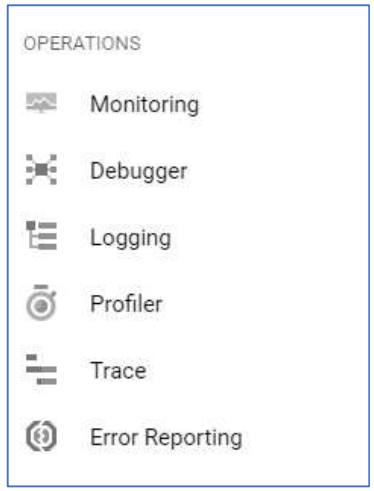
What is Stackdriver?

- Suite of tools for monitoring, logging, and tracking diagnostics for your applications
- Native monitoring of both GCP and AWS
- Connect via service accounts/APIs on other platforms/on-premises
- Dynamically discover all GCP resources
- Install Stackdriver client on VMs for even greater levels of monitoring

Stackdriver : 주요기능

Five Different Products

- Monitoring
  : Monitor metrics, health checks, dashboards, and alerts
- Debugger
  : Find/fix code errors in production
- Logging
  : Central collection for all activity logs
- Profiler
  : Collect CPU/memory data? optimize performance
- Trace
  : Find latency bottlenecks in applications
- Error Reporting
  : Identify and understand application errors



Stackdriver : Logging

Pricing

- First 50 GB/project/mo free—$0.50/GB after that
- Admin and system event logs exempt
- Formerly Standard and Premium tiers—gone as of July 2018

Retention

Log Type	Retention Period
Admin Activity	400 days
Data Access	30 days
System Event	400 days
Access Transparency	400 days
All other logs	30 days

Exporting logs

- After retention period, logs are deleted and cannot be recovered!
- Export logs for long term retention
- Long term storage (Cloud Storage), big data analysis (BigQuery), stream to other sources (Pub/Sub)

Stackdriver : Monitoring

What is it?

- Full-stack monitoring, powered by Google
- What is up? What is down? What is overloaded?
- Native monitoring of GCP, AWS, and third-party applications
- Monitor system and application metrics
- Interacts with Stackdriver Logging
- Easy to view insights with dashboards and alerts
- Uptime checks on external applications
  - Must be public and allow firewall traffic from uptime source IP addresses

PRICING

Feature	Price	Free allotment per month
Monitoring Data	$0.2580/MB: 150–100,000 MB $0.1510/MB: 100,000–250,000 MB $0.0610/MB: above 250,000 MB	All GCP Metrics First 150 MB per billing account
Monitoring API calls	$0.01/1000 API calls	First million API calls

Stackdriver : 모니터링과 로깅 실습

- 강사 안내 참조